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(54) **CANINIZED ANTIBODIES**

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(57) **ABSTRACT**

The present invention discloses caninized antibodies with specific properties. The present invention also discloses caninized murine antibodies against canine PD-1 that have a high binding affinity for canine PD-1. The invention further discloses the use of the caninized antibodies of the present invention in the treatment of cancer in dogs.

Specification includes a Sequence Listing.

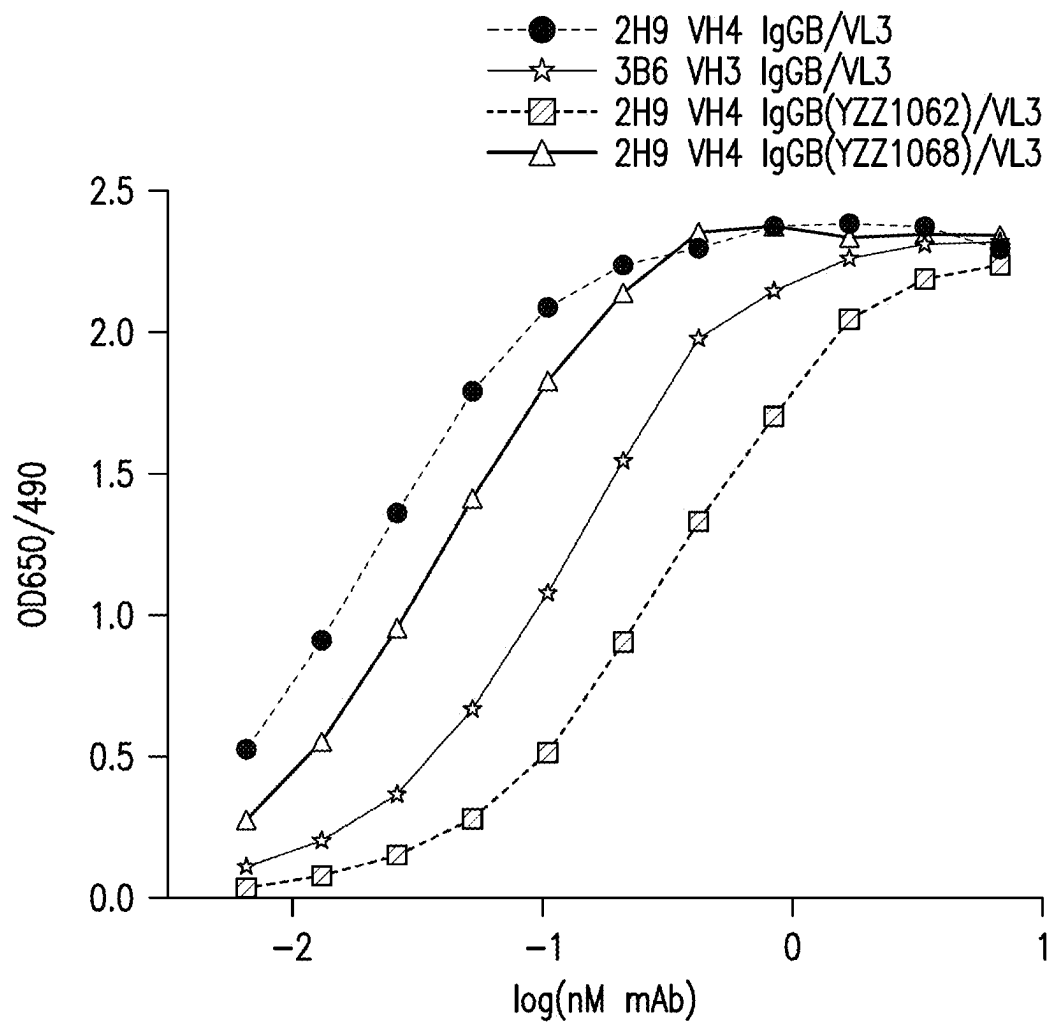


FIG. 1

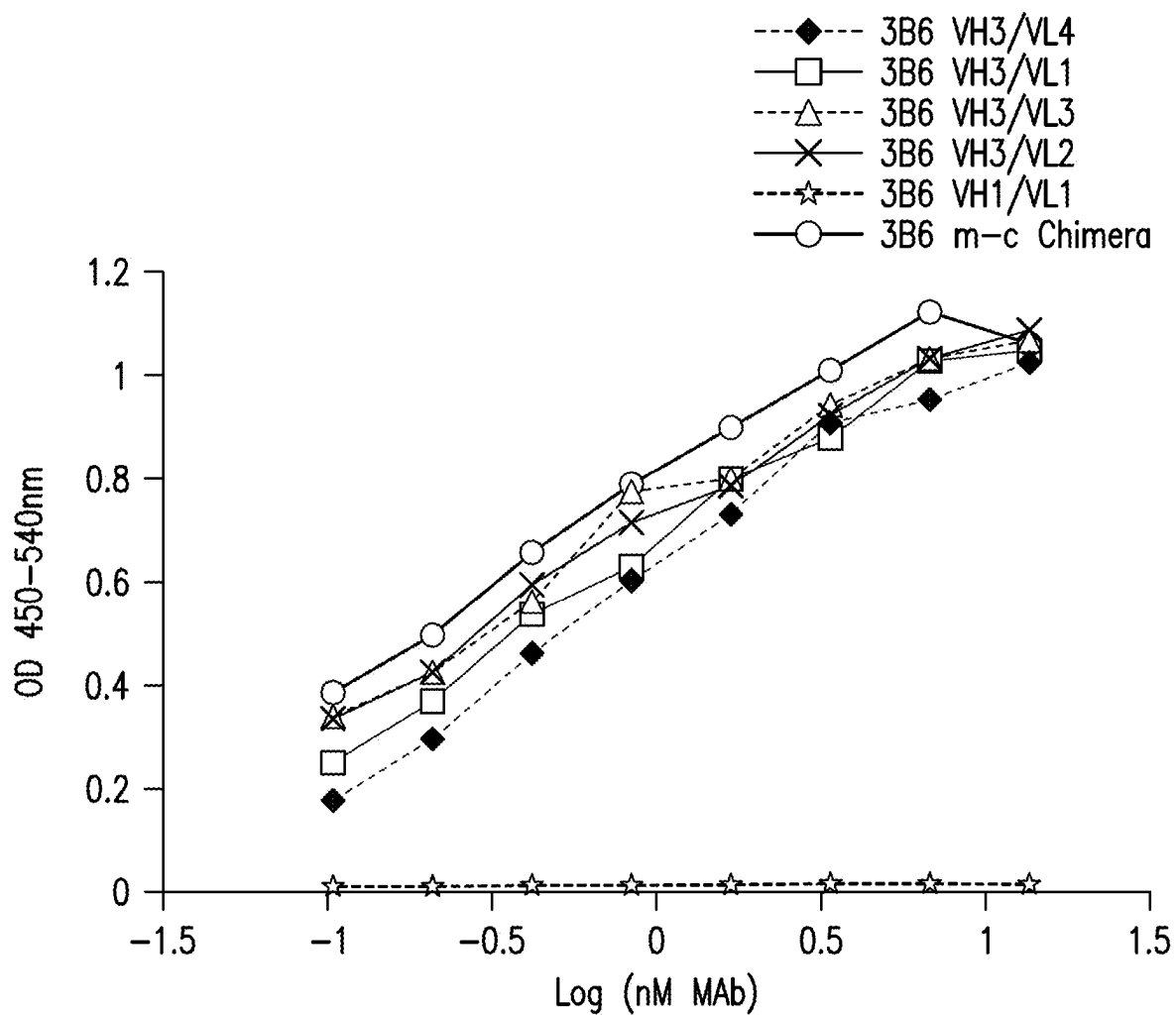


FIG.2

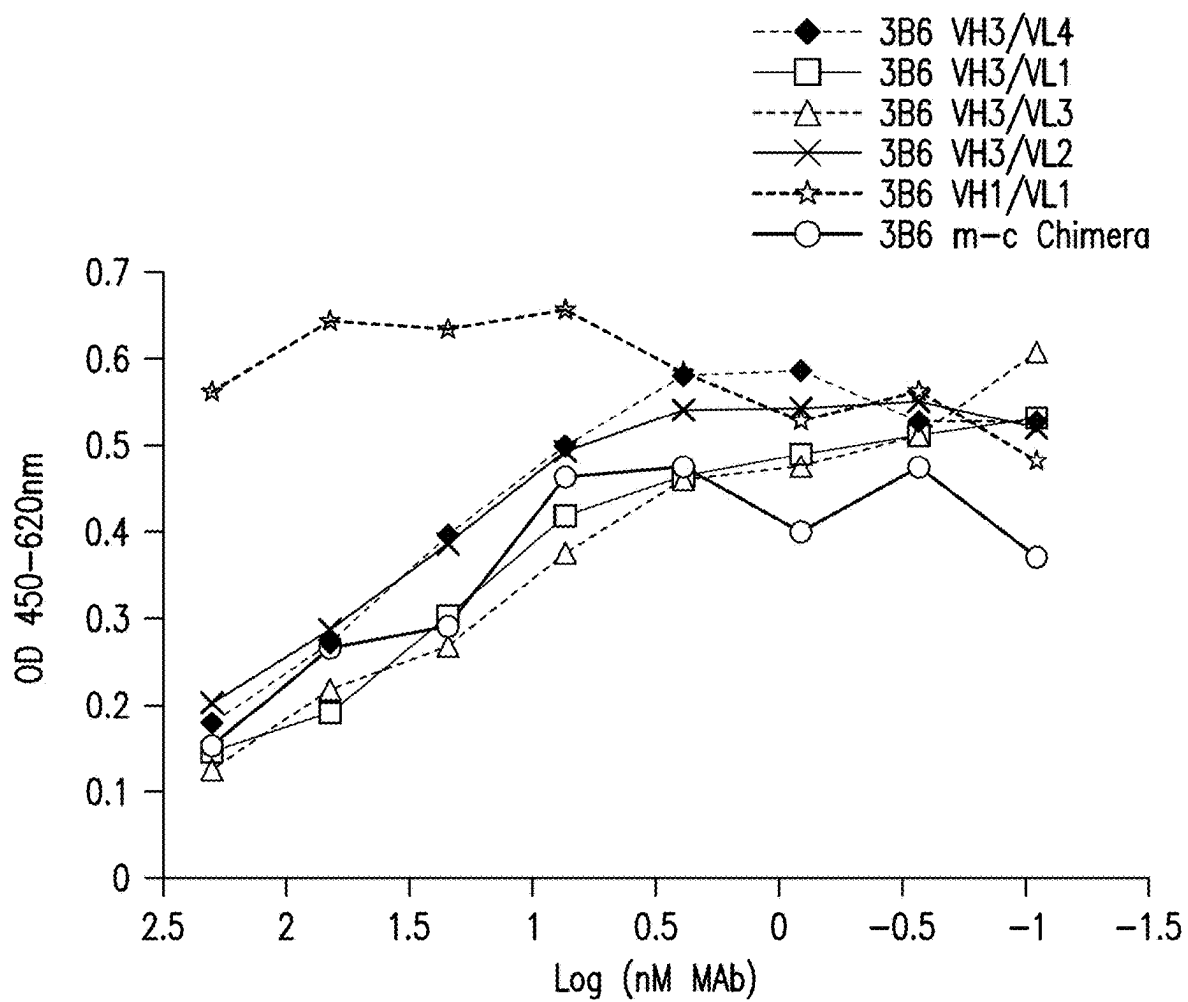


FIG.3

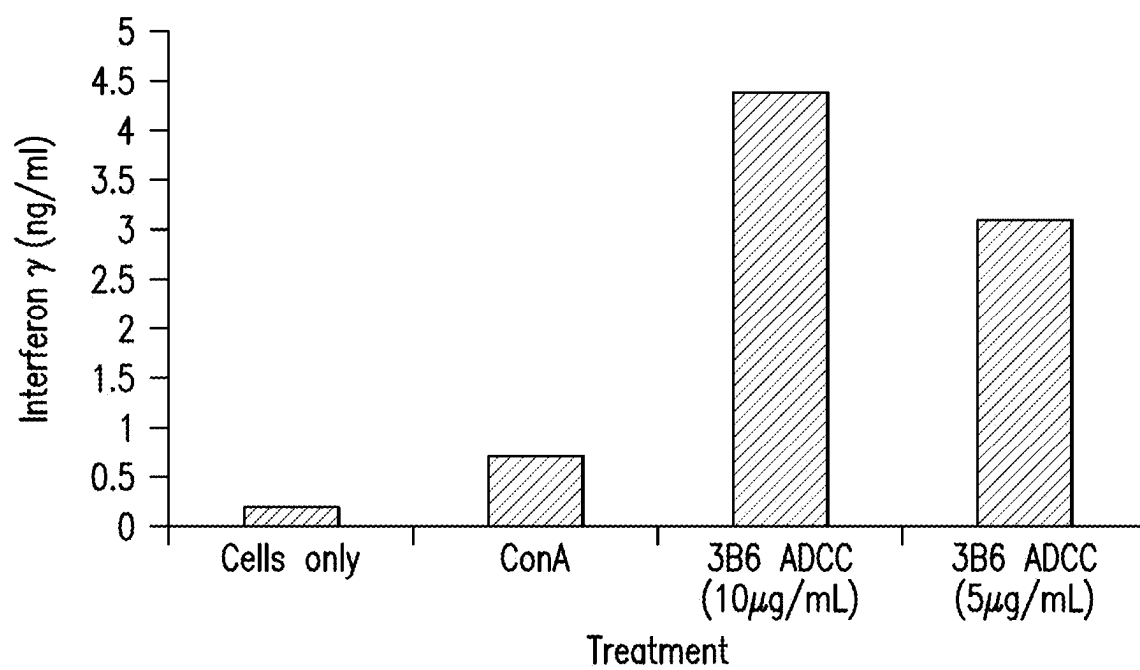


FIG.4

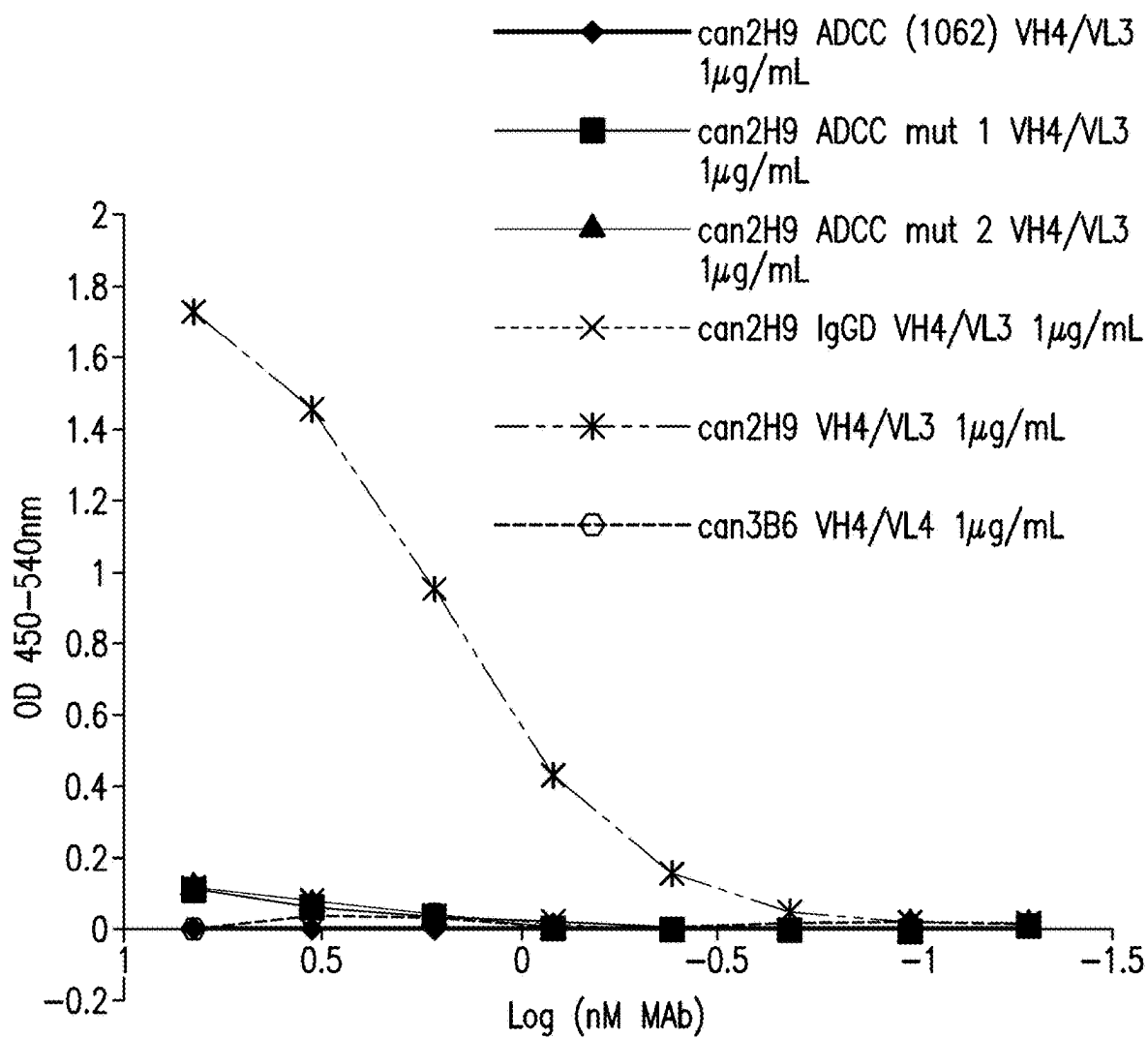


FIG.5A

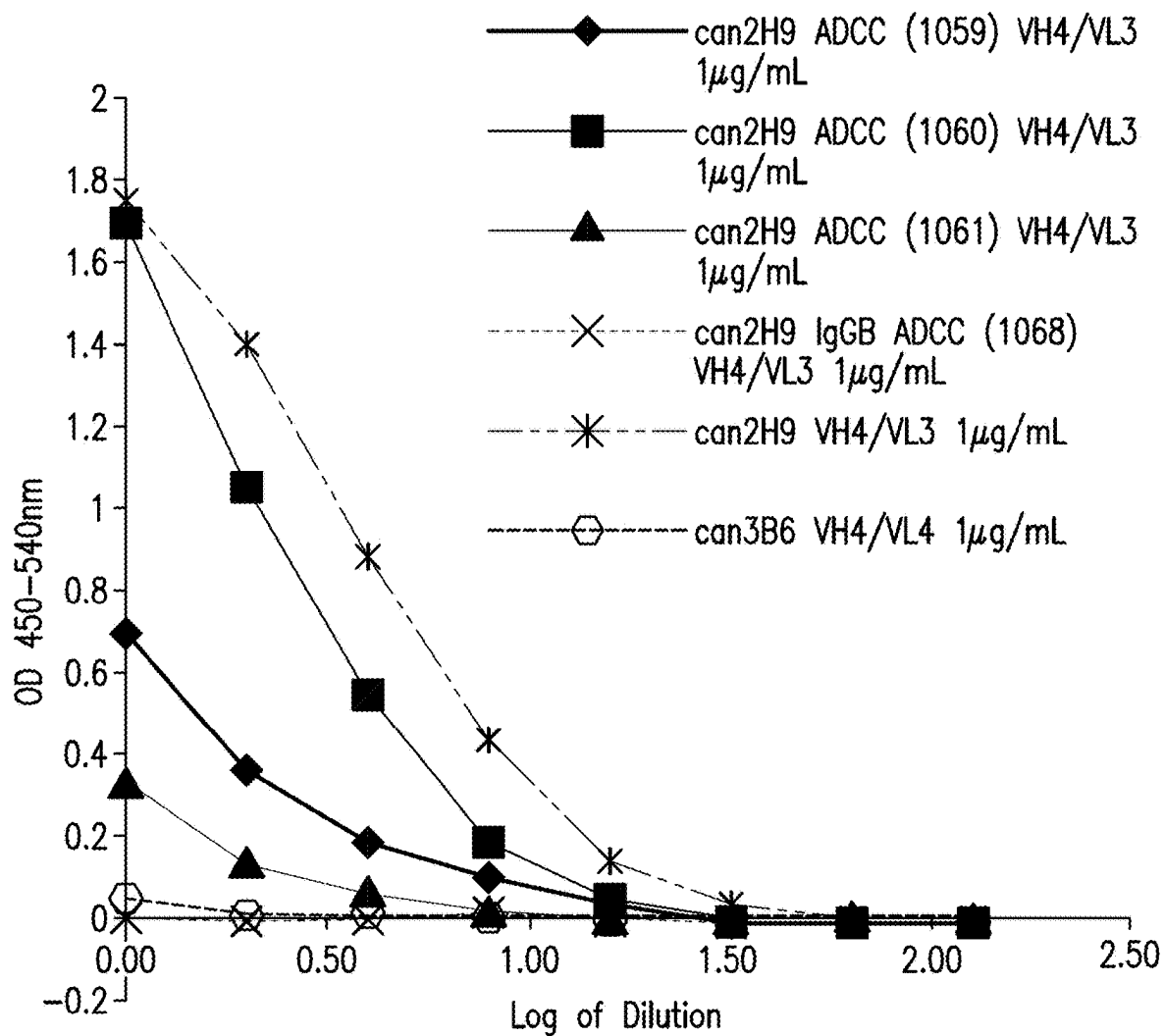


FIG.5B

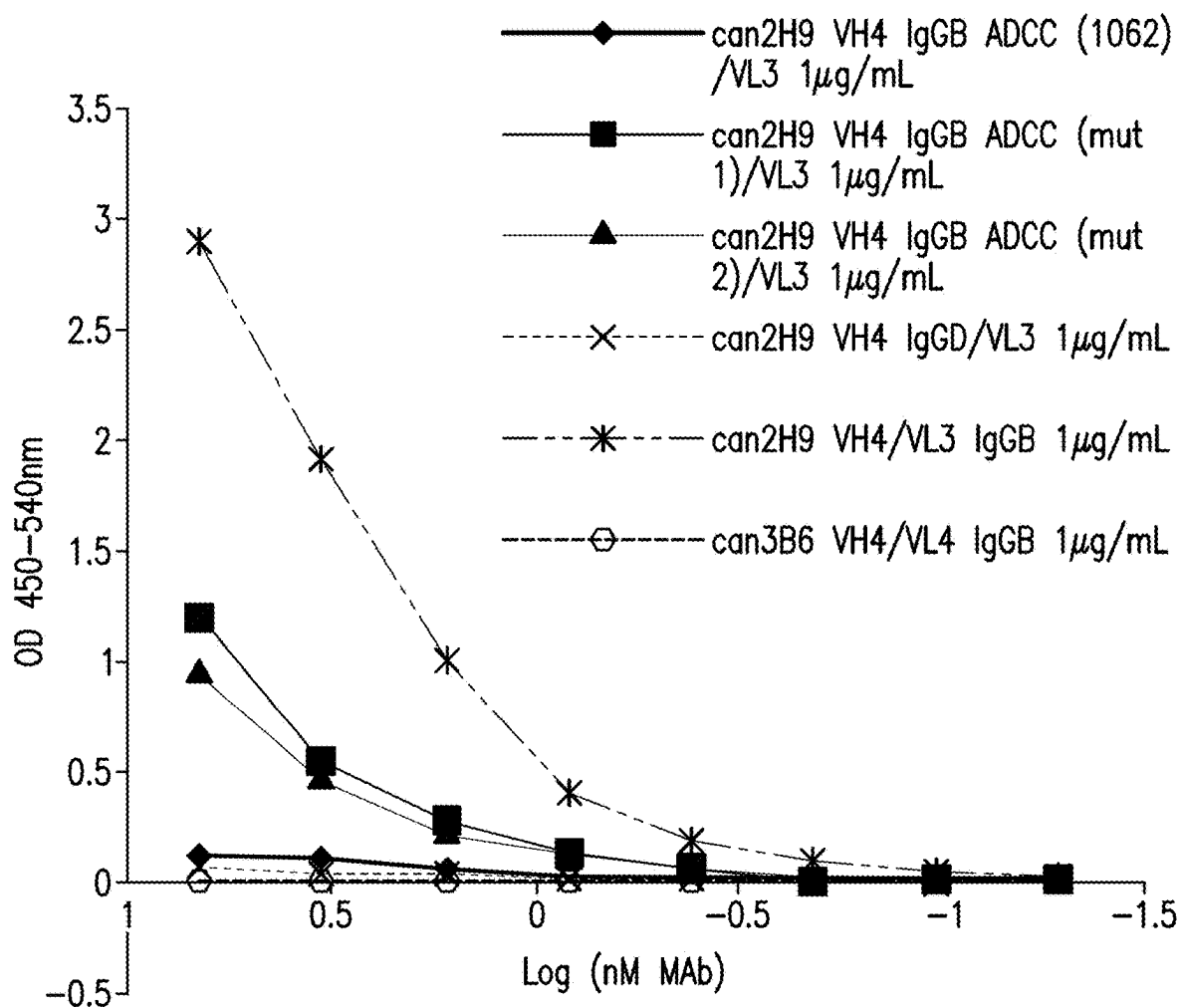


FIG.6A

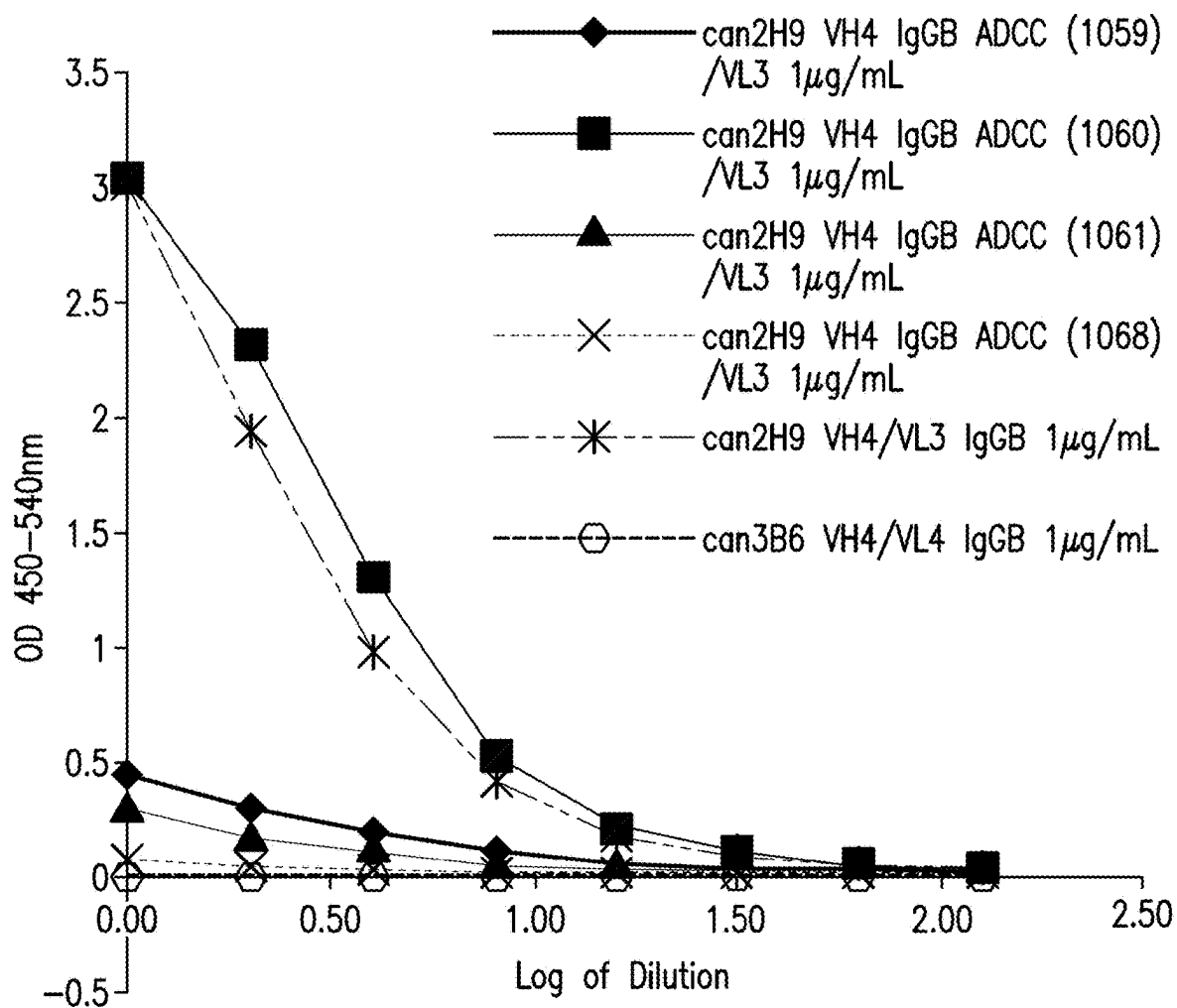


FIG.6B

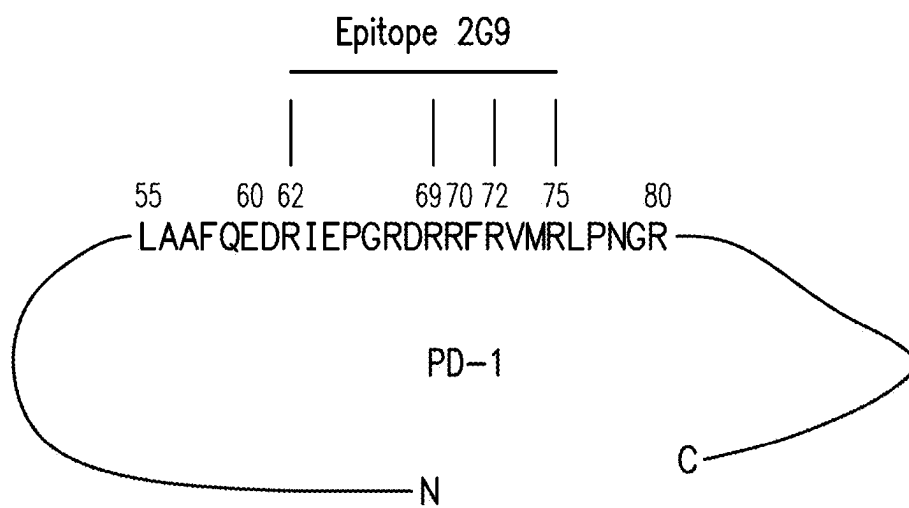


FIG.7A

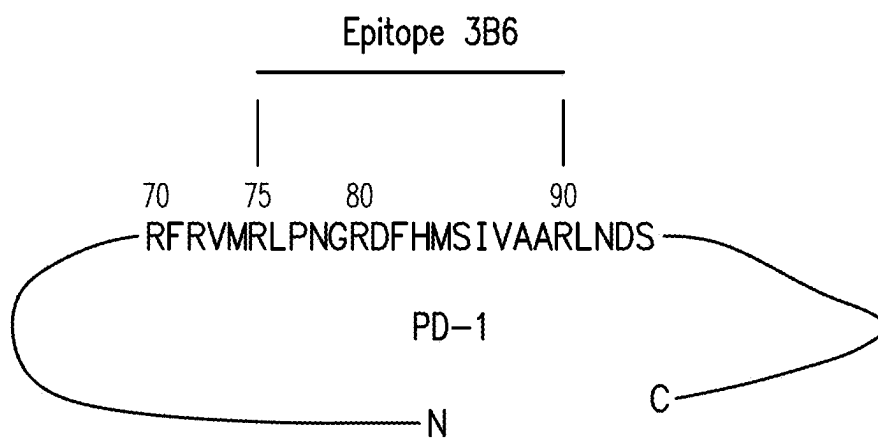


FIG.7B

CANINIZED ANTIBODIES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/573,870, filed Jan. 12, 2022, now allowed, which is a division of U.S. patent application Ser. No. 16/130,578, filed Sep. 13, 2018, now U.S. Pat. No. 11,248,047, which is a division of U.S. patent application Ser. No. 15/105,211 filed Jun. 16, 2016, now U.S. Pat. No. 10,106,607, which is a National Stage Entry Under 371 of PCT/EP2014/078653, filed Dec. 19, 2014, which claims benefit of U.S. Provisional Application Ser. No. 62/030,812, filed on Jul. 30, 2014, U.S. Provisional Application Ser. No. 61/918,847, filed on Dec. 20, 2013, and U.S. Provisional Application Ser. No. 61/918,946, filed on Dec. 20, 2013, the contents of all of which are hereby incorporated by reference in their entireties.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

[0002] The contents of the electronic sequence listing (23799USCNT_SL.xml; Size: 308,735 bytes; and Date of Creation: Jan. 31, 2025) are herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0003] The present invention relates to caninized antibodies with specific properties. The present invention also relates to caninized antibodies against canine PD-1 that have specific sequences and a high binding affinity for canine PD-1. The invention further relates to use of the antibodies of the present invention in the treatment of dogs, including cancer treatment.

BACKGROUND OF THE INVENTION

[0004] Canine antibodies (also referred to as immunoglobulin G or IgG) are large tetrameric proteins of about 150 Kd. Each IgG protein is composed of two identical light chains of about 25 Kd each, and two identical heavy chains of about 50 Kd each. There are four known IgG heavy chain subclasses of canine IgG and they are referred to as IgGA, IgGB, IgGC, and IgGD. There are two types of light chains; kappa and lambda chains. Each of the kappa or lambda light chains is composed of one variable domain (VL) and one constant domain (CL). Each of the two heavy chains consists of one variable domain (VH) and three constant domains referred to as CH-1, CH-2, and CH-3. The CH-1 domain is connected to the CH-2 domain via an amino acid sequence referred to as the “hinge” or alternatively as the “hinge region”. In humans, IgG exists in one of four subclasses referred to as IgG1, IgG2, IgG3, and IgG4. The subclass of IgG is determined largely by the sequence of the hinge region, which differs among the four subclasses of IgG. The two heavy chains are linked to each other by disulfide bonds and each heavy chain is linked to one of the light chains also through a disulfide bond.

[0005] Digestion of IgG antibodies with the enzyme papain breaks the antibody molecule in the hinge region and results in the formation of three fragments. Two of these fragments are identical and each consists of the light chain held together with the VH and CH1 domains of the heavy chain. These fragments are called the “Fab” fragments and

they contain the antigen binding sites of the antibody. The third fragment that results from digestion with papain is called the “Fc” and it contains the remainder of the two heavy chains held together by disulfide bonds. The Fc thus contains a dimer consisting of the CH2 and CH3 domain of each of the two heavy chains. While the Fab enables the antibody to bind to its cognate epitope, the Fc enables the antibody to mediate immune effector functions such as antibody dependent cellular cytotoxicity (ADCC), antibody-dependent phagocytosis (ADCP) and complement dependent cytotoxicity (CDC).

[0006] It is well known in the art that IgG antibodies mediate effector functions such as ADCC and ADCP through binding of their Fc portion to a family of proteins known as Fc γ receptors, whereas CDC is mediated through the binding of the Fc to the first component of complement, C1q. It is also well known in the art that different IgG sub-classes differ in their capacity to mediate these effector functions. For example, human IgG1 displays strong ADCC and CDC, whereas IgG4 displays a weak to no ADCC and CDC. In addition, methods for identification of which IgG sub-classes display or lack effector functions are well known in the art.

[0007] Approaches that rely on use of monoclonal antibodies for therapeutic purposes require the design of fit-for-purpose antibodies or antibody fragments to achieve the desired therapeutic response. For example, some therapeutic approaches for cancer require the therapeutic antibodies to have enhanced effector functions, while others require the effector functions to be significantly reduced or eliminated altogether. Enhancement or elimination of effector functions may be achieved through introduction of one or more amino acid mutations (substitutions) in the Fc portion of the antibody so as to enhance or reduce binding to Fc γ receptors and the first component of complement. There are numerous reports in the prior art describing amino acid substitutions that may be introduced into an antibody molecule in order to modulate its effector functions. For example, Shields et al., [*J. of Biol. Chem.*, 276 (9): 6591-6604 (2001)] disclosed that an asparagine to alanine (N297A) substitution, which result in a non-glycosylated antibody, significantly reduced antibody binding to several Fc γ receptors. Additionally, Shields et al., disclosed that an aspartic acid-to-alanine (D265A) substitution also significantly reduced binding of the antibody to Fc γ receptors. Each of the N297A and D265A substitutions were also shown to significantly impair CDC. There are other similar reports identifying potential substitutions to reduce or eliminate effector function in antibodies [e.g., Sazinsky et al., *Proc. Nat. Acad. Sci.*, 105:20167-20172 (2008), Alegre et al., *Transplantation*, 57:1537-1543 (1994), Hutchins et al., *Proc. Nat. Acad. Sci.* 92:11980-11984 (1994), McEarchem et al., *Blood*, 109:1185-1192 (2007)].

[0008] An immunoinhibitory receptor that is primarily expressed on activated T and B cells, Programmed Cell Death Receptor 1, also referred to as Programmed Death Receptor 1 (PD-1), is a member of the immunoglobulin superfamily related to CD28 and CTLA-4. PD-1 and like family members are type I transmembrane glycoproteins containing an extracellular Ig Variable-type (V-type) domain that binds its ligands and a cytoplasmic tail that binds signaling molecules. The cytoplasmic tail of PD-1 contains two tyrosine-based signaling motifs, an ITIM (immunore-

ceptor tyrosine-based inhibition motif) and an ITSM (immunoreceptor tyrosine-based switch motif).

[0009] PD-1 attenuates T-cell responses when bound to Programmed Cell Death Ligand 1, also referred to as Programmed Death Ligand 1 (PD-L1), and/or Programmed Cell Death Ligand 2, also referred to as Programmed Death Ligand 2 (PD-L2). The binding of either of these ligands to PD-1 negatively regulates antigen receptor signaling. Blocking the binding of PD-L1 to PD-1 enhances tumor-specific CD8⁺ T-cell immunity, while aiding the clearance of tumor cells by the immune system. The three-dimensional structure of murine PD-1, as well as the co-crystal structure of mouse PD-1 with human PD-L1 have been reported [Zhang et al., *Immunity* 20:337-347 (2004); Lin et al., *Proc. Natl. Acad. Sci. USA* 105:3011-3016 (2008)].

[0010] PD-L1 and PD-L2 are type I transmembrane ligands that contain both IgV- and IgC-like domains in the extracellular region along with short cytoplasmic regions with no known signaling motifs. Both PD-L1 and PD-L2 are either constitutively expressed or can be induced in a variety of cell types, including non-hematopoietic tissues as well as various tumor types. PD-L1 is not only expressed on B, T, myeloid and dendritic cells (DCs), but also on peripheral cells, such as microvascular endothelial cells and non-lymphoid organs e.g., heart or lung. In contrast, PD-L2 is only found on macrophages and DCs. The expression pattern of PD-1 ligands suggests that PD-1 plays a role in maintaining peripheral tolerance and may further serve to regulate self-reactive T- and B-cell responses in the periphery.

[0011] In any case, it is now abundantly clear that PD-1 plays a critical role in at least certain human cancers, presumably by mediating immune evasion. Accordingly, PD-L1 has been shown to be expressed on a number of mouse and human tumors and is inducible by IFN- γ in the majority of PD-L1 negative tumor cell lines [Iwai et al., *Proc. Natl. Acad. Sci. U.S.A.* 99:12293-12297 (2002); Strome et al., *Cancer Res.*, 63:6501-6505 (2003)]. Furthermore, the expression of PD-1 on tumor infiltrating lymphocytes and/or PD-L1 on tumor cells has been identified in a number of primary human tumor biopsies. Such tumor tissues include cancers of the lung, liver, ovary, cervix, skin, colon, glioma, bladder, breast, kidney, esophagus, stomach, oral squamous cell, urothelial cell, and pancreas, as well as tumors of the head and neck [Brown et al., *J. Immunol.* 170:1257-1266 (2003); Dong et al., *Nat. Med.* 8:793-800 (2002); Wintterle et al., *Cancer Res.* 63:7462-7467 (2003); Strome et al., *Cancer Res.*, 63:6501-6505 (2003); Thompson et al., *Cancer Res.* 66:3381-5 (2006); Thompson et al., *Clin. Cancer Res.* 13:1757-1761 (2007); Nomi et al., *Clin. Cancer Res.* 13:2151-2157. (2007)]. More strikingly, PD-ligand expression on tumor cells has been correlated to poor prognosis of human cancer patients across multiple tumor types [reviewed in Okazaki and Honjo, *Int. Immunol.* 19:813-824 (2007)].

[0012] Moreover, Nomi et al. [*Clin. Cancer Res.* 13:2151-2157 (2007)] demonstrated the therapeutic efficacy of blocking the binding of PD-L1 to PD-1 in a murine model of aggressive pancreatic cancer through administering either PD-1 or PD-L1 directed antibody. These antibodies effectively promoted tumor reactive CD8⁺ T cell infiltration into the tumor resulting in the up-regulation of anti-tumor effectors including IFN- γ , granzyme B, and perforin. Similarly, the use of antibodies to block the binding of PD-L1 and

PD-1 significantly inhibited tumor growth in a model of mouse squamous cell carcinoma [Tsushima et al., *Oral Oncol.* 42:268-274 (2006)].

[0013] In other studies, transfection of a murine mastocytoma line with PD-L1 led to decreased lysis of the tumor cells when co-cultured with a tumor-specific CTL clone. Lysis was restored when anti-PD-L1 monoclonal antibody was added 12293-12297 (2002). In vivo, blocking the PD1/PD-L1 interaction was shown to increase the efficacy of adoptive T cell transfer therapy in a mouse tumor model Strome et al., *Cancer Res.*, 63:6501-6505 (2003). Further evidence for the role of PD-1 in cancer treatment comes from experiments performed with PD-1 knockout mice in which PD-L1 expressing myeloma cells grew only in wild-type animals (resulting in tumor growth and associated animal death), but not in PD-1 deficient mice [Iwai Y. et al., *Proc. Natl. Acad. Sci. U.S.A.* 99:12293-12297 (2002)]. More recently, antibodies against PD-1 (including humanized murine monoclonal antibodies against human PD-1) have shown at least initial success in cancer therapy in humans [see e.g., U.S. Pat. No. 8,354,509 B2, U.S. Pat. No. 8,008,449 B2, and U.S. Pat. No. 7,595,048 B2].

[0014] Anti-PD-1 antibodies may also be useful in chronic viral infection. Memory CD8⁺ T cells generated after an acute viral infection are highly functional and constitute an important component of protective immunity. In contrast, chronic infections are often characterized by varying degrees of functional impairment (exhaustion) of virus-specific T-cell responses, and this defect is a principal reason for the inability of the host to eliminate the persisting pathogen. Although functional effector T cells are initially generated during the early stages of infection, they gradually lose function during the course of a chronic infection. Barber et al. [*Nature* 439:682-687 (2006)] showed that mice infected with a laboratory strain of LCMV developed chronic infection resulted in high levels of virus in the blood and other tissues. These mice initially developed a robust T cell response, but eventually succumbed to the infection upon T cell exhaustion. Barber et al. found that the decline in number and function of the effector T cells in chronically infected mice could be reversed by injecting an antibody that blocked the interaction between PD-1 and PD-L1.

[0015] The citation of any reference herein should not be construed as an admission that such reference is available as "prior art" to the instant application.

SUMMARY OF THE INVENTION

[0016] The present invention provides a canine fragment crystallizable region (cFc region) of an antibody in which the cFc has been genetically modified to augment, decrease, or eliminate one or more effector functions. In one aspect of the present invention, the genetically modified cFc decreases or eliminates one or more effector functions. In another aspect of the invention the genetically modified cFc augments one or more effector function.

[0017] In certain embodiments, the genetically modified cFc region is a genetically modified canine IgGB Fc region. In another such embodiment, the genetically modified cFc region is a genetically modified canine IgGC Fc region. In a particular embodiment the effector function is antibody-dependent cytotoxicity (ADCC) that is augmented, decreased, or eliminated. In another embodiment the effector function is complement-dependent cytotoxicity (CDC) that is augmented, decreased, or eliminated. In yet another

embodiment, the cFc region has been genetically modified to augment, decrease, or eliminate both the ADCC and the CDC.

[0018] The present invention further provides canine frames and/or full length heavy chains that comprise the genetically modified cFc regions. Accordingly, the present invention provides full length heavy chains of antibodies in which the full length heavy chains comprise the genetically modified cFc regions of the present invention. Such full length heavy chains can also be combined with corresponding canine light (kappa or lambda) chains to form a complete antibody. In particular embodiments of this type, the resulting antibody binds to a particular canine antigen with specificity. In certain such embodiments the canine antigen is canine PD-1. In yet other embodiments the canine antigen is canine PD-L1. In still other embodiments, the canine antigen is the IL-4 receptor alpha chain. In yet other embodiments the canine antigen is canine thymic stromal lymphopoietin protein (cTSLP) [see, U.S. Pat. No. 7,718,772 B2, the contents of which are hereby incorporated by reference in their entireties.]

[0019] In certain embodiments, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 (or SEQ ID NO: 132) in which one to seven of the following amino acid residues are replaced by another amino acid residue at the indicated positions: P4, D31, N63, G64, T65, A93, or P95. The amino acid substituting for P4, D31, N63, G64, T65, A93, and/or P95 are individually selected from one of the other 19 standard naturally occurring amino acids, as listed in Table 1 below. The present invention further provides variants of the genetically modified cFc regions that comprise an amino acid sequence that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of such genetically modified cFc regions and retain at least 50%, 75%, 90%, 95%, or more of the augmentation, decrease, or elimination of the ADCC and/or the CDC as the genetically modified cFc regions comprising the amino acid sequence of SEQ ID NO: 130 (or SEQ ID NO: 132) in which one or more of the following amino acid residues were replaced: i.e., at P4, D31, N63, G64, T65, A93, or P95.

[0020] In other embodiments two to five of the following amino acid residues are replaced by another amino acid residue at the indicated positions: P4, D31, N63, G64, T65, A93, or P95. In particular embodiments of this type, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 or SEQ ID NO: 132 with the following substitutions: P4A, D31A, N63A, A93G, and P95A. In related embodiments, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 or SEQ ID NO: 132 with the following substitutions: P4A, D31A, N63A, and P95A. In other embodiments, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 or SEQ ID NO: 132 with substitutions at D31 and N63. In particular embodiments of this type, the aspartic acid residue at position 31 is replaced with a glutamic acid residue, an asparagine residue, or an alanine residue, whereas the asparagine residue at position 63 is replaced with a glutamine residue, a histidine residue, or an alanine residue. In a more particular embodiment of this type, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 or SEQ ID NO: 132 with the following substitutions: D31A and N63A. In particular embodiments, the genetically modified cFc region is encoded by the nucleotide sequence of SEQ ID NO: 129

or SEQ ID NO: 131 comprising nucleotide changes that correspond to the amino acid sequences that they encode.

[0021] In another embodiments, the genetically modified cFc region comprises the amino acid sequence of SEQ ID NO: 130 or SEQ ID NO: 132 with the substitution at A93. In a particular embodiment of this type, the substitution is A93G. In a related embodiment the the substitution is A93S. As shown below in Example 4, the substitution of A93G leads to an enhancement in complement C1q binding, which is indicative of increasing CDC activity.

[0022] In related embodiments the genetically modified cFc region further comprises a hinge region that comprises the amino acid sequence of SEQ ID NO: 109. In other embodiments the genetically modified Fc region further comprises a hinge region that comprises the amino acid sequence of SEQ ID NO: 110. In still other embodiments the genetically modified Fc region further comprises a hinge region that comprises the amino acid sequence of SEQ ID NO: 111. In yet other embodiments the genetically modified Fc region further comprises a genetically modified hinge region that comprises the amino acid sequence of SEQ ID NO: 112.

[0023] In alternative embodiments, the present invention provides a canine IgGD Fc region with a genetically modified hinge region from a canine IgGD antibody, a hinge region from a canine IgGA antibody, a hinge region from a canine IgGB antibody, or a hinge region from a canine IgGC antibody. Moreover, the present invention provides full length heavy chains of antibodies in which the full length heavy chains comprise the canine IgGD Fc region of the present invention with a genetically modified hinge region from a canine IgGD antibody, a hinge region from a canine IgGA antibody, a hinge region from a canine IgGB antibody, or a hinge region from a canine IgGC antibody. Such full length heavy chains also can be combined with corresponding canine light (kappa or lambda) chains to form a complete antibody.

[0024] Accordingly, the present invention provides a canine IgGD Fc region that further comprises a genetically modified hinge region from a canine IgGD antibody. In particular embodiments of this type the canine IgGD Fc region and genetically modified hinge region comprise the amino acid sequence of SEQ ID NO: 6 or an amino acid sequence that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 6, which comprises a proline residue at position 10 (P10). In a more particular embodiment the canine IgGD Fc region and genetically modified hinge region is encoded by the nucleotide sequence of SEQ ID NO: 5. In other embodiments, the canine IgGD Fc region further comprises a hinge region from a canine IgGA antibody. In particular embodiments of this type the canine IgGD Fc region and hinge region comprise the amino acid sequence of SEQ ID NO: 8 or an amino acid sequence that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 8. In a more particular embodiment the canine IgGD Fc region and hinge region is encoded by the nucleotide sequence of SEQ ID NO: 7. In still other embodiments, the canine IgGD Fc region further comprises a hinge region from a canine IgGB antibody. In particular embodiments of this type the canine IgGD Fc region and hinge region comprise the amino acid sequence of SEQ ID NO: 10 or an amino acid sequence that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 10. In a more particular embodiment the canine IgGD Fc

region and hinge region is encoded by the nucleotide sequence of SEQ ID NO: 9. In yet other embodiments, the canine IgGD Fc region further comprises a hinge region from a canine IgGC antibody. In particular embodiments of this type the canine IgGD cFc region and hinge region comprise the amino acid sequence of SEQ ID NO: 12 or an amino acid sequence that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 12. In a more particular embodiment the canine IgGD cFc region and hinge region is encoded by the nucleotide sequence of SEQ ID NO: 11. The present invention further provides caninized antibodies that comprise these canine IgGD Fc regions and hinge regions. In a particular embodiment the caninized antibody or antigen binding fragment thereof binds canine Programmed Death Receptor 1 (canine PD-1) with specificity.

[0025] The present invention therefore provides caninized anti-canine PD-1 antibodies with specificity and/or that have a high binding affinity for canine PD-1. In particular embodiments, the caninized anti-canine PD-1 antibodies also have the ability to block the binding of canine PD-1 to canine PD-L1. In specific embodiments the caninized anti-canine PD-1 antibodies have a high binding affinity to canine PD-1, as well as have the ability to also block the binding of canine PD-1 to canine PD-L2. The caninized antibodies or antigen binding fragments thereof that specifically bind canine PD-1 can comprise a canine IgG heavy chain of the present invention and a canine kappa or lambda light chain. In particular embodiments the caninized anti-canine PD-1 antibodies are caninized murine anti-canine PD-1 antibodies. The present invention also relates to use of such caninized antibodies in the treatment of disease such as cancer and/or those due to infections.

[0026] In particular embodiments the caninized anti-canine PD-1 antibody comprises a genetically modified cFc region of the present invention. In alternative embodiments the caninized anti-canine PD-1 antibody comprises the canine IgGD Fc region with a genetically modified hinge region from a canine IgGD antibody, a hinge region from a canine IgGA antibody, a hinge region from a canine IgGB antibody, or a hinge region from a canine IgGC antibody. The present invention further provides such caninized anti-canine PD-1 antibodies comprising the canine frames of the present invention in combination with CDRs obtained from mouse anti-canine PD-1 antibodies, i.e., three light chain CDRs: CDR light 1 (CDRL1), CDR light 2 (CDRL2), and CDR light 3 (CDRL3) and three heavy chain CDRs CDR heavy 1 (CDRH1), CDR heavy 2 (CDRH2) and CDR heavy 3 (CDRH3).

[0027] In particular embodiments, the caninized murine anti-canine PD-1 antibodies comprise the genetically modified cFc region of IgGB or IgGC of the present invention or alternatively, the canine IgGD Fc region, together with a genetically modified hinge region from a canine IgGD antibody, a hinge region from a canine IgGA antibody, a hinge region from a canine IgGB antibody, or a hinge region from a canine IgGC antibody in combination with CDRs obtained from mouse anti-canine PD-1 antibodies. Moreover, the present invention not only provides caninized mouse anti-canine PD-1 antibodies with specific CDRs as detailed herein, but further provides caninized mouse anti-canine PD-1 antibodies comprising conservatively modified variants of those CDRs as well as variants that comprise (e.g., share) the same canonical structure.

[0028] Accordingly in particular embodiments the caninized anti-canine PD-1 antibody further comprises complementary determining regions (CDRs) in which the CDRs have canonical structures of: H1-1, H2-1, and H3-6, respectively for CDR1, CDR2, and CDR3 of the heavy chain, i.e., CDR1 of the heavy chain has the canonical structure class 1, CDR2 of the heavy chain has the canonical structure class 1, and CDR3 of the heavy chain has the canonical structure class 6. In even more particular embodiments, the CDRs for the corresponding light chains have canonical structures of: L1-3, L2-1, and L3-1, respectively for CDR1, CDR2, and CDR3 of the light chain. In other embodiments the caninized anti-canine PD-1 antibody further comprises complementary determining regions (CDRs) in which the CDRs have canonical structures of: H1-1, H2-1, and H3-11, respectively for CDR1, CDR2, and CDR3 of the heavy chain. In even more particular embodiments of this type, the CDRs for the corresponding light chains have canonical structures of: L1-2A, L2-1, and L3-1, respectively for CDR1, CDR2, and CDR3 of the light chain. In still other embodiments the caninized anti-canine PD-1 antibody further comprises complementary determining regions (CDRs) in which the CDRs have canonical structures of: H1-1, H2-2A, and H3-11, respectively for CDR1, CDR2, and CDR3 of the heavy chain. In even more particular embodiments of this type, the CDRs for the corresponding light chains have canonical structures of: L1-2A, L2-1, and L3-1, respectively for CDR1, CDR2, and CDR3 of the light chain. In yet other embodiments the caninized anti-canine PD-1 antibody further comprises complementary determining regions (CDRs) in which the CDRs have canonical structures of: H1-1, H2-2A, and H3-13, respectively for CDR1, CDR2, and CDR3 of the heavy chain. In even more particular embodiments of this type, the CDRs for the corresponding light chains have canonical structures of: L1-4, L2-1, and L3-1, respectively for CDR1, CDR2, and CDR3 of the light chain.

[0029] In more particular embodiments, the caninized antibody of the present invention or antigen binding fragment thereof comprises one or more of the heavy chain complementary determining region 1 (VH CDR1) with an amino acid sequence of SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, or SEQ ID NO: 30. In another embodiment, the heavy chain complementary determining region 2 (VH CDR2) comprises an amino acid sequence of SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, or SEQ ID NO: 35. In still another embodiment the heavy chain complementary determining region 3 (VH CDR3) comprises an amino acid sequence of SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, or SEQ ID NO: 146. In a particular embodiment of this type, the caninized antibody or antigen binding fragment comprises both a VH CDR1 comprising an amino acid sequence of SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, or SEQ ID NO: 30 and a VH CDR2 comprising an amino acid sequence of SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, or SEQ ID NO: 35. In another such embodiment, the caninized antibody or antigen binding fragment comprises both a VH CDR1 comprising an amino acid sequence of SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, or SEQ ID NO: 30 and a VH CDR3 comprising an amino acid sequence of SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, or SEQ ID NO: 146. In yet another such embodiment, the caninized antibody or antigen binding fragment comprises both a VH CDR2 comprising an amino acid sequence of SEQ ID NO:

31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, or SEQ ID NO: 35 and a VH CDR3 comprising an amino acid sequence of SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, or SEQ ID NO: 146. In still another such embodiment, the caninized antibody or antigen binding fragment comprises a VH CDR1 comprising an amino acid sequence of SEQ ID NO: 27, SEQ ID NO: 28, SEQ ID NO: 29, or SEQ ID NO: 30, a VH CDR2 comprising an amino acid sequence of SEQ ID NO: 31, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, or SEQ ID NO: 35, and a VH CDR3 comprising an amino acid sequence of SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, or SEQ ID NO: 146.

[0030] In particular embodiments, the caninized antibody or antigen binding fragment also comprises a light chain complementary determining region 1 (VL CDR1) comprising an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, or SEQ ID NO: 15. In related embodiments the light chain complementary determining region 2 (VL CDR2) comprises an amino acid sequence of SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, or SEQ ID NO: 21. In still another embodiment the light chain complementary determining region 3 (VL CDR3) comprises an amino acid sequence of SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, or SEQ ID NO: 26. In a particular embodiment of this type, the caninized antibody or antigen binding fragment comprises both a VL CDR1 comprising an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, or SEQ ID NO: 15 and a VL CDR2 comprising an amino acid sequence of SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, or SEQ ID NO: 21.

[0031] In another such embodiment, the caninized antibody or antigen binding fragment comprises both a VL CDR1 comprising an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, or SEQ ID NO: 15 and a VL CDR3 comprising an amino acid sequence of SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, or SEQ ID NO: 26. In yet another such embodiment, the caninized antibody or antigen binding fragment comprises both a VL CDR2 comprising an amino acid sequence of SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, or SEQ ID NO: 21 and a VL CDR3 comprising an amino acid sequence of SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, or SEQ ID NO: 26. In still another such embodiment, the caninized antibody or antigen binding fragment comprises a VL CDR1 comprising an amino acid sequence of SEQ ID NO: 13, SEQ ID NO: 14, or SEQ ID NO: 15, a VL CDR2 comprising an amino acid sequence of SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18, SEQ ID NO: 19, SEQ ID NO: 20, or SEQ ID NO: 21, and a VL CDR3 comprising an amino acid sequence of SEQ ID NO: 22, SEQ ID NO: 23, SEQ ID NO: 24, SEQ ID NO: 25, or SEQ ID NO: 26.

[0032] The present invention further provides caninized antibodies that comprise the amino acid sequence of SEQ ID NO: 40 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 40, SEQ ID NO: 42 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 42, SEQ ID NO: 44 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 44, SEQ ID NO: 46 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 46, SEQ ID NO: 48 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 48,

SEQ ID NO: 50 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 50, SEQ ID NO: 52 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 52, SEQ ID NO: 54 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 54, SEQ ID NO: 56 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 58 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 58, SEQ ID NO: 60 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 60, SEQ ID NO: 62 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 62, SEQ ID NO: 64 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 64, or SEQ ID NO: 66 or that is 90%, 95%, 98%, or 99% identical to the amino acid sequence of SEQ ID NO: 66, or antigen binding fragments of these caninized antibodies.

[0033] In particular embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 40, 52, 56, or 64 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 40, 52, 56, or 64) that comprises (i) P, A, G, or S at position 239, (ii) A, G, or S at position 266, (iii) A, G, or S at position 298, (iv) G, P, or A at position 299, (v) T, A, G, or S at position 300, (vi) A, G, or S at position 328, and (vii) P, A, G, or S at position 330. In other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 42, 54, 58, or 66 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 42, 54, 58, or 66) that comprises (i) P, A, G, or S at position 237, (ii) A, G, or S at position 264, (iii) A, G, or S at position 296, (iv) G, P, or A at position 297, (v) T, A, G, or S at position 298, (vi) A, G, or S at position 326, and (vii) P, A, G, or S at position 328. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 44, 50, or 60 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 44, 50, or 60) that comprises (i) P, A, G, or S at position 244, (ii) A, G, or S at position 271, (iii) A, G, or S at position 303, (iv) G, P, or A at position 304, (v) T, A, G, or S at position 305, (vi) A, G, or S at position 333, and (vii) P, A, G, or S at position 335. In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 46 or 62 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 46 or 62) that comprises (i) P, A, G, or S at position 242, (ii) A, G, or S at position 269, (iii) A, G, or S at position 301, (iv) G, P, or A at position 302, (v) T, A, G, or S at position 303, (vi) A, G, or S at position 331, and (vii) P, A, G, or S at position 333. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 48 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 48) that comprises (i) P, A, G, or S at position 246, (ii) A, G, or S at position 273, (iii) A, G, or S at position 305, (iv) G, P, or A at position 306, (v) T, A, G, or S at position 307, (vi) A, G, or S at position 335, and (vii) P, A, G, or S at position 337.

[0034] In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 40, 52, 56, or 64 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 40, 52, 56, or 64) that comprises (i) P, A, G, or S at position 239, (ii) A at position 266, (iii) A at position 298, (iv) G, P, or A at position 299, (v) T, A, G, or S at position 300, (vi) A, G, or S at position 328, and (vii) P, A, G, or S at position 330. In other embodiments, the heavy chain of an antibody comprises the amino acid sequence of

at position 298, (iv) G at position 299, (v) T at position 300, (vi) A, G, or S at position 328, and (vii) P, A, G, or S at position 330. In other such embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 42, 54, 58, or 66 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 42, 54, 58, or 66) that comprises (i) P, A, G, or S at position 237, (ii) A, G, or S at position 264, (iii) A, G, or S at position 296, (iv) G at position 297, (v) T at position 298, (vi) A, G, or S at position 326, and (vii) P, A, G, or S at position 328. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 44, 50, or 60 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 44, 50, or 60) that comprises (i) P, A, G, or S at position 244, (ii) A, G, or S at position 271, (iii) A, G, or S at position 303, (iv) G at position 304, (v) T at position 305, (vi) A, G, or S at position 333, and (vii) P, A, G, or S at position 335. In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 46 or 62 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 46 or 62) that comprises (i) P, A, G, or S at position 242, (ii) A, G, or S at position 269, (iii) A, G, or S at position 301, (iv) G at position 302, (v) T at position 303, (vi) A, G, or S at position 331, and (vii) P, A, G, or S at position 333. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 48 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 48) that comprises (i) P, A, G, or S at position 246, (ii) A, G, or S at position 273, (iii) A, G, or S at position 305, (iv) G at position 306, (v) T at position 307, (vi) A, G, or S at position 335, and (vii) P, A, G, or S at position 337.

[0039] In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 40, 52, 56, or 64 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 40, 52, 56, or 64) that comprises (i) P, A, G, or S at position 239, (ii) A at position 266, (iii) A at position 298, (iv) G at position 299, (v) T at position 300, (vi) A, G, or S at position 328, and (vii) P, A, G, or S at position 330. In other such embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 42, 54, 58, or 66 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 42, 54, 58, or 66) that comprises (i) P, A, G, or S at position 237, (ii) A at position 264, (iii) A at position 296, (iv) G at position 297, (v) T at position 298, (vi) A, G, or S at position 326, and (vii) P, A, G, or S at position 328. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 44, 50, or 60 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 44, 50, or 60) that comprises (i) P, A, G, or S at position 244, (ii) A at position 271, (iii) A at position 303, (iv) G at position 304, (v) T at position 305, (vi) A, G, or S at position 333, and (vii) P, A, G, or S at position 335. In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 46 or 62 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 46 or 62) that comprises (i) P, A, G, or S at position 242, (ii) A at position 269, (iii) A at position 301, (iv) G at position 302, (v) T at position 303, (vi) A, G, or S at position 331, and (vii) P, A, G, or S at position 333. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 48 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 48) that comprises (i) P, A, G, or S at position 246, (ii) A at position 273, (iii) A at position 305, (iv) G at position 306, (v) T at position 307, (vi) A, G, or S at position 335, and (vii) P, A, G, or S at position 337.

[0040] In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 40, 52, 56, or 64 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 40, 52, 56, or 64) that comprises (i) A at position 239, (ii) A at position 266, (iii) A at position 298, (iv) G at position 299, (v) T at position 300, (vi) G at position 328, and (vii) A at position 330. In other such embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 42, 54, 58, or 66 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 42, 54, 58, or 66) that comprises (i) A at position 237, (ii) A at position 264, (iii) A at position 296, (iv) G at position 297, (v) T at position 298, (vi) G at position 326, and (vii) A at position 328. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 44, 50, or 60 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 44, 50, or 60) that comprises (i) A at position 244, (ii) A at position 271, (iii) A at position 303, (iv) G at position 304, (v) T at position 305, (vi) G at position 333, and (vii) A at position 335. In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 46 or 62 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 46 or 62) that comprises (i) A at position 242, (ii) A at position 269, (iii) A at position 301, (iv) G at position 302, (v) T at position 303, (vi) G at position 331, and (vii) A at position 333. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 48 (or 90%, 95%, 98%, or 99% identical to SEQ ID NO: 48) that comprises (i) A at position 246, (ii) A at position 273, (iii) A at position 305, (iv) G at position 306, (v) T at position 307, (vi) G at position 335, and (vii) A at position 337.

[0041] In addition, the present invention provides caninized antibody or antigen binding fragment thereof that further comprise a canine light chain that comprises the amino acid sequence of SEQ ID NO: 72, SEQ ID NO: 78, SEQ ID NO: 84, SEQ ID NO: 90, SEQ ID NO: 96, SEQ ID NO: 102, or SEQ ID NO: 108.

[0042] Accordingly, the present invention further provides a caninized antibody or antigen binding fragment thereof that comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 68 and a light chain comprising the amino acid sequence of SEQ ID NO: 72. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 70 and a light chain comprising the amino acid sequence of SEQ ID NO: 72. In another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 74 and a light chain comprising the amino acid sequence of SEQ ID NO: 78. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 76 and a light chain comprising the amino acid sequence of SEQ ID NO: 78. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 80 and a light chain comprising the amino acid sequence of SEQ ID NO: 84. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 82 and a light chain comprising the amino acid sequence of SEQ ID NO: 84. In still another embodiment, the caninized antibody

or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 86 and a light chain comprising the amino acid sequence of SEQ ID NO: 90. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 88 and a light chain comprising the amino acid sequence of SEQ ID NO: 90. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 92 and a light chain comprising the amino acid sequence of SEQ ID NO: 96. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 94 and a light chain comprising the amino acid sequence of SEQ ID NO: 96. In still another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 98 and a light chain comprising the amino acid sequence of SEQ ID NO: 102. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 100 and a light chain comprising the amino acid sequence of SEQ ID NO: 102. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 104 and a light chain comprising the amino acid sequence of SEQ ID NO: 108. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 106 and a light chain comprising the amino acid sequence of SEQ ID NO: 108.

[0043] The present invention further provides a caninized antibody or antigen binding fragment thereof that comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 40 and a light chain comprising the amino acid sequence of SEQ ID NO: 72. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 42 and a light chain comprising the amino acid sequence of SEQ ID NO: 72. In another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 44 and a light chain comprising the amino acid sequence of SEQ ID NO: 78. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 46 and a light chain comprising the amino acid sequence of SEQ ID NO: 78. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 48 and a light chain comprising the amino acid sequence of SEQ ID NO: 84. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 50 and a light chain comprising the amino acid sequence of SEQ ID NO: 84. In still another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 52 and a light chain comprising the amino acid sequence of SEQ ID NO: 90. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 54 and

a light chain comprising the amino acid sequence of SEQ ID NO: 90. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 56 and a light chain comprising the amino acid sequence of SEQ ID NO: 96. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 58 and a light chain comprising the amino acid sequence of SEQ ID NO: 96.

[0044] In still another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 60 and a light chain comprising the amino acid sequence of SEQ ID NO: 102. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 62 and a light chain comprising the amino acid sequence of SEQ ID NO: 102. In yet another embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 64 and a light chain comprising the amino acid sequence of SEQ ID NO: 108. In a related embodiment, the caninized antibody or antigen binding fragment thereof comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 66 and a light chain comprising the amino acid sequence of SEQ ID NO: 108.

[0045] The present invention further provides nucleic acids that encode any of the amino acid sequences of the present invention including the CDRs, cFc regions, the cFc regions with the hinge regions, and the heavy chains, and the light chains of the caninized antibodies of the present invention. The present invention further provides expression vectors that comprise one or more of the nucleic acids of the present invention. The present invention further provides host cells that comprise one or more expression vectors of the present invention and methods for expressing the CDRs, and/or cFc regions, and/or the cFc regions with the hinge regions, and/or the heavy chains, and/or the light chains of the caninized antibodies of the present invention using such host cells. The present invention also provides host cells that have been genetically engineered to express the CDRs, and/or cFc regions, and/or the cFc regions with the hinge regions, and/or the heavy chains, and/or the light chains of the caninized antibodies of the present invention in the absence of such vectors. In particular embodiments, these nucleic acids, expression vectors, polypeptides, or host cells of the invention are useful in methods of making an antibody.

[0046] In particular embodiments, the antibody is a recombinant antibody or an antigen binding fragment thereof. In related embodiments, the variable heavy chain domain and variable light chain domain are connected by a flexible linker to form a single-chain antibody.

[0047] In particular embodiments, the antibody or antigen binding fragment is a Fab fragment.

[0048] In other embodiments, the antibody or antigen binding fragment is a Fab' fragment. In other embodiments, the antibody or antigen binding fragment is a (Fab')₂ fragment. In still other embodiments, the antibody or antigen binding fragment is a diabody. In particular embodiments, the antibody or antigen binding fragment is a domain antibody. In particular embodiments, the antibody or antigen binding fragment is a camelized single domain antibody.

[0049] In particular embodiments, a caninized murine anti-canine PD-1 antibody or antigen binding fragment increases the immune response of the canine subject being treated.

[0050] In certain embodiments when bound to canine PD-1, the caninized antibody or antigen binding fragment thereof binds to at least one amino acid residue within one or more amino acid sequences of the following: SEQ ID NO: 138, SEQ ID NO: 139, SEQ ID NO: 140, SEQ ID NO: 141, of SEQ ID NO: 142, SEQ ID NO: 143, SEQ ID NO: 144, and/or SEQ ID NO: 145.

[0051] Furthermore, the present invention provides caninized antibodies to canine PD-1, that comprise variants of the CDRs of the present invention that have the corresponding canonical structures provided herein and/or that bind to the amino acid sequence of SEQ ID NO: 144. In particular embodiments of this type, the dissociation constant (Kd) for caninized antibody-canine PD-1 binding is 1×10^{-5} to 1×10^{-12} M. In more particular embodiments the caninized antibodies to canine PD-1, comprise variants of the CDRs of the present invention that have the corresponding canonical structures provided herein and bind to the amino acid sequence of SEQ ID NO: 145. The present invention therefore includes caninized antibodies and antigen binding fragments thereof that bind canine PD-1 with specificity, that when they are bound to canine PD-1, the antibody binds to at least one amino acid residue within SEQ ID NO: 144. In particular embodiments of this type, the antibodies and antigen binding fragments thereof bind canine PD-1 and block the binding of canine PD-1 to canine Programmed Death Ligand 1 (PD-L1).

[0052] Accordingly, in particular embodiments when bound to canine PD-1, the caninized antibody (including the antibodies with one or more variant CDR, e.g., a variant including a conservatively modified variant and/or a variant that comprises a defined canonical structure class) binds to at least one amino acid residue within one or more amino acid sequences of the following: SEQ ID NO: 138, SEQ ID NO: 139, SEQ ID NO: 140, SEQ ID NO: 141, of SEQ ID NO: 142, SEQ ID NO: 143, and/or SEQ ID NO: 145. In even more particular embodiments when bound to canine PD-1, the caninized antibodies or antigen binding fragments thereof bind to one or more amino acid residues of the following arginine residues: R₆₂, R₆₉, R₇₂, R₇₅, and R₉₀ of SEQ ID NO: 114. In specific embodiments when bound to canine PD-1, the caninized antibodies or antigen binding fragments thereof bind to at least one amino acid residue within SEQ ID NO: 145. In more specific embodiments when bound to canine PD-1, the antibodies or antigen binding fragments thereof bind to one or more amino acid residues of the following arginine residues: R₆₂, R₆₉, R₇₂, and R₇₅ of SEQ ID NO: 114. In even more specific embodiments when bound to canine PD-1, the antibodies or antigen binding fragments thereof bind to R₇₅ of SEQ ID NO: 114.

[0053] The present invention further provides caninized antibodies or antigen binding fragments thereof that bind to canine PD-1 with a dissociation constant (Kd) that is lower (e.g., 1×10^{-13} M, or lower) than 1×10^{-12} M. In particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with a dissociation constant of 1×10^{-5} M to 1×10^{-12} M. In more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with a dissociation constant of 1×10^{-7} M to 1×10^{-11} M. In still more particular

embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with a dissociation constant of 1×10^{-8} M to 1×10^{-11} M. In yet more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with a dissociation constant of 1×10^{-8} M to 1×10^{-10} M.

[0054] The present invention also provides caninized antibodies or antigen binding fragments thereof that bind to canine PD-1 with an on rate (k_{on}) that is greater than 1×10^7 M⁻¹s⁻¹. In particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an on rate of 1×10^2 M⁻¹s⁻¹ to 1×10^7 M⁻¹s⁻¹. In more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an on rate of 1×10^3 M⁻¹s⁻¹ to 1×10^6 M⁻¹s⁻¹. In still more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an on rate of 1×10^3 M⁻¹s⁻¹ to 1×10^5 M⁻¹s⁻¹. In yet more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 on rate of 1×10^4 M⁻¹s⁻¹ to 1×10^5 M⁻¹s⁻¹.

[0055] The present invention further provides caninized antibodies or antigen binding fragments thereof that bind to canine PD-1 with an off rate (k_{off}) slower than 1×10^{-7} s⁻¹. In particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an off rate of 1×10^{-3} s⁻¹ to 1×10^{-8} s⁻¹. In more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an off rate of 1×10^{-4} s⁻¹ to 1×10^{-7} s⁻¹. In still more particular embodiments the caninized antibodies or antigen binding fragments thereof bind to canine PD-1 with an off rate of 1×10^{-5} s⁻¹ to 1×10^{-7} s⁻¹.

[0056] In related embodiments, the caninized antibodies or antigen binding fragments thereof stimulate antigen-specific memory responses to a tumor or pathogen. In particular embodiments, the caninized antibodies or antigen binding fragments thereof stimulate an antibody response in vivo. In other particular embodiments, the caninized antibodies or antigen binding fragments thereof stimulate an immune response in an animal subject. In more specific embodiments the animal subject is a canine. In a related embodiment, the animal subject is a feline.

[0057] Accordingly, any of the caninized antibodies of the present invention can exhibit one, two, three, four, five, or all these properties, i.e., the aforesaid dissociation constants with canine PD-1, the aforesaid on rates for binding with canine PD-1, the aforesaid off rates for dissociating from the caninized antibody-canine PD-1 binding complex, stimulating an antigen-specific memory responses to a tumor or pathogen, stimulating an antibody response in vivo, and/or stimulating an immune response in an animal subject.

[0058] In more particular embodiments the caninized antibodies and antigen binding fragments thereof of the present invention bind canine PD-1 and also block the binding of canine PD-1 to PD-L1. In even more particular embodiments the caninized antibodies and antigen binding fragments thereof of the present invention bind canine PD-1, block the binding of canine PD-1 to PD-L1, and also block the binding of canine PD-1 to PD-L2.

[0059] The present invention further provides nucleic acids that encode the caninized murine anti-canine PD-1 antibodies or portions thereof of the present invention. In related embodiments such antibodies or antigen binding fragments can be used for the preparation of a medicament

to treat cancer in a canine subject. Alternatively, or in conjunction, the present invention provides for the use of any of the antibodies or antibody fragments of the present invention for diagnostic use. In yet additional embodiments, a kit is provided comprising any of the caninized antibodies or antigen binding fragments disclosed herein.

[0060] The present invention further includes pharmaceutical compositions comprising an anti-canine antigen antibody or binding fragment thereof (e.g., an anti-canine PD-1 antibody or antigen binding fragment thereof) together with a pharmaceutically acceptable carrier or diluent. The present invention also provides a method of increasing the activity of an immune cell, comprising administering to a subject (e.g., a canine) in need thereof a therapeutically effective amount of the pharmaceutical composition of the present invention. In certain embodiments the method is used for the treatment of cancer. In other embodiments, the method is used in the treatment of an infection or infectious disease. In still other embodiments, a caninized antibody of the present invention or antigen binding fragment thereof is used as a vaccine adjuvant. In yet another embodiment, a caninized anti-TSLP antibody is administered to a canine to treat atopic dermatitis.

[0061] These and other aspects of the present invention will be better appreciated by reference to the following Brief Description of the Drawings and the Detailed Description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0062] FIG. 1 shows the reactivity of caninized monoclonal antibodies (mAbs) against extracellular domain of canine PD-1, as a function of OD 650/490 versus the log mAb (nM). Various caninized mAbs were tested for their binding to extracellular domain of canine PD-1 by ELISA. The four mAbs tested were designated as: 2H9 VH4 IgGB/VL3, 3B6 VH3 IgGB/VL4, 2H9 VH4 IgGB (YZZ1062)/VL3, and 2H9 VH4 IgGB (YZZ1068)/VL3.

[0063] FIG. 2 shows the reactivity of caninized mAbs against cell surface-expressed canine PD-1. Various mouse mAbs were tested for their binding to canine PD-1 expressed on CHO cells by CELISA as a function of OD 450/540 versus the log mAb (nM). The six mAbs tested were designated as: 3B6 VH3/VL4, 3B6 VH3/VL1, 3B6 VH3/VL3, 3B6 VH3/VL2, 3B6 VH1/VL1, and 3B6 m-c Chimera.

[0064] FIG. 3 shows ligand blockade with caninized mAbs against canine PD-1. Various caninized mAbs were tested for their ability to inhibit binding of PD-1 expressed on CHO cells to PD-L1 as a function of OD 450/540 versus the log mAb (nM). The six mAbs tested were designated as: 3B6 VH3/VL4, 3B6 VH3/VL1, 3B6 VH3/VL3, 3B6 VH3/VL2, 3B6 VH1/VL1, and 3B6 m-c Chimera.

[0065] FIG. 4 shows the cytokine secretion induced by caninized mAbs against canine PD-1. Various caninized mAbs and their variants were tested for their ability to induce cytokine secretion from PBMC from healthy dogs.

[0066] FIGS. 5A and 5B show the binding of caninized mAbs and their variants (beginning at 1 µg/ml) to FcγRI. Various mAbs were tested for their ability to bind to FcγRI. Antibodies are designated as: can 2H9 ADCC (1062) VH4/VL3, can 2H9 ADCC mut 1 VH4/VL3, can 2H9 ADCC mut 2 VH4/VL3, can 2H9 IgGD VH4/VL3, can 2H9 VH4/VL3, and can 3B6 VH4/VL4 in FIG. 5A; and can 2H9 ADCC (1059) VH4/VL3, can 2H9 ADCC (1060) VH4/VL3, can

2H9 ADCC (1061) VH4/VL3, can 2H9 IgGB ADCC (1068) VH4/VL3, can 2H9 VH4/VL3, and can 3B6 VH4/VL4 in FIG. 5B.

[0067] FIGS. 6A and 6B show the binding of caninized mAbs and their variants (beginning at 1 µg/ml) to C1Q. Various mAbs were tested for their ability to bind to C1Q. Antibodies are designated as: can 2H9 VH4 IgGB ADCC (1062)/VL3, can 2H9 VH4 IgGB ADCC (mut 1)/VL3, can 2H9 VH4 IgGB ADCC (mut 2)/VL3, can 2H9 VH4 IgGD/VL3, can 2H9 VH4/VL3, and can 3B6 VH4/VL4 IgGB in FIG. 6A; and can 2H9 VH4 IgGB ADCC (1059)/VL3, can 2H9 VH4 IgGB ADCC (1060)/VL3, can 2H9 VH4 IgGB ADCC (1061)/VL3, can 2H9 VH4 IgGB ADCC (1068)/VL3, can 2H9 VH4/VL3 IgGB, and can 3B6 VH4/VL4 IgGB in FIG. 6B.

[0068] FIG. 7A shows the characterization of the interface between canine PD-1 and the caninized antibody 2G9. The amino acid positions are with respect to the PD-1 amino acid sequence without the signal sequence, i.e., SEQ ID NO: 114. The determination was performed by chemical cross-linking, High-Mass MALDI mass spectrometry and nLC-Orbitrap mass spectrometry.

[0069] FIG. 7B shows the characterization of the interface between canine PD-1 and the caninized antibody 3B6. The amino acid positions are with respect to the PD-1 amino acid sequence without the signal sequence, i.e., SEQ ID NO: 114. The determination was performed by chemical cross-linking, High-Mass MALDI mass spectrometry and nLC-Orbitrap mass spectrometry.

DETAILED DESCRIPTION

Abbreviations

[0070] Throughout the detailed description and examples of the invention the following abbreviations will be used:

[0071] ADCC Antibody-dependent cellular cytotoxicity

[0072] CDC Complement-dependent cytotoxicity

[0073] CDR Complementarity determining region in the immunoglobulin variable regions, defined for human antibodies using the Kabat numbering system

[0074] CHO Chinese hamster ovary

[0075] EC50 concentration resulting in 50% efficacy or binding

[0076] ELISA Enzyme-linked immunosorbent assay

[0077] FR Antibody framework region: the immunoglobulin variable regions excluding the CDR regions.

[0078] HRP Horseradish peroxidase

[0079] IFN interferon

[0080] IC50 concentration resulting in 50% inhibition

[0081] IgG Immunoglobulin G

[0082] Kabat An immunoglobulin alignment and numbering system for human antibodies pioneered by Elvin A. Kabat [*Sequences of Proteins of Immunological Interest*, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)]

[0083] mAb Monoclonal antibody (also Mab or MAb)

[0084] MES 2-(N-morpholino) ethanesulfonic acid

[0085] MOA Mechanism of action

[0086] NHS Normal human serum

[0087] PCR Polymerase chain reaction

[0088] PK Pharmacokinetics

[0089] SEB *Staphylococcus* Enterotoxin B

[0090] TT Tetanus toxoid

[0091] V region The segment of human IgG chains which is variable in sequence between different antibodies. It extends to Kabat residue 109 in the light chain and 113 in the heavy chain.

[0092] VH Immunoglobulin heavy chain variable region

[0093] VK Immunoglobulin kappa light chain variable region

Definitions

[0094] So that the invention may be more readily understood, certain technical and scientific terms are specifically defined below. Unless specifically defined elsewhere in this document, all other technical and scientific terms used herein have the meaning commonly understood by one of ordinary skill in the art to which this invention belongs.

[0095] As used herein, including the appended claims, the singular forms of words such as “a,” “an,” and “the,” include their corresponding plural references unless the context clearly dictates otherwise.

[0096] “Activation” as it applies to cells or to receptors refers to the activation or treatment of a cell or receptor with a ligand, unless indicated otherwise by the context or explicitly. “Ligand” encompasses natural and synthetic ligands, e.g., cytokines, cytokine variants, analogues, muteins, and binding compounds derived from antibodies. “Ligand” also encompasses small molecules, e.g., peptide mimetics of cytokines and peptide mimetics of antibodies. “Activation” can refer to cell activation as regulated by internal mechanisms as well as by external or environmental factors.

[0097] “Activity” of a molecule may describe or refer to the binding of the molecule to a ligand or to a receptor, to catalytic activity; to the ability to stimulate gene expression or cell signaling, differentiation, or maturation; to antigenic activity, to the modulation of activities of other molecules, and the like. “Activity” of a molecule may also refer to activity in modulating or maintaining cell-to-cell interactions, e.g., adhesion, or activity in maintaining a structure of a cell, e.g., cell membranes or cytoskeleton. “Activity” can also mean specific activity, e.g., [catalytic activity]/[mg protein], or [immunological activity]/[mg protein], concentration in a biological compartment, or the like. “Activity” may refer to modulation of components of the innate or the adaptive immune systems.

[0098] “Administration” and “treatment,” as it applies to an animal, e.g., a canine experimental subject, cell, tissue, organ, or biological fluid, refers to contact of an exogenous pharmaceutical, therapeutic, diagnostic agent, or composition to the animal e.g., a canine subject, cell, tissue, organ, or biological fluid. Treatment of a cell encompasses contact of a reagent to the cell, as well as contact of a reagent to a fluid, where the fluid is in contact with the cell. “Administration” and “treatment” also means in vitro and ex vivo treatments, e.g., of a cell, by a reagent, diagnostic, binding compound, or by another cell.

[0099] The term “subject” includes any organism, preferably an animal, more preferably a mammal (e.g., canine, feline, or human) and most preferably a canine.

[0100] As used herein, the term “feline” refers to any member of the Felidae family. Members of this family include wild, zoo, and domestic members, such as any member of the subfamilies Felinae, e.g., cats, lions, tigers, pumas, jaguars, leopards, snow leopards, panthers, North

American mountain lions, cheetahs, lynx, bobcats, caracals or any cross breeds thereof. Cats also include domestic cats, pure-bred and/or mongrel companion cats, show cats, laboratory cats, cloned cats, and wild or feral cats.

[0101] As used herein, a “substitution of an amino acid residue” with another amino acid residue in an amino acid sequence is equivalent to “replacing an amino acid residue” with another amino acid residue and denotes that a particular amino acid residue at a specific position in the amino acid sequence has been replaced by (or substituted for) by a different amino acid. For example, one such substitution (replacement) is denoted as P4A of an Fc region of an IgGB or IgGC amino acid sequence, in which case, the proline residue at amino acid position 4 of the amino acid sequence of the Fc region of an IgGB or the Fc region of an IgGC has been substituted for (replaced) by an alanine residue.

[0102] Accordingly, such amino acid substitutions can be particularly designed i.e., purposefully replacing an alanine with a serine at a specific position in the amino acid sequence by e.g., recombinant DNA technology. Alternatively, a particular amino acid residue or string of amino acid residues of an antibody can be replaced by one or more amino acid residues through more natural selection processes e.g., based on the ability of the antibody produced by a cell to bind to a given region on that antigen, e.g., one containing an epitope or a portion thereof, and/or for the antibody to comprise a particular CDR that retains the same canonical structure as the CDR it is replacing. Such substitutions/replacements can lead to “variant” CDRs and/or antibodies.

[0103] “Treat” or “treating” means to administer a therapeutic agent, such as a composition containing any of the antibodies or antigen binding fragments of the present invention, internally or externally to a canine subject or patient having one or more disease symptoms, or being suspected of having a disease, for which the agent has therapeutic activity.

[0104] Typically, the agent is administered in an amount effective to alleviate and/or ameliorate one or more disease symptom in the treated subject or population, whether by inducing the regression of or inhibiting the progression of such symptom(s) by any clinically measurable degree. The amount of a therapeutic agent that is effective to alleviate any particular disease symptom (also referred to as the “therapeutically effective amount”) may vary according to factors such as the disease state, age, and weight of the subject (e.g., canine), and the ability of the pharmaceutical composition to elicit a desired response in the subject. Whether a disease symptom has been alleviated or ameliorated can be assessed by any clinical measurement typically used by veterinarians or other skilled healthcare providers to assess the severity or progression status of that symptom. While an embodiment of the present invention (e.g., a treatment method or article of manufacture) may not be effective in alleviating the target disease symptom(s) in every subject, it should alleviate the target disease symptom (s) in a statistically significant number of subjects as determined by any statistical test known in the art such as the Student’s t-test, the chi2-test, the U-test according to Mann and Whitney, the Kruskal-Wallis test (H-test), Jonckheere-Terpstra-test and the Wilcoxon-test.

[0105] “Treatment,” as it applies to a human, veterinary (e.g., canine) or research subject, refers to therapeutic treatment, as well as research and diagnostic applications.

“Treatment” as it applies to a human, veterinary (e.g., canine), or research subject, or cell, tissue, or organ, encompasses contact of the caninized antibodies or antigen binding fragments of the present invention to a canine or other animal subject, a cell, tissue, physiological compartment, or physiological fluid.

[0106] Canine PD-1 has been found to comprise the amino acid sequence of SEQ ID NO: 114 [U.S. provisional application No. 61/918,946, filed on Dec. 20, 2013, the contents of which are hereby incorporated herein in their entireties]. In a specific embodiment canine PD-1 is encoded by a nucleic acid that comprises the nucleotide sequence of SEQ ID NO: 113.

[0107] Canine PD-L1 has been found to comprise the amino acid sequence of SEQ ID NO: 120 [U.S. provisional application No. 61/918,946, filed on Dec. 20, 2013, *supra*]. In a specific embodiment canine PD-L1 is encoded by a nucleotide sequence comprising SEQ ID NO: 119.

[0108] The term “immune response” refers to the action of, for example, lymphocytes, antigen presenting cells, phagocytic cells, granulocytes, and soluble macromolecules produced by the above cells or the liver (including antibodies, cytokines, and complement) that results in selective damage to, destruction of, or elimination from the mammalian body (e.g., canine body) of cancerous cells, cells or tissues infected with pathogens, or invading pathogens.

Caninized Anti-Canine Antigen Antibodies

[0109] As used herein, the term “canine” includes all domestic dogs, *Canis lupus familiaris* or *Canis familiaris*, unless otherwise indicated.

[0110] As used herein, an antibody is said to bind specifically to a polypeptide comprising a given antigen sequence (in this case a portion of the amino acid sequence of a canine antigen, e.g., canine PD-1) if it binds to polypeptides comprising that portion of the amino acid sequence of the canine antigen, e.g., canine PD-1, but does not bind to other canine proteins lacking that portion of the sequence of the canine antigen, e.g., canine PD-1. For example, an antibody that specifically binds to a polypeptide comprising canine PD-1 may bind to a FLAG®-tagged form of canine PD-1, but will not bind to other FLAG®-tagged canine proteins with specificity. An antibody, or binding compound derived from the antigen-binding site of an antibody, binds to its canine antigen, or a variant or mutein thereof, “with specificity” when it has an affinity for that canine antigen or a variant or mutein thereof which is at least ten-times greater, more preferably at least 20-times greater, and even more preferably at least 100-times greater than its affinity for any other canine antigen tested.

[0111] As used herein, the term “antibody” refers to any form of antibody that exhibits the desired biological activity. Thus, it is used in the broadest sense and specifically covers, but is not limited to, monoclonal antibodies (including full length monoclonal antibodies), polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies), caninized antibodies, fully canine antibodies, chimeric antibodies and camelized single domain antibodies. “Parental antibodies” are antibodies obtained by exposure of an immune system to an antigen prior to modification of the antibodies for an intended use, such as caninization of an antibody for use as a canine therapeutic antibody.

[0112] As used herein, unless otherwise indicated, “antibody fragment” or “antigen binding fragment” refers to

antigen binding fragments of antibodies, i.e. antibody fragments that retain the ability to bind specifically to the antigen bound by the full-length antibody, e.g. fragments that retain one or more CDR regions. Examples of antigen binding fragments include, but are not limited to, Fab, Fab', F(ab')₂, and Fv fragments; diabodies; linear antibodies; single-chain antibody molecules, e.g., sc-Fv; nanobodies and multispecific antibodies formed from antibody fragments.

[0113] A “Fab fragment” is comprised of one light chain and the CH1 and variable regions of one heavy chain. The heavy chain of a Fab molecule cannot form a disulfide bond with another heavy chain molecule. A “Fab fragment” can be the product of papain cleavage of an antibody.

[0114] A “fragment crystallizable” (“Fc”) region contains two heavy chain fragments (i.e., two identical polypeptides) comprising the CH2 and CH3 domains of an antibody. The two heavy chain fragments are held together by two or more disulfide bonds and by hydrophobic interactions of the CH3 domains. In the present invention, the amino acid sequence for each of the four canine IgG Fc fragments is based on the identified boundary of CH1 and CH2 domains as determined by Tang et al. [Vet. Immunol. Immunopathol. 80:259-270 (2001)].

[0115] A “Fab' fragment” contains one light chain and a portion or fragment of one heavy chain that contains the VH domain and the CH1 domain and also the region between the CH1 and CH2 domains, such that an interchain disulfide bond can be formed between the two heavy chains of two Fab' fragments to form a F(ab')₂ molecule.

[0116] A “F(ab')₂ fragment” contains two light chains and two heavy chains containing a portion of the constant region between the CH1 and CH2 domains, such that an interchain disulfide bond is formed between the two heavy chains. A F(ab')₂ fragment thus is composed of two Fab' fragments that are held together by a disulfide bond between the two heavy chains. An “F(ab')₂ fragment” can be the product of pepsin cleavage of an antibody.

[0117] The “Fv region” comprises the variable regions from both the heavy and light chains, but lacks the constant regions.

[0118] The term “single-chain Fv” or “scFv” antibody refers to antibody fragments comprising the VH and VL domains of an antibody, wherein these domains are present in a single polypeptide chain. Generally, the Fv polypeptide further comprises a polypeptide linker between the VH and VL domains which enables the scFv to form the desired structure for antigen binding. [See, Pluckthun, *The Pharmacology of Monoclonal Antibodies*, vol. 113 Rosenberg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994); WO 88/01649; and U.S. Pat. Nos. 4,946,778 and 5,260,203.]

[0119] As used herein, the term “canonical structure” refers to the local conformation that can be adopted by each of the hypervariable regions of the heavy and light chain of an antibody within the framework that they reside. For each hypervariable region, there are a small number of canonical structures (generally denoted by simple integers such as 1 or 2 etc.), which can be predicted with great accuracy from the amino acid sequences of the corresponding hypervariable region (particularly within the context of the amino acid sequence of its framework, as provided below for the corresponding caninized murine anti-canine PD-1 variable domains). These canonical structures can be determinative regarding whether a modification of the amino acid

sequence of a given CDR will result in the retention or loss of the ability to bind to its antigen binding partner [See, Chothia and Lesk, *Canonical Structures for the hypervariable regions of immunoglobulins*, *J. Mol. Biol.* 196:901-917 (1987); Chothia et al., *Conformation of immunoglobulin hypervariable regions*, *Nature*, 34:877-883 (1989); and Al-Lazikani et al., *Standard Conformations for the canonical structures of immunoglobulins*, *J. Mol. Biol.* 273:927-948 (1997)].

[0120] A “domain antibody” is an immunologically functional immunoglobulin fragment containing only the variable region of a heavy chain or the variable region of a light chain. In some instances, two or more VH regions are covalently joined with a peptide linker to create a bivalent domain antibody. The two V_H regions of a bivalent domain antibody may target the same or different antigens.

[0121] A “bivalent antibody” comprises two antigen binding sites. In some instances, the two binding sites have the same antigen specificities. However, bivalent antibodies may be bispecific (see below).

[0122] In certain embodiments, monoclonal antibodies herein also include camelized single domain antibodies. [See, e.g., Muyldermans et al., *Trends Biochem. Sci.* 26:230 (2001); Reichmann et al., *J. Immunol. Methods* 231:25 (1999); WO 94/04678; WO 94/25591; U.S. Pat. No. 6,005, 079]. In one embodiment, the present invention provides single domain antibodies comprising two VH domains with modifications such that single domain antibodies are formed.

[0123] As used herein, the term “diabodies” refers to small antibody fragments with two antigen-binding sites, which fragments comprise a heavy chain variable domain (V_H) connected to a light chain variable domain (V_L) in the same polypeptide chain (V_H - V_L or V_L - V_H). By using a linker that is too short to allow pairing between the two domains on the same chain, the domains are forced to pair with the complementary domains of another chain and create two antigen-binding sites. [See, EP 0 404 097 B1; WO 93/11161; and Holliger et al., *Proc. Natl. Acad. Sci. USA* 90:6444-6448 (1993)]. For a review of engineered antibody variants [generally see Holliger and Hudson *Nat. Biotechnol.* 23:1126-1136 (2005)].

[0124] Typically, an antibody or antigen binding fragment of the invention retains at least 10% of its canine PD-1 binding activity (when compared to the parental antibody) when that activity is expressed on a molar basis. Preferably, an antibody or antigen binding fragment of the invention retains at least 20%, 50%, 70%, 80%, 90%, 95% or 100% or more of the canine antigen, (e.g., PD-1) binding affinity as the parental antibody. It is also intended that a caninized antibody or antigen binding fragment of the invention can include conservative or non-conservative amino acid substitutions (referred to as “conservative variants” or “function conserved variants” of the antibody) that do not substantially alter its biologic activity.

[0125] “Isolated antibody” refers to the purification status and in such context means the molecule is substantially free of other biological molecules such as nucleic acids, proteins, lipids, carbohydrates, or other material such as cellular debris and growth media. Generally, the term “isolated” is not intended to refer to a complete absence of such material or to an absence of water, buffers, or salts, unless they are

present in amounts that substantially interfere with experimental or therapeutic use of the binding compound as described herein.

[0126] As used herein, a “chimeric antibody” is an antibody having the variable domain from a first antibody and the constant domain from a second antibody, where the first and second antibodies are from different species. [U.S. Pat. No. 4,816,567; and Morrison et al., *Proc. Natl. Acad. Sci. USA* 81:6851-6855 (1984)]. Typically the variable domains are obtained from an antibody from an experimental animal (the “parental antibody”), such as a rodent, and the constant domain sequences are obtained from the animal subject antibodies, e.g., canine, so that the resulting chimeric antibody will be less likely to elicit an adverse immune response in a canine subject, than the parental (e.g., rodent) antibody.

[0127] As used herein, the term “caninized antibody” refers to forms of antibodies that contain sequences from both canine and non-canine (e.g., murine) antibodies. In general, the caninized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable loops correspond to those of a non-canine immunoglobulin (e.g., comprising 6 murine anti-canine PD-1 CDRs as exemplified below), and all or substantially all of the canine frame.

[0128] The term “fully canine antibody” refers to an antibody that comprises canine immunoglobulin protein sequences only. A fully canine antibody may contain murine carbohydrate chains if produced in a mouse, in a mouse cell, or in a hybridoma derived from a mouse cell. Similarly, “mouse antibody” refers to an antibody that comprises mouse immunoglobulin sequences only. Alternatively, a fully canine antibody may contain rat carbohydrate chains if produced in a rat, in a rat cell, or in a hybridoma derived from a rat cell. Similarly, “rat antibody” refers to an antibody that comprises rat immunoglobulin sequences only.

[0129] The variable regions of each light/heavy chain pair form the antibody binding site. Thus, in general, an intact antibody has two binding sites. Except in bifunctional or bispecific antibodies, the two binding sites are, in general, the same.

[0130] Typically, the variable domains of both the heavy and light chains comprise three hypervariable regions, also called complementarity determining regions (CDRs), located within relatively conserved framework regions (FR). The CDRs are usually flanked by the framework regions, enabling binding to a specific epitope. In general, from N-terminal to C-terminal, both light and heavy chains variable domains comprise FR1, CDR1, FR2, CDR2, FR3, CDR3 and FR4. The assignment of amino acids to each domain for human antibodies is, generally, in accordance with the definitions of *Sequences of Proteins of Immunological Interest*, Kabat, et al.; National Institutes of Health, Bethesda, Md.; 5th ed.; NIH Publ. No. 91-3242 (1991); Kabat, *Adv. Prot. Chem.* 32:1-75 (1978); Kabat, et al., *J. Biol. Chem.* 252:6609-6616 (1977); Chothia, et al., *J. Mol. Biol.* 196:901-917 (1987) or Chothia, et al., *Nature* 342:878-883 (1989)].

[0131] As used herein, the term “hypervariable region” refers to the amino acid residues of an antibody that are responsible for antigen-binding. The hypervariable region comprises amino acid residues from a “complementarity determining region” or “CDR” (i.e. CDRL1, CDRL2 and CDRL3 in the light chain variable domain and CDRH1, CDRH2 and CDRH3 in the heavy chain variable domain).

[See Kabat et al. *Sequences of Proteins of Immunological Interest*, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991), defining the CDR regions of a human antibody by sequence; see also Chothia and Lesk, *J. Mol. Biol.* 196:901-917 (1987) defining the CDR regions of an antibody by structure]. As used herein, the term “framework” or “FR” residues refers to those variable domain residues other than the hypervariable region residues defined herein as CDR residues.

[0132] As used herein the term “canine frame” refers to the amino acid sequence of the heavy chain and light chain of a canine antibody other than the hypervariable region residues defined herein as CDR residues. In both chains, the amino acid sequences of the native canine CDRs are replaced with the corresponding foreign CDRs (e.g., those from a mouse antibody). Optionally the heavy and/or light chains of the canine antibody may contain some foreign non-CDR residues, e.g., so as to preserve the conformation of the foreign CDRs within the canine antibody, and/or to modify the Fc function, as exemplified below.

[0133] As used herein, an “anti-canine PD-1 antibody” refers to an antibody that was raised against canine PD-1 (in a mammal such as a mouse or rabbit) and that specifically binds to canine PD-1. An antibody that “specifically binds to canine PD-1,” or an antibody that “specifically binds to a polypeptide comprising the amino acid sequence of SEQ ID NO: 114”, is an antibody that exhibits preferential binding to canine PD-1 as compared to other antigens, e.g., binds canine PD-1 “with specificity”. The binding does not require absolute binding specificity. An anti-canine PD-1 antibody is considered “specific” for canine PD-1 if its binding is determinative of the presence of canine PD-1 in a sample, or if it is capable of altering the activity of canine PD-1 without unduly interfering with the activity of other molecules in a canine sample, e.g., without producing undesired results such as false positives in a diagnostic context or side effects in a therapeutic context. The degree of specificity necessary for an anti-canine PD-1 antibody may depend on the intended use of the antibody, and at any rate is defined by its suitability for use for an intended purpose.

[0134] Accordingly the present invention provides caninized anti-canine PD-1 antibodies or antigen binding fragments thereof (including in isolated form) that bind canine PD-1 (e.g., with specificity) and uses of such antibodies or fragments thereof. In specific embodiments murine anti-canine PD-1 CDRs from murine anti-canine PD-1 antibodies are provided that have been shown to both bind canine PD-1 and to block the binding of canine PD-1 to at least one of its ligands, e.g., canine PD-L1. These CDRs can be inserted into a modified canine frame of the present invention to make a caninized murine anti-canine PD-1 antibody, as exemplified herein.

[0135] More specifically, a “caninized murine anti-PD-1 antibody” of the present invention refers to an antibody that comprises the three heavy chain CDRs and the three light chain CDRs from a murine anti-canine PD-1 antibody together with a canine frame or a modified canine frame. A modified canine frame comprises one or more amino acids changes as exemplified herein that further optimize the effectiveness of the caninized antibody, e.g., to augment, reduce, or eliminate antibody effector functions, to increase its binding to the canine antigen, e.g., canine PD-1, and/or increase its ability to block the binding of the canine antigen,

e.g., canine PD-1, to its natural binding partner, (e.g., canine PD-L1 in the case where the antigen is canine PD-1).

[0136] “Homology” refers to sequence similarity between two polynucleotide sequences or between two polypeptide sequences when they are optimally aligned. When a position in both of the two compared sequences is occupied by the same base or amino acid monomer subunit, e.g., if a position in each of two DNA molecules is occupied by adenine, then the molecules are homologous at that position. The percent of homology is the number of homologous positions shared by the two sequences divided by the total number of positions compared $\times 100$. For example, if 6 of 10 of the positions in two sequences are matched or homologous when the sequences are optimally aligned then the two sequences are 60% homologous. Generally, the comparison is made when two sequences are aligned to give maximum percent homology.

[0137] “Isolated nucleic acid molecule” means a DNA or RNA of genomic, mRNA, cDNA, or synthetic origin or some combination thereof which is not associated with all or a portion of a polynucleotide in which the isolated polynucleotide is found in nature, or is linked to a polynucleotide to which it is not linked in nature. For purposes of this disclosure, it should be understood that “a nucleic acid molecule comprising” a particular nucleotide sequence does not encompass intact chromosomes. Isolated nucleic acid molecules “comprising” specified nucleic acid sequences may include, in addition to the specified sequences, coding sequences for up to ten or even up to twenty or more other proteins or portions or fragments thereof, or may include operably linked regulatory sequences that control expression of the coding region of the recited nucleic acid sequences, and/or may include vector sequences.

[0138] The phrase “control sequences” refers to DNA sequences necessary for the expression of an operably linked coding sequence in a particular host organism. The control sequences that are suitable for prokaryotes, for example, include a promoter, optionally an operator sequence, and a ribosome binding site. Eukaryotic cells are known to use promoters, polyadenylation signals, and enhancers.

[0139] A nucleic acid is “operably linked” when it is placed into a functional relationship with another nucleic acid sequence. For example, DNA for a presequence or secretory leader is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence; or a ribosome binding site is operably linked to a coding sequence if it is positioned so as to facilitate translation. Generally, “operably linked” means that the DNA sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading phase. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites. If such sites do not exist, the synthetic oligonucleotide adaptors or linkers are used in accordance with conventional practice.

[0140] As used herein, the expressions “cell,” “cell line,” and “cell culture” are used interchangeably and all such designations include progeny. Thus, the words “transformants” and “transformed cells” include the primary subject cell and cultures derived therefrom without regard for the number of transfers. It is also understood that not all progeny will have precisely identical DNA content, due to deliberate

or inadvertent mutations. Mutant progeny that have the same function or biological activity as screened for in the originally transformed cell are included. Where distinct designations are intended, it will be clear from the context.

[0141] As used herein, “germline sequence” refers to a sequence of unrearranged immunoglobulin DNA sequences. Any suitable source of unrearranged immunoglobulin sequences may be used. Human germline sequences may be obtained, for example, from JOINSOLVER® germline databases on the website for the National Institute of Arthritis and Musculoskeletal and Skin Diseases of the United States National Institutes of Health. Mouse germline sequences may be obtained, for example, as described in Giudicelli et al. [*Nucleic Acids Res.* 33: D256-D261 (2005)].

Properties of Caninized Antibodies

[0142] In canine, there are four IgG heavy chains referred to as A, B, C, and D. These heavy chains represent four different subclasses of dog IgG, which are referred to as IgGA, IgGB, IgGC and IgGD. The DNA and amino acid sequences of these four heavy chains were first identified by Tang et al. [*Vet. Immunol. Immunopathol.* 80:259-270 (2001)]. The amino acid and DNA sequences for these heavy chains are also available from the GenBank data bases. For example, the amino acid sequence of IgGA heavy chain has accession number AAL35301.1, IgGB has accession number AAL35302.1, IgGC has accession number AAL35303.1, and IgGD has accession number (AAL35304.1). Canine antibodies also contain two types of light chains, kappa and lambda. The DNA and amino acid sequence of these light chains can be obtained from GenBank Databases. For example the kappa light chain amino acid sequence has accession number ABY 57289.1 and the lambda light chain has accession number ABY 55569.1. In the present invention, the amino acid sequence for each of the four canine IgG Fc fragments is based on the identified boundary of CH1 and CH2 domains as determined by Tang et al, supra.

[0143] The development of a therapeutic monoclonal antibody is a complex process that entails coordination of a complex set of activities to generate the desired antibody. These include optimization of the antibody specificity, affinity, functional activity, expression level in engineered cell lines, long-term stability, elimination or enhancement of effector functions and development of commercially viable manufacturing and purification methods. Considering the objectives of the present invention and aside from the capacity to activate cells of the immune systems, a caninized or canine monoclonal antibody against canine PD-1 optimally has three additional attributes:

[0144] 1. lack of effector functions such as antibody-dependent cytotoxicity (ADCC) and complement-dependent cytotoxicity (CDC),

[0145] 2. relatively long half-life in vivo; and

[0146] 3. be readily purified on a large scale using industry standard technologies such as that based on protein A chromatography.

[0147] None of the naturally occurring canine IgG subclasses satisfy all these criteria. For example, IgGB can be purified using protein A, but has a high level of ADCC activity. IgGC also has considerable ADCC activity. On the other hand, IgGA binds weakly to protein A, but displays undesirable ADCC activity. Moreover, neither IgGC nor IgGD can be purified on protein A columns, although IgGD displays no ADCC activity. Additionally IgGC has short

serum half-life as it does not bind to the canine FcRn receptor. The present invention overcomes this difficulty by providing modified canine IgG antibodies specific to canine antigens, e.g., canine PD-1; such antibodies lack effector functions such as ADCC and CDC, display relatively long half-life, and can be easily purified using industry standard protein A chromatography.

[0148] Heretofore, genetically modified canine IgGs that lacked both ADCC and CDC effector functions and in addition, could be purified by protein A chromatography had not been previously described. As disclosed herein, a single substitution at a position in canine IgG that is analogous to that of human and mouse IgG, such as N297A or D265A, does not completely eliminate both ADCC and CDC effector functions in the corresponding canine antibody. For example, while each of the substitutions N297 and D265 in human or murine antibodies results in abrogation of binding to Fc receptor and C1q, neither substitution alone completely abrogated the binding of canine antibodies to C1q. Instead, as further disclosed below, in order to eliminate both ADCC and CDC in canine antibodies of IgGB or IgGC sub-classes, it proved necessary to make a double substitution in the Fc of the canine antibody combining both an asparagine-to-alanine and an aspartic acid-to-alanine substitution. Moreover, completely unexpectedly, one substitution that had been shown to reduce effector functions in human antibodies actually resulted in an increase in binding of corresponding canine IgG to Fc_γR and C1q.

[0149] In order to generate variants of canine IgGB and IgGC that lack effector functions, modified canine IgGB or modified canine IgGC heavy chains can be generated. A total of seven amino acid residues which are present in both of these canine fragment crystallizable regions (cFc) were identified for such possible substitution. These seven amino acid residues are: P4, D31, N63, G64, T65, A93, and P95 for both the amino acid sequence of SEQ ID NO: 130 for the Fc of canine IgGB; and the amino acid sequence of SEQ ID NO: 132 for the Fc of canine IgGC. Accordingly, the amino acid sequence of SEQ ID NO: 2 differs from that of SEQ ID NO: 130 by having the amino acid residues at positions: 4, 31, 63, 64, 65, 93, and 95, which are proline (P), aspartic acid (D), asparagine (N), glycine (G), threonine (T), alanine (A), and proline (P), respectively, in the amino acid sequence of SEQ ID NO: 130 as “X” (or “Xaa” in the three letter code) for all seven positions, signifying that these seven amino acid positions can be any of the twenty natural amino acids (see list in column 1 of Table 1 below). Similarly, the amino acid sequence of SEQ ID NO: 4 differs from that of SEQ ID NO: 132 by having the amino acid residues at positions 4, 31, 63, 64, 65, 93, and 95 are listed as “X” (or “Xaa” in the three letter code) for all seven positions, signifying that these seven amino acid positions can be any of the twenty natural amino acids. The amino acid sequence of SEQ ID NO: 2 is encoded by the nucleotide sequence of SEQ ID NO: 1, whereas the amino acid sequence of SEQ ID NO: 4 is encoded by the nucleotide sequence of SEQ ID NO: 3.

[0150] In one embodiment, the cFc comprises the amino acid sequence of SEQ ID NO: 130 with the following substitutions P4 (A, G, or S), D31 (A, G, or S) N63 (A, G, or S), G64 (A or P), T65 (A, G, or S), A93 (G or S), and P95 (A, G, or S); in which P4 (A, G, or S) signifies that the proline residue at position 4 is replaced by either an alanine, glycine, or serine residue, and similarly G64 (P or A) signifies that

the glycine residue at position 64 is replaced by either a proline or an alanine residue, etc.). In a particular embodiment, the cFc comprises the amino acid sequence of SEQ ID NO: 130 with the following substitutions: P4A, D31A, N63A, G64P, T65A, A93G, and P95A.

[0151] In a related embodiment, the cFc comprises the amino acid sequence of SEQ ID NO: 4, which contains 7 amino acids designated as Xaa, with the following amino acid residues: A4, A31, A63, G64, T65, G93, and A95, i.e., the amino acid sequence of SEQ ID NO: 132 with the following five (5) amino acid residue changes: P4A, D31A, N63A, A93G, and P95A and the remaining two amino acid residues of the seven, G64 and T65, being retained from the amino acid sequence of SEQ ID NO: 132.

[0152] The amino acid sequences of SEQ ID NO: 40, SEQ ID NO: 42, SEQ ID NO: 44, SEQ ID NO: 46, SEQ ID NO: 48, SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 58, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64, and SEQ ID NO: 66 all contain "X" (or "Xaa" in the three letter code) at seven amino acid positions, signifying that these seven amino acid positions can be any of the twenty natural amino acids listed in column 1 of Table 1 below. Notably SEQ ID NO: 40, SEQ ID NO: 42, SEQ ID NO: 44, SEQ ID NO: 46, SEQ ID NO: 48, SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 58, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64, and SEQ ID NO: 66 comprise either the amino acid sequence of SEQ ID NO: 2 or that of SEQ ID NO: 4 within their respective sequences. Specific examples of the amino acid residues at one or more of these seven positions of the amino acid sequences are delineated above and below, and are therefore included within the genus of the individual amino acid sequences of SEQ ID NO: 2, SEQ ID NO: 4, SEQ ID NO: 40, SEQ ID NO: 42, SEQ ID NO: 44, SEQ ID NO: 46, SEQ ID NO: 48, SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 58, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64, and SEQ ID NO: 66, as well as within the caninized antibodies that comprise these sequences.

[0153] Table 10 provided below, specifically correlates the seven amino acid positions that can be replaced, as disclosed herein, of the cIgGB Fc (SEQ ID NO: 130 and SEQ ID NO: 2) and the cIgGC Fc (SEQ ID NO: 132 and SEQ ID NO: 4) with that of the full length canine heavy chains that comprises these cFc amino acid sequences, i.e., SEQ ID NO: 40, SEQ ID NO: 42, SEQ ID NO: 44, SEQ ID NO: 46, SEQ ID NO: 48, SEQ ID NO: 50, SEQ ID NO: 52, SEQ ID NO: 54, SEQ ID NO: 56, SEQ ID NO: 58, SEQ ID NO: 60, SEQ ID NO: 62, SEQ ID NO: 64, and SEQ ID NO: 66. Accordingly, the actual position in the full length sequence IgGB or IgGC can be readily coordinated with that of the cFc that it comprises through the use of Table 10 below.

[0154] In particular embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 40, 52, 56, or 64 comprising (i) P, A, G, or S at position 239, (ii) D, A, G, or S at position 266, (iii) N, A, G, or S at position 298, (iv) G, P, or A at position 299, (v) T, A, G, or S at position 300, (vi) A, G, or S at position 328, and (vii) P, A, G, or S at position 330. In other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 42, 54, 58, or 66 comprising (i) P, A, G, or S at position 237, (ii) D, A, G, or S at position 264, (iii) N, A, G, or S at position 296, (iv) G, P, or A at position 297, (v) T, A, G, or S at position 298, (vi) A, G, or S at

position 326, and (vii) P, A, G, or S at position 328. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 44, 50, or 60 comprising (i) P, A, G, or S at position 244, (ii) D, A, G, or S at position 271, (iii) N, A, G, or S at position 303, (iv) G, P, or A at position 304, (v) T, A, G, or S at position 305, (vi) A, G, or S at position 333, and (vii) P, A, G, or S at position 335. In still other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 46 or 62 comprising (i) P, A, G, or S at position 242, (ii) D, A, G, or S at position 269, (iii) N, A, G, or S at position 301, (iv) G, P, or A at position 302, (v) T, A, G, or S at position 303, (vi) A, G, or S at position 331, and (vii) P, A, G, or S at position 333. In yet other embodiments, the heavy chain of an antibody comprises the amino acid sequence of SEQ ID NO: 48 comprising (i) P, A, G, or S at position 246, (ii) D, A, G, or S at position 273, (iii) N, A, G, or S at position 305, (iv) G, P, or A at position 306, (v) T, A, G, or S at position 307, (vi) A, G, or S at position 335, and (vii) P, A, G, or S at position 337.

[0155] The present invention also provides modified canine IgGDs which comprise a hinge region from either IgGA, IgGB, or IgGC in place of its natural IgGD hinge region. Alternatively, the IgGD hinge region can be genetically modified by replacing a serine residue with a proline residue as shown in Table 5. Such modifications can lead to a canine IgGD lacking fab arm exchange. The modified canine IgGDs can be constructed using standard methods of recombinant DNA technology [e.g., Maniatis et al., *Molecular Cloning, A Laboratory Manual* (1982)]. In order to construct these variants, the nucleic acids encoding the amino acid sequence of canine IgGD can be modified so that it encodes the modified IgGDs. The modified nucleic acid sequences are then cloned into expression plasmids for protein expression. The nucleic acids encoding the canine IgGD Fcs with the substitute hinge region are exemplified by nucleotide sequences of SEQ ID NOs: 7, 9, and 11 which encode the amino acid sequences of SEQ ID NOs: 8, 10, and 12. A nucleic acid encoding a canine IgGD Fc with a modified IgGD hinge region comprises the nucleotide sequence of SEQ ID NO: 5 which encodes the amino acid sequence of SEQ ID NO: 6.

[0156] The present invention further provides full length canine heavy chains that can be matched with corresponding light chains to make a caninized antibody. Accordingly, the present invention further provides caninized murine anti-canine antigen antibodies (including isolated caninized murine anti-canine PD-1 antibodies) and methods of use of the antibodies or antigen binding fragments thereof in the treatment of disease e.g., the treatment of cancer in canines.

[0157] Moreover, the present invention provides caninized murine anti-canine PD-1 antibodies or antigen binding fragments that bind to canine PD-1 and block the binding of canine PD-1 to canine PD-L1. In certain embodiments the caninized murine anti-canine PD-1 antibodies comprise a modified canine IgGB Fc, modified canine IgGC Fc, or a modified canine IgGD lacking fab arm exchange as described herein.

[0158] The antibody or antigen binding fragment thereof that binds the canine antigen, e.g., canine PD-1, can comprise one, two, three, four, five, or six of the complementarity determining regions (CDRs) of the murine anti-canine antibody as described herein. The one, two, three, four, five, or six CDRs may be independently selected from the CDR

sequences of those provided below. In a further embodiment, the antibody or antigen-binding fragment thereof that binds canine PD-1 comprises a canine antibody kappa light chain comprising a murine light chain CDR-1, CDR-2 and/or CDR-3 and a canine antibody heavy chain IgG comprising a murine heavy chain CDR-1, CDR-2 and/or CDR-3. Accordingly, the present invention further provides full length canine heavy chains then can be matched e.g., with the corresponding light chains to make a caninized antibody [see Table 2 below, in which the sequences of seven sets of CDRs of murine anti-canine PD-1, e.g., 1B5, 2G9, 2H9, 3B6, 4D12, 5G5, and 7C9 are provided].

[0159] In other embodiments, the invention provides antibodies or antigen binding fragments thereof that bind PD-1 with specificity and have canine antibody kappa light chains comprising one to six different CDRs comprising at least 80%, 85%, 90%, 95%, 98% or 99% sequence identity with the amino acid sequences of SEQ ID NOs: 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, and/or 26 and canine antibody heavy chain IgG comprising one to six different CDRs comprising at least 80%, 85%, 90%, 95%, 98% or 99% sequence identity with the amino acid sequences of SEQ ID NOs: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, and/or 146, while still exhibiting the desired binding and functional properties. In another embodiment the antibody or antigen binding fragment of the present invention comprises a canine frame comprising of a combination of IgG heavy chain sequence with a kappa light chain having one or more of the above-mentioned CDR amino acid sequences with 0, 1, 2, 3, 4, or 5 conservative or non-conservative amino acid substitutions, while still exhibiting the desired binding and functional properties.

[0160] Sequence identity refers to the degree to which the amino acids of two polypeptides are the same at equivalent positions when the two sequences are optimally aligned. As used herein one amino acid sequence is 100% “identical” to a second amino acid sequence when the amino acid residues of both sequences are identical. Accordingly, an amino acid sequence is 50% “identical” to a second amino acid sequence when 50% of the amino acid residues of the two amino acid sequences are identical. The sequence comparison is performed over a contiguous block of amino acid residues comprised by a given protein, e.g., a protein, or a portion of the polypeptide being compared. In a particular embodiment, selected deletions or insertions that could otherwise alter the correspondence between the two amino acid sequences are taken into account.

[0161] Sequence similarity includes identical residues and nonidentical, biochemically related amino acids. Biochemically related amino acids that share similar properties and may be interchangeable are discussed

[0162] “Conservatively modified variants” or “conservative substitution” refers to substitutions of amino acids in a protein with other amino acids having similar characteristics (e.g. charge, side-chain size, hydrophobicity/hydrophilicity, backbone conformation and rigidity, etc.), such that the changes can frequently be made without altering the biological activity of the protein. Those of skill in this art recognize that, in general, single amino acid substitutions in non-essential regions of a polypeptide do not substantially alter biological activity [see, e.g., Watson et al., *Molecular Biology of the Gene*, The Benjamin/Cummings Pub. Co., p. 224 (4th Ed.; 1987)]. In addition, substitutions of structurally or functionally similar amino acids are less likely to

disrupt biological activity. Exemplary conservative substitutions are set forth in Table I directly below.

TABLE 1

EXEMPLARY CONSERVATIVE AMINO ACID SUBSTITUTIONS	
Original residue	Conservative substitution
Ala (A)	Gly; Ser
Arg (R)	Lys; His
Asn (N)	Gln; His
Asp (D)	Glu; Asn
Cys (C)	Ser; Ala
Gln (Q)	Asn
Glu (E)	Asp; Gln
Gly (G)	Ala
His (H)	Asn; Gln
Ile (I)	Leu; Val
Leu (L)	Ile; Val
Lys (K)	Arg; His
Met (M)	Leu; Ile; Tyr
Phe (F)	Tyr; Met; Leu
Pro (P)	Ala; Gly
Ser (S)	Thr
Thr (T)	Ser
Trp (W)	Tyr; Phe
Tyr (Y)	Trp; Phe
Val (V)	Ile; Leu

[0163] Function-conservative variants of the antibodies of the invention are also contemplated by the present invention. “Function-conservative variants,” as used herein, refers to antibodies or fragments in which one or more amino acid residues have been changed without altering a desired property, such an antigen affinity and/or specificity. Such variants include, but are not limited to, replacement of an amino acid with one having similar properties, such as the conservative amino acid substitutions of Table I above.

Nucleic Acids,

[0164] The present invention further comprises the nucleic acids encoding the immunoglobulin chains of caninized murine anti-canine PD-1 antibodies and antigen binding fragments thereof disclosed herein (see Examples below).

[0165] Also included in the present invention are nucleic acids that encode immunoglobulin polypeptides comprising amino acid sequences that are at least about 70% identical, preferably at least about 80% identical, more preferably at least about 90% identical and most preferably at least about 95% identical (e.g., 95%, 96%, 97%, 98%, 99%, 100%) to the amino acid sequences of the CDRs and/or canine cFc’s and/or antibodies provided herein when the comparison is performed by a BLAST algorithm wherein the parameters of the algorithm are selected to give the largest match between the respective sequences over the entire length of the respective reference sequences. The present invention further provides nucleic acids that encode immunoglobulin polypeptides comprising amino acid sequences that are at least about 70% similar, preferably at least about 80% similar, more preferably at least about 90% similar and most preferably at least about 95% similar (e.g., 95%, 96%, 97%, 98%, 99%, 100%) to any of the reference amino acid sequences when the comparison is performed with a BLAST algorithm, wherein the parameters of the algorithm are selected to give the largest match between the respective sequences over the entire length of the respective reference sequences.

[0166] As used herein, nucleotide and amino acid sequence percent identity can be determined using C, MacVector (MacVector, Inc. Cary, NC 27519), Vector NTI (Informax, Inc. MD), Oxford Molecular Group PLC (1996) and the Clustal W algorithm with the alignment default parameters, and default parameters for identity. These commercially available programs can also be used to determine sequence similarity using the same or analogous default parameters. Alternatively, an Advanced Blast search under the default filter conditions can be used, e.g., using the GCG (Genetics Computer Group, Program Manual for the GCG Package, Version 7, Madison, Wisconsin) pileup program using the default parameters.

[0167] The following references relate to BLAST algorithms often used for sequence analysis: BLAST ALGORITHMS: Altschul, S. F., et al., *J. Mol. Biol.* 215:403-410 (1990); Gish, W., et al., *Nature Genet.* 3:266-272 (1993); Madden, T. L., et al., *Meth. Enzymol.* 266:131-141 (1996); Altschul, S. F., et al., *Nucleic Acids Res.* 25:3389-3402 (1997); Zhang, J., et al., *Genome Res.* 7:649-656 (1997); Wootton, J. C., et al., *Comput. Chem.* 17:149-163 (1993); Hancock, J. M. et al., *Comput. Appl. Biosci.* 10:67-70 (1994); ALIGNMENT SCORING SYSTEMS: Dayhoff, M. O., et al., "A model of evolutionary change in proteins." in *Atlas of Protein Sequence and Structure*, vol. 5, suppl. 3. M. O. Dayhoff (ed.), pp. 345-352, (1978); *Natl. Biomed. Res. Found.*, Washington, DC; Schwartz, R. M., et al., "Matrices for detecting distant relationships." in *Atlas of Protein Sequence and Structure*, vol. 5, suppl. 3." (1978), M. O. Dayhoff (ed.), pp. 353-358 (1978), *Natl. Biomed. Res. Found.*, Washington, DC; Altschul, S. F., *J. Mol. Biol.* 219:555-565 (1991); States, D. J., et al., *Methods* 3:66-70 (1991); Henikoff, S., et al., *Proc. Natl. Acad. Sci. USA* 89:10915-10919 (1992); Altschul, S. F., et al., *J. Mol. Evol.* 36:290-300 (1993); ALIGNMENT STATISTICS: Karlin, S., et al., *Proc. Natl. Acad. Sci. USA* 87:2264-2268 (1990); Karlin, S., et al., *Proc. Natl. Acad. Sci. USA* 90:5873-5877 (1993); Dembo, A., et al., *Ann. Prob.* 22:2022-2039 (1994); and Altschul, S. F. "Evaluating the statistical significance of multiple distinct local alignments." in *Theoretical and Computational Methods in Genome Research* (S. Suhai, ed.), pp. 1-14, Plenum, New York (1997).

[0168] This present invention also provides expression vectors comprising the nucleic acids (including isolated nucleic acids) of the invention, wherein the nucleic acid is operably linked to control sequences that are recognized by a host cell when the host cell is transfected with the vector. Also provided are host cells comprising an expression vector of the present invention and methods for producing the antibody or antigen binding fragment thereof disclosed herein comprising culturing a host cell harboring an expression vector encoding the antibody or antigen binding fragment in culture medium, and isolating the antigen or antigen binding fragment thereof from the host cell or culture medium.

[0169] A caninized murine anti-canine PD-1 antibody for example, can be produced recombinantly by methods that are known in the field. Mammalian cell lines available as hosts for expression of the antibodies or fragments disclosed herein are well known in the art and include many immortalized cell lines available from the American Type Culture Collection (ATCC). These include, inter alia, Chinese hamster ovary (CHO) cells, NSO, SP2 cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS),

human hepatocellular carcinoma cells (e.g., Hep G2), A549 cells, 3T3 cells, HEK-293 cells and a number of other cell lines. Mammalian host cells include human, mouse, rat, dog, monkey, pig, goat, bovine, horse, and hamster cells. Cell lines of particular preference are selected through determining which cell lines have high expression levels. Other cell lines that may be used are insect cell lines, such as Sf9 cells, amphibian cells, bacterial cells, plant cells and fungal cells. When recombinant expression vectors encoding the heavy chain or antigen-binding portion or fragment thereof, the light chain and/or antigen-binding fragment thereof are introduced into mammalian host cells, the antibodies are produced by culturing the host cells for a period of time sufficient to allow for expression of the antibody in the host cells or, more preferably, secretion of the antibody into the culture medium in which the host cells are grown.

[0170] Antibodies can be recovered from the culture medium using standard protein purification methods. Further, expression of antibodies of the invention (or other moieties therefrom) from production cell lines can be enhanced using a number of known techniques. For example, the glutamine synthetase gene expression system (the GS system) is a common approach for enhancing expression under certain conditions. The GS system is discussed in whole or part in connection with European Patent Nos. 0 216 846, 0 256 055, and 0 323 997 and European Patent Application No. 89303964.4.

[0171] In general, glycoproteins produced in a particular cell line or transgenic animal will have a glycosylation pattern that is characteristic for glycoproteins produced in the cell line or transgenic animal. Therefore, the particular glycosylation pattern of an antibody will depend on the particular cell line or transgenic animal used to produce the antibody. However, all antibodies encoded by the nucleic acid molecules provided herein, or comprising the amino acid sequences provided herein, comprise the instant invention, independent of the glycosylation pattern that the antibodies may have. Similarly, in particular embodiments, antibodies with a glycosylation pattern comprising only non-fucosylated N-glycans may be advantageous, because these antibodies have been shown to typically exhibit more potent efficacy than their fucosylated counterparts both in vitro and in vivo [See for example, Shinkawa et al., *J. Biol. Chem.* 278:3466-3473 (2003); U.S. Pat. Nos. 6,946,292 and 7,214,775].

[0172] The present invention further includes antibody fragments of the caninized murine anti-canine PD-1 antibodies disclosed herein. The antibody fragments include F(ab)₂ fragments, which may be produced by enzymatic cleavage of an IgG by, for example, pepsin. Fab fragments may be produced by, for example, reduction of F(ab)₂ with dithiothreitol or mercaptoethylamine. A Fab fragment is a V_L-C_L chain appended to a V_H-C_{H1} chain by a disulfide bridge. A F(ab)₂ fragment is two Fab fragments which, in turn, are appended by two disulfide bridges. The Fab portion of an F(ab)₂ molecule includes a portion of the F_c region between which disulfide bridges are located. An Fv fragment is a V_L or V_H region.

[0173] In one embodiment, the antibody or antigen binding fragment comprises a heavy chain constant region, e.g., a canine constant region, such as IgGA, IgGB, IgGC and IgGD canine heavy chain constant region or a variant thereof. In another embodiment, the antibody or antigen binding fragment comprises a light chain constant region,

e.g., a canine light chain constant region, such as lambda or kappa canine light chain region or variant thereof. By way of example, and not limitation the canine heavy chain constant region can be from IgGB and the canine light chain constant region can be from kappa.

Antibody Engineering

[0174] Caninized murine anti-canine PD-1 antibodies of the present invention can be engineered to include modifications in the canine frame of a parental (i.e., canine) monoclonal antibody, e.g. to improve the properties of the antibody, as detailed below.

[0175] The cross-blocking caninized antibodies and antigen-binding fragments thereof discussed herein can be identified based on their ability to cross-compete with any of IB5, 3B6, 4D12, 7C9, 2H9, 5G5, and/or 2G9 in standard binding assays (e.g., BIAcore®, ELISA, as exemplified below, or flow cytometry). For example, standard ELISA assays can be used in which a recombinant canine PD-1 protein is immobilized on the plate, one of the antibodies is fluorescently labeled and the ability of non-labeled antibodies to compete off the binding of the labeled antibody is evaluated. Additionally or alternatively, BIAcore® analysis can be used to assess the ability of the antibodies to cross-compete. The ability of a test antibody to inhibit the binding of, for example, IB5, 3B6, 4D12, 7C9, 2H9, 5G5, and/or 2G9, to canine PD-1 demonstrates that the test antibody can compete with IB5, 3B6, 4D12, 7C9, 2H9, 5G5, and/or 2G9 for binding to canine PD-1 and thus, may, in some cases, bind to the same epitope on canine PD-1 as IB5, 3B6, 4D12, 7C9, 2H9, 5G5, and/or 2G9. As stated above, antibodies and fragments that bind to the same epitope as any of the anti-canine PD-1 antibodies or fragments of the present invention also form part of the present invention.

Pharmaceutical Compositions and Administration

[0176] To prepare pharmaceutical or sterile compositions of a caninized murine anti-canine PD-1 antibody or antigen binding fragment thereof it can be admixed with a pharmaceutically acceptable carrier or excipient. [See, e.g., *Remington's Pharmaceutical Sciences and U.S. Pharmacopeia: National Formulary*, Mack Publishing Company, Easton, PA (1984)].

[0177] Formulations of therapeutic and diagnostic agents may be prepared by mixing with acceptable carriers, excipients, or stabilizers in the form of, e.g., lyophilized powders, slurries, aqueous solutions or suspensions [see, e.g., Hardman, et al. (2001) *Goodman and Gilman's The Pharmacological Basis of Therapeutics*, McGraw-Hill, New York, NY; Gennaro (2000) *Remington: The Science and Practice of Pharmacy*, Lippincott, Williams, and Wilkins, New York, NY; Avis, et al. (eds.) (1993) *Pharmaceutical Dosage Forms: Parenteral Medications*, Marcel Dekker, NY; Lieberman, et al. (eds.) (1990) *Pharmaceutical Dosage Forms: Tablets*, Marcel Dekker, NY; Lieberman, et al. (eds.) (1990) *Pharmaceutical Dosage Forms: Disperse Systems*, Marcel Dekker, NY; Weiner and Kotkoskie (2000) *Excipient Toxicity and Safety*, Marcel Dekker, Inc., New York, NY]. In one embodiment, anti-PD-1 antibodies of the present invention are diluted to an appropriate concentration in a sodium acetate solution pH 5-6, and NaCl or sucrose is added for tonicity. Additional agents, such as polysorbate 20 or polysorbate 80, may be added to enhance stability.

[0178] Toxicity and therapeutic efficacy of the antibody compositions, administered alone or in combination with another agent, can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD₅₀ (the dose lethal to 50% of the population) and the ED₅₀ (the dose therapeutically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index (LD₅₀/ED₅₀). In particular aspects, antibodies exhibiting high therapeutic indices are desirable. The data obtained from these cell culture assays and animal studies can be used in formulating a range of dosage for use in canines. The dosage of such compounds lies preferably within a range of circulating concentrations that include the ED₅₀ with little or no toxicity. The dosage may vary within this range depending upon the dosage form employed and the route of administration.

[0179] The mode of administration can vary. Suitable routes of administration include oral, rectal, transmucosal, intestinal, parenteral; intramuscular, subcutaneous, intradermal, intramedullary, intrathecal, direct intraventricular, intravenous, intraperitoneal, intranasal, intraocular, inhalation, insufflation, topical, cutaneous, transdermal, or intra-arterial.

[0180] In particular embodiments, the caninized murine anti-canine PD-1 antibody or antigen binding fragment thereof can be administered by an invasive route such as by injection. In further embodiments of the invention, a murine anti-canine PD-1 antibody or antigen binding fragment thereof, or pharmaceutical composition thereof, is administered intravenously, subcutaneously, intramuscularly, intraarterially, intratumorally, or by inhalation, aerosol delivery. Administration by non-invasive routes (e.g., orally; for example, in a pill, capsule or tablet) is also within the scope of the present invention.

[0181] The pharmaceutical compositions disclosed herein may also be administered by infusion. Examples of well-known implants and modules for administering pharmaceutical compositions include: U.S. Pat. No. 4,487,603, which discloses an implantable micro-infusion pump for dispensing medication at a controlled rate; U.S. Pat. No. 4,447,233, which discloses a medication infusion pump for delivering medication at a precise infusion rate; U.S. Pat. No. 4,447,224, which discloses a variable flow implantable infusion apparatus for continuous drug delivery; U.S. Pat. No. 4,439,196, which discloses an osmotic drug delivery system having multi-chamber compartments. Many other such implants, delivery systems, and modules are well known to those skilled in the art.

[0182] Alternately, one may administer a caninized murine anti-canine PD-1 antibody in a local rather than systemic manner, for example, via injection of the antibody directly into an arthritic joint or pathogen-induced lesion characterized by immunopathology, often in a depot or sustained release formulation. Furthermore, one may administer the antibody in a targeted drug delivery system, for example, in a liposome coated with a tissue-specific antibody, targeting, for example, arthritic joint or pathogen-induced lesion characterized by immunopathology. The liposomes will be targeted to and taken up selectively by the afflicted tissue.

[0183] The administration regimen depends on several factors, including the serum or tissue turnover rate of the therapeutic antibody, the level of symptoms, the immunogenicity of the therapeutic antibody, and the accessibility of

the target cells in the biological matrix. Preferably, the administration regimen delivers sufficient therapeutic antibody to effect improvement in the target disease state, while simultaneously minimizing undesired side effects. Accordingly, the amount of biologic delivered depends in part on the particular therapeutic antibody and the severity of the condition being treated. Guidance in selecting appropriate doses of therapeutic antibodies is available [see, e.g., Wawrzynczak *Antibody Therapy*, Bios Scientific Pub. Ltd, Oxfordshire, U K (1996); Kresina (ed.) *Monoclonal Antibodies, Cytokines and Arthritis*, Marcel Dekker, New York, NY (1991); Bach (ed.) *Monoclonal Antibodies and Peptide Therapy in Autoimmune Diseases*, Marcel Dekker, New York, NY (1993); Baert, et al. *New Engl. J. Med.* 348:601-608 (2003); Milgrom et al. *New Engl. J. Med.* 341:1966-1973 (1999); Slamon et al. *New Engl. J. Med.* 344:783-792 (2001); Beniaminovitz et al. *New Engl. J. Med.* 342:613-619 (2000); Ghosh et al. *New Engl. J. Med.* 348:24-32 (2003); Lipsky et al. *New Engl. J. Med.* 343:1594-1602 (2000)].

[0184] Determination of the appropriate dose is made by the veterinarian, e.g., using parameters or factors known or suspected in the art to affect treatment. Generally, the dose begins with an amount somewhat less than the optimum dose and it is increased by small increments thereafter until the desired or optimum effect is achieved relative to any negative side effects. Important diagnostic measures include those of symptoms of, e.g., the inflammation or level of inflammatory cytokines produced.

[0185] Antibodies or antigen binding fragments thereof disclosed herein may be provided by continuous infusion, or by doses administered, e.g., daily, 1-7 times per week, weekly, biweekly, monthly, bimonthly, quarterly, semiannually, annually etc. Doses may be provided, e.g., intravenously, subcutaneously, topically, orally, nasally, rectally, intramuscular, intracerebrally, intraspinally, or by inhalation. A total weekly dose is generally at least 0.05 µg/kg body weight, more generally at least 0.2 µg/kg, 0.5 µg/kg, 1 µg/kg, 10 µg/kg, 100 µg/kg, 0.25 mg/kg, 1.0 mg/kg, 2.0 mg/kg, 5.0 mg/ml, 10 mg/kg, 25 mg/kg, 50 mg/kg or more [see, e.g., Yang, et al. *New Engl. J. Med.* 349:427-434 (2003); Herold, et al. *New Engl. J. Med.* 346:1692-1698 (2002); Liu, et al. *J. Neurol. Neurosurg. Psych.* 67:451-456 (1999); Portielji, et al. *Cancer Immunol. Immunother.* 52:133-144 (2003)]. Doses may also be provided to achieve a pre-determined target concentration of a caninized murine anti-canine PD-1 antibody in the subject's serum, such as 0.1, 0.3, 1, 3, 10, 30, 100, 300 µg/ml or more. In other embodiments, a caninized murine anti-canine PD-1 antibody of the present invention is administered subcutaneously or intravenously, on a weekly, biweekly, "every 4 weeks," monthly, bimonthly, or quarterly basis at 10, 20, 50, 80, 100, 200, 500, 1000 or 2500 mg/subject.

[0186] As used herein, "inhibit" or "treat" or "treatment" includes a postponement of development of the symptoms associated with a disorder and/or a reduction in the severity of the symptoms of such disorder. The terms further include ameliorating existing uncontrolled or unwanted symptoms, preventing additional symptoms, and ameliorating or preventing the underlying causes of such symptoms. Thus, the terms denote that a beneficial result has been conferred on a vertebrate subject with a disorder, disease or symptom, or with the potential to develop such a disorder, disease or symptom.

[0187] As used herein, the terms "therapeutically effective amount", "therapeutically effective dose" and "effective amount" refer to an amount of a caninized murine anti-canine PD-1 antibody or antigen binding fragment thereof of the present invention that, when administered alone or in combination with an additional therapeutic agent to a cell, tissue, or subject, is effective to cause a measurable improvement in one or more symptoms of a disease or condition or the progression of such disease or condition. A therapeutically effective dose further refers to that amount of the binding compound sufficient to result in at least partial amelioration of symptoms, e.g., treatment, healing, prevention or amelioration of the relevant medical condition, or an increase in rate of treatment, healing, prevention or amelioration of such conditions. When applied to an individual active ingredient administered alone, a therapeutically effective dose refers to that ingredient alone. When applied to a combination, a therapeutically effective dose refers to combined amounts of the active ingredients that result in the therapeutic effect, whether administered in combination, serially or simultaneously. An effective amount of a therapeutic will result in an improvement of a diagnostic measure or parameter by at least 10%; usually by at least 20%; preferably at least about 30%; more preferably at least 40%, and most preferably by at least 50%. An effective amount can also result in an improvement in a subjective measure in cases where subjective measures are used to assess disease severity.

Other Combination Therapies

[0188] As previously described, a caninized murine anti-canine PD-1 antibody or antigen binding fragment thereof may be coadministered with one or other more therapeutic agents (such as a chemotherapeutic agent). The antibody may be linked to the agent (as an immunocomplex) or can be administered separately from the agent. In the latter case (separate administration), the antibody can be administered before, after or concurrently with the agent or can be coadministered with other known therapies.

Kits

[0189] Further provided are kits comprising one or more components that include, but are not limited to, an antibody or antigen binding fragment, as discussed herein, which specifically binds PD-1 (e.g., a caninized murine anti-canine PD-1 antibody or antigen binding fragment thereof) in association with one or more additional components including, but not limited to a pharmaceutically acceptable carrier and/or a chemotherapeutic agent, as discussed herein. The binding composition and/or the chemotherapeutic agent can be formulated as a pure composition or in combination with a pharmaceutically acceptable carrier, in a pharmaceutical composition.

[0190] In one embodiment, the kit includes a binding composition of the present invention (e.g., a caninized murine anti-canine PD-1 or a pharmaceutical composition thereof in one container (e.g., in a sterile glass or plastic vial) and a pharmaceutical composition thereof and/or a chemotherapeutic agent in another container (e.g., in a sterile glass or plastic vial).

[0191] If the kit includes a pharmaceutical composition for parenteral administration to a subject, the kit can also include a device for performing such administration. For

example, the kit can include one or more hypodermic needles or other injection devices as discussed above. The kit can also include a package insert including information concerning the pharmaceutical compositions and dosage forms in the kit. Generally, such information aids pet owners and veterinarians in using the enclosed pharmaceutical compositions and dosage forms effectively and safely. For example, the following information regarding a combination of the invention may be supplied in the insert: pharmacokinetics, pharmacodynamics, clinical studies, efficacy parameters, indications and usage, contraindications, warnings, precautions, adverse reactions, overdosage, proper dosage and administration, how supplied, proper storage conditions, references, manufacturer/distributor information and patent information.

[0192] As a matter of convenience, an antibody or specific binding agent disclosed herein can be provided in a kit, i.e., a packaged combination of reagents in predetermined amounts with instructions for performing the diagnostic or detection assay. Where the antibody is labeled with an enzyme, the kit will include substrates and cofactors required by the enzyme (e.g., a substrate precursor which provides the detectable chromophore or fluorophore). In addition, other additives may be included such as stabilizers, buffers (e.g., a block buffer or lysis buffer) and the like. The relative amounts of the various reagents may be varied widely to provide for concentrations in solution of the reagents which substantially optimize the sensitivity of the assay. Particularly, the reagents may be provided as dry powders, usually lyophilized, including excipients which on dissolution will provide a reagent solution having the appropriate concentration.

EXAMPLES

Example 1

Canine Pd-1 and Pd-L1

Canine PD-1 and PD-L1:

[0193] U.S. provisional application No. 61/918,946, filed on Dec. 20, 2013, hereby incorporated by reference in its entirety, provides: the full length nucleotide sequence for canine PD-1 (cPD-1) of SEQ ID NO: 113 [SEQ ID NO: 133 includes the signal sequence]; the corresponding translated amino acid sequence of SEQ ID NO: 114 [SEQ ID NO: 134 includes the signal sequence]; the nucleotide sequence encoding the extra-cellular domain (ECD) of canine PD-1, SEQ ID NO: 115; the amino acid sequence of the ECD of canine PD-1, SEQ ID NO: 116; the nucleotide sequence of canine PD-1 ECD plus a GT linker and the Fc part of human IgG1 Fc gene, of SEQ ID NO: 117; and the amino acid sequence of the canine PD-1 ECD plus a GT linker and the Fc part of human IgG1 Fc gene, SEQ ID NO: 118 [SEQ ID NO: 137 includes the signal sequence].

[0194] U.S. provisional application No. 61/918,946 further provides: the full length nucleotide sequence for canine PD-L1 (cPD-L1) of SEQ ID NO: 119 [SEQ ID NO: 135 includes the signal sequence]; the corresponding translated amino acid sequence of SEQ ID NO: 120 [SEQ ID NO: 136 includes the signal sequence]; the nucleotide sequence encoding the extra-cellular domain (ECD) of canine PD-L1, SEQ ID NO: 121; the amino acid sequence of the ECD of canine PD-L1, SEQ ID NO: 122; the nucleotide sequence of

canine PD-L1 ECD plus a GT linker and the Fc part of human IgG1 Fc gene, SEQ ID NO: 123; and the amino acid sequence of canine PD-L1 ECD plus a GT linker and the Fc part of human IgG1 Fc gene, SEQ ID NO: 124.

Example 2

Murine Anti-Canine Pd-1 Antibodies

Generation of Anti-Canine PD1 Monoclonal Antibodies:

[0195] A total of three Balb/c mice were immunized multiple times (with 10 µg each time) over a 17 day period. The immunizing antigen was the canine PD-1 ECD-Fc fusion protein. Following immunization, serum was collected from each mouse and tested for reactivity with canine PD-1 ECD-HIS tagged protein. The spleen cells of the mouse with the highest serum anti-PD-1 ECD-HIS titer were fused to the myeloma P3×63Ag8.653 cell line. Approximately 2 weeks following fusion, supernatant from putative hybridoma cells were tested by ELISA for their reactivity to the PD-1 ECD-HIS tagged protein. Hybridomas producing strong positive signals in the ELISA were subcloned by limiting dilution and tested again for reactivity to canine PD-1 ECD-HIS tagged protein.

Confirmation of Monoclonal Antibodies Reactivity Against Canine PD-1:

[0196] The reactivity of antibodies secreted by hybridomas to ECD of canine PD-1 was confirmed by ELISA. Hybridoma cells were cultured using CELLline bioreactors (Integra-biosciences) for 10-30 days. Cells were initially maintained in DMEM supplemented with 4 mM L-glutamine and 10% Ultra Low IgG fetal bovine serum (FBS) from Gibco. Hybridoma cells were seeded in CELLline bioreactor cell chambers at a cell density of approximately 2×10^6 cells/mL in 15 mL of the same medium with the FBS concentration increased to 20%. The outer chamber was filled with 1 L of nutrient medium (DMEM with 4 mM L-glutamine and 2% standard FBS). Hybridoma cells in the cell chamber were expanded to approximately 2.5×10^7 cells/mL over 3-7 days. Then, 10 mL of cell suspension was harvested from the cell chamber and replaced with fresh media to allow for re-expansion of cells and subsequent harvests. This procedure was repeated as necessary to obtain adequate amounts of mAb from each hybridoma clone. Harvested cell suspensions were centrifuged and the supernatants were filtered through 0.2 micron filter membranes. For antibody purification, each clone's supernatant was purified using a Protein G Sepharose 4 Fast flow 5 mL column (GE Healthcare) by gravity flow. After washing with Tris-EDTA (TE) buffer pH 8.0, bound antibodies were eluted using 0.1 M glycine buffer, pH 2.7, followed by pH neutralization using 1 M Tris, pH 8.0. Antibodies were concentrated and buffer exchanged into phosphate-buffered saline (PBS) using Centrprep® YM-10.10 kDa NMWL centrifugal filter units (Millipore). Antibody concentrations were quantified using spectrophotometry.

[0197] Purified anti-canine PD-1 mAbs were tested for reactivity with the HIS-tagged ECD domain of canine PD-1 by ELISA as follows: HIS-tagged canine PD-1 ECD protein is diluted to 10 µg/mL in coating buffer (Carbonate/Bicarbonate pH 9.0) and dispensed at 100 µL/well in 96-well flat bottomed ELISA plates (NUNC). The plates are incubated at 4° C. overnight. The plates are then washed three times with

TABLE 2-continued

AMINO ACID SEQUENCES OF THE CDRs									
									SEQ ID NO.
5G5	Gln Thr	Gln	Gly	Gln	Ser	Tyr	Pro	Leu	26
7C9	Gln Thr	Gln	Tyr	Phe	Ser	Thr	Pro	Leu	22
VH CDR1									
1B5	Gly Met	Tyr Ser	Thr	Phe	Thr	Thr	Tyr	Gly	27
2G9	Gly Met	Tyr His	Thr	Phe	Thr	Arg	Tyr	Asn	28
2H9	Gly Met	Phe His	Asn	Ile	Lys	Asn	Thr	Tyr	29
3B6	Gly Met	Tyr Ser	Thr	Phe	Thr	Thr	Tyr	Gly	27
4D12	Gly Met	Tyr Ser	Thr	Phe	Thr	Thr	Tyr	Gly	27
5G5	Gly Met	Phe His	Asn	Ile	Lys	Asn	Thr	Tyr	29
7C9	Gly Val	Phe His	Ser	Leu	Thr	Ser	Tyr	Gly	30
VH CDR2									
1B5	Trp Pro Gly	Ile Thr	Asn Tyr	Ile Ala	Tyr Asp	Ser Asp	Gly Phe	Ile Lys	31
2G9	Thr Thr Gly	Ile Ser	Tyr Tyr	Pro Asn	Gly Gln	Tyr Lys	Gly Phe	Asp Lys	32
2H9	Arg Thr Gly	Ile Lys	Ala Tyr	Pro Ala	Ala Pro	Asn Lys	Val Phe	Asp Gln	33
3B6	Trp Pro Gly	Ile Thr	Asn Tyr	Ile Ala	Tyr Asp	Ser Asp	Gly Phe	Ile Lys	31
4D12	Trp Pro Gly	Ile Thr	Asn Tyr	Ile Ala	Tyr Asp	Ser Asp	Gly Phe	Met Lys	34
5G5	Arg Thr Gly	Ile Lys	Asp Tyr	Pro Ala	Ala Pro	Asn Lys	Val Phe	Asn Gln	35
7C9	Trp Pro Gly	Ile Thr	Asn Tyr	Ile Ala	Tyr Asp	Ser Asp	Gly Phe	Ile Lys	31
VHCDR3									
1B5	Phe	Asp	Gly	Pro	Asp	Tyr			36
2G9	Glu Pro	Phe Pro	Ala Phe	Asp Asp	Asp Tyr	Tyr	Pro	Ile	37
2H9	Ile Ile	Tyr Asp	Tyr Val	Asp	Tyr	Asp	Gly	Asp	38
3B6	Phe	Asp	Gly	Pro	Asp	Tyr			36
4D12	Phe	Asp	Gly	Pro	Asp	Tyr			36

TABLE 2-continued

AMINO ACID SEQUENCES OF THE CDRs									
								SEQ ID NO.	
5G5	Ile	Phe	Tyr	Asp	Tyr	Asp	Gly	Asp	146
	Ile	Asp	Val						
7C9	Phe	Asp	Gly	Pro	Asp	Tyr			36
mAbs: 4D12, 3B6, 7C9, and 1B5:				CDR: H1-1; CDR2: H2-1; CDR3: H3-6					
mAb: 5G5:				CDR: H1-1; CDR2: H2-1; CDR3: H3-11					
mAb: 2H9				CDR: H1-1; CDR2: H2-2A; CDR3: H3-11					
mAb: 2G9				CDR: H1-1; CDR2: H2-2A; CDR3: H3-13					
mAbs: 4D12, 3B6, 7C9, 1B5:				CDRL: L1-3; CDR2: L2-1; CDR3: L3-1					
mAb: 5G5:				CDR: L1-2A; CDR2: L2-1; CDR3: L3-1					
mAb: 2H9				CDR: L1-2A; CDR2: L2-1; CDR3: L3-1					
mAb: 2G9				CDR: L1-4; CDR2: L2-1; CDR3: L3-1					

Example 3

Caninization and Characterization of Caninized Antibodies

[0201] In order to produce caninized antibodies it was necessary to identify the DNA sequence encoding the heavy and light chains of canine IgG. The nucleotide and amino acid sequences of the canine heavy chain can be obtained from the NCBI gene and protein databases. There are four known IgG subclasses of canine IgG: IgGA, IgGB, IgGC, and IgGD and two types of light chains: kappa and lambda. Table 7 lists the amino acid and nucleotide SEQ ID NOs of the unmodified canine Fc fragments.

[0202] Without being bound by any specific approach, the process of producing variants of anti-PD-1 monoclonal antibodies with various contents of canine and mouse sequences involved the general following scheme:

[0203] i) Determine the nucleotide sequence of VH and VL chains of mouse mAbs;

[0204] ii) Identify the H and L chain CDRs of mouse mAbs;

[0205] iii) Identify a suitable H and L chain of canine IgG;

[0206] iv) Determine the nucleotide sequence of canine IgG H and L chains;

[0207] v) Replace the nucleotide sequence encoding endogenous canine H and L chain CDRs with nucleotide sequences encoding the respective mouse CDRs. Also, optionally replace some canine framework residues with selected residues from the mouse framework regions;

[0208] vi) Synthesize the nucleotide from step (v) and insert it into a suitable expression plasmid; Transfect plasmids into appropriate cells, e.g., HEK 293 cells;

[0209] vii) Purify the expressed antibody from HEK 293 supernatant; and

[0210] viii) Test purified antibody for binding to canine PD-1.

[0211] A set of experiments was conducted following the above steps which resulted in a set of variant caninized antibodies with various contents of canine and mouse sequences.

[0212] Reactivity of caninized mAbs against canine PD-1 proteins expressed on CHO cells: The reactivity of caninized anti-canine PD-1 mAbs with canine PD-1 on CHO cells was determined by a cell-based assay using CHO cells that express canine PD-1. Briefly, the CHO cells expressing canine PD-1 were cultured to 80-100% confluency in 50 μ l media (DMEM/HAM's F12, 10% FBS). Next, 50 μ l of media containing various concentrations of purified mAbs were added for 1 hour at 37° C. Following three washes with PBS-Tween, 100 μ l of goat anti-dog horse raddish peroxidase (HRP)-labelled antibody diluted 1:1000 in culture media was added for one hour at 37° C. After three additional washes with PBS-Tween, bound mAbs were visualized with a peroxidase substrate (TMB). The absorbance increase due to peroxidase activity at 450 nm was measured in a microplate reader.

Binding Studies of Mouse Anti-Canine PD-1 mAbs and Caninized Mouse Anti-Canine PD-1 mAbs with Canine PD-1

[0213] Approximately 70 resonance units (RU) of the canine PD-1 antigen was immobilized directly by amine coupling. Affinity measurements were made via label-free surface plasmon resonance based technology (e.g., Biacore® T200) with an association time of 300 seconds, a dissociation time of 1200 seconds, and at concentrations of 50, 100, 200 ($\times$ 2) 400, and 800 nanomolar (nM). A fitting model of 1:1 binding was used. The antigen (canine PD-1) was immobilized on the sensor chip through amine coupling and the four antibodies as indicated in Table 14 below, were used as analytes that flowed through the antigen surface. The results demonstrated that the binding affinities of the anti-canine PD-1 antibodies of the present invention for the canine PD-1 antigen were strong, having nanomolar and even subnanomolar dissociation constants (Kd). Moreover, the mouse anti-canine PD-1 monoclonal antibody and the corresponding caninized mouse anti-canine PD-1 monoclonal antibody from the same clone yielded strikingly similar Kd values (see Table 14 below).

TABLE 14

Binding Constant Determinations					
Antibody	k_{on} (k^1) $M^{-1}s^{-1}$	k_{off} (k^{-1}) s^{-1}	Kd M	Chi ² (RU ²)	Rmax (RU)
Murine 2H9	2.3×10^{-4}	$\leq 5 \times 10^{-6\#}$	$\leq 2.0 \times 10^{-10\#}$	0.19	25.6
Caninized 2H9	1.0×10^{-4}	5.9×10^{-6}	5.9×10^{-10}	0.10	27.7
Murine 3B6	1.8×10^{-4}	3.4×10^{-5}	2.0×10^{-9}	0.13	48.7
Caninized 3B6	1.6×10^{-4}	4.7×10^{-5}	2.9×10^{-9}	0.07	49.9

[#]The off-rate was so slow that it was below the detection limit of the instrument used.

Ligand Blockade by Caninized Anti-Canine PD1 mAbs:

[0214] For caninized antibodies which react with canine PD-1 (cPD-1), a cell-based ELISA (CELISA) assay was used which is based on the CHO cell line expressing canine PD-1. Briefly, cPD-1 CHO cells were placed in 96-well plates at 4×10^4 cells per well and the cells were incubated at 37° C. for 18-24 hours till they are 95-100% confluent. The cell culture media was aspirated off, the plates were washed 3 $\times$ with PBS plus 0.05% Tween® 20 and 1 $\times$ CHO media. 3-fold serial dilutions of caninized anti-cPD1 mAbs were

made in CHO media, starting at 30 μ g/mL, and 50 μ L/well of each antibody dilution was added down the plate. The plates were then incubated at 37° C., 5% CO₂ for 30 min, with shaking. Human PD-L1-Fc was added to 4 μ g/ml in CHO media, 50 μ L/well without removing or washing the incubated anti-PD1 mAbs, then incubated at 37° C., 5% CO₂ for 45 min, with shaking. The plates were washed with 6 $\times$ with PBS plus 0.05% Tween® 20. 100 μ l/well of anti-human Fc-HRP (Calbiochem) (1:2500) in CHO media was added and incubated at 37° C./5% CO₂ for 30-60 min. (anti-human Fc-HRP does not bind canine Fc.) The plates were washed with 5 $\times$ PBS plus 0.05% Tween® 20. 100 μ l/well TMB microwell substrate was added and then incubated at room temp for 10 minutes. The reaction was stopped with 100 μ l/well 1.5 M phosphoric acid. Measure A450-A620 on the ELISA reader.

Cytokine Release from Dog PBMC:

[0215] PBMC were prepared from EDTA blood samples obtained from healthy dogs and dogs with cancer, using Ficoll separation. Cells were washed 3 times, and resuspended in complete tissue culture medium at a concentration of 2.5×10^5 cells per well in triplicate wells in 96-well plates. Cells were activated with concanavalin A at 1 μ g/ml. Test antibodies were added at various concentrations and the cultures were incubated for 96 hours. Controls included cells incubated with conA and no antibody, or conA and irrelevant isotype-matched antibodies. After 96 hours in culture, supernatants were collected and assayed for IFN- γ release, using a commercial canine IFN- γ ELISA kit (R & D Systems) [see, FIG. 4].

Example 4

Genetically Modified Canine Iggs

[0216] In order to generate variants of canine IgG that lack effector functions, a number of mutant canine IgGB heavy chains were generated. These variants may include one of the following single or combined substitutions in the Fc portion of the heavy chain amino acid sequence: P4A, D31A, N63A, G64P, T65A, A93G, and P95A. Variant heavy chains (i.e., containing such amino acid substitutions) were cloned into expression plasmids and transfected into HEK 293 cells along with a plasmid containing the gene encoding a light chain. Intact antibodies expressed and purified from HEK 293 cells were evaluated for binding to Fc γ RI and C1q to assess their potential for mediation of immune effector functions. Table 3 lists examples of the plasmids encoding the genetically modified caninized heavy chains, the caninized heavy chains; and the genetic modifications in these heavy chains. The variant heavy chains were used for assessment of effector function in the genetically modified mAbs. All of the heavy chains comprised the CDRs from the 2H9 murine anti-canine PD-1 antibody.

TABLE 3

Plasmid	Heavy chain	Modification	AA position in native Fc
YZZ1057/ Mut-1	can2H9VH4	D31 to A	D31
YZZ1058/ Mut-2	can2H9VH4	N63 to A	N63
YZZ1062	can2H9VH4	D31 to A + N63 to A	D31 and N63
YZZ1059	can2H9VH4	P4 to A	P4

TABLE 3-continued

Plasmid	Heavy chain	Modification	AA position in native Fc
YZZ1060	can2H9VH4	A93 to G	A93
YZZ1061	can2H9VH4	P95 to A	P95
YZZ1068	can2H9VH4	D31 to A, N63 to A, P4 to A, A93 to G, and P95 to A	D31, N63, P4, A93, P95

Fc_γRI Binding:

[0217] Binding to Fc_γRI is a measure of the ability of an antibody to mediate ADCC. In order to assess this property for the caninized antibodies an assay to measure binding of caninized antibodies to Fc_γRI was conducted as follows: Coat 96-well plates with 100 μl per well of 2.5 μg/mL PD-1 HIS. Incubate at 2-7° C. overnight. Equilibrate the plates to room temp for 15 minutes. Wash plates 3× with phosphate buffered saline containing 0.05% Tween® 20 (PBST) and then block the wells using 200 μL/well of 5% NFDm (Non Fat Dried Milk). Incubate for 60 minutes at 36-38° C. Wash 3× with PBST. Make 2-fold dilution of antibodies starting at 1 μg/mL in 5% NFDm. Add 100 μL/well of diluted antibodies. Incubate for 60 minutes at 36-38° C. Wash 6× with PBST. Add 100 μL/well of recombinant human CD64 protein (R&D systems) diluted to 1 μg/mL. Incubate for 60 minutes at 36-38° C. Wash 6× with PBST. Add 100 μL/well biotinylated-anti-CD64 antibody (R&D systems) diluted to 1:3000. Incubate for 60 minutes at 36-38° C. Wash 6× with PBST. Add 100 μL/well Streptavidin-HRP antibody (R&D systems) diluted to 1:7500. Incubate for 60 minutes at 36-38° C. Wash 6× with PBST. Add 100 μL/well TMB substrate. Incubate for 10 minutes at 15-30° C. Read plates using ELISA plate reader at 450-540 nm.

[0218] Results: FIG. 5A shows that caninized mAb designated can2H9 ADCC mut-1 VH4/VL3 which has the genetic modification of D31A or the mAb designated can2H9 ADCC mut-2 VH4/VL3 which has the genetic modification of N63A display a near complete reduction in binding to Fc_γRI. On the other hand, the mAb designated can2H9 ADCC (1062) VH4/VL3 which contains the combined D31A plus N63A genetic modifications lacks detectable binding to Fc_γRI. In FIG. 5A, can2H9 IgGD VH4/VL3 is a caninized antibody which contains the Fc from canine IgGD and can3B6 VH4/VL4 IgGB is a caninized antibody that does not bind to the coating antigen (PD-1 HIS), and caninized mAb designated can2H9 VH4/VL3 is an antibody that contains un-modified IgGB Fc. FIG. 5B shows that the caninized mAb designated can2H9 ADCC (1059) VH4/VL3 which contains the genetic modification of P4A and the mAb designated can2H9 ADCC (1061) VH4/VL3 which contain the genetic modification of P95A display considerable reduction in binding to Fc_γRI, whereas the mAb designated can2H9 ADCC (1060) VH4/VL3 which contains the genetic modification of A93G displays a slight reduction in binding to Fc_γRI. On the other hand, the mAb designated can2H9 IgGB ADCC (1068) VH4/VL3 which contains five genetic modifications (D31A, N63A, P4A, A93G, P95A) is completely lacking in binding to Fc_γRI.

C1q Binding:

[0219] Binding to the first component of complement, C1q, is a measure of the ability of an antibody to mediate

CDC. In order to assess this property for the caninized antibodies an assay to measure binding of caninized antibodies to C1q was conducted as follows: Coat 96-well plates with 2.5 μg/mL PD-1 HIS. Incubate at 2-7° C. overnight. Equilibrate to room temperature for 15 minutes. Wash with PBST 3×. Block with 200 μL/well with 5% BSA. Incubate for 60 minutes at 36-38° C. Wash with PBST 3×. Make a 2-fold dilution of antibodies starting at 1 μg/mL in 5% BSA. Add 100 μL/well diluted antibodies. Incubate for 60 minutes at 36-38° C. Wash with PBST 6×. Add 100 μL/well C1q protein diluted to 4 μg/mL. Incubate for 60 minutes at 36-38° C. Wash with PBST 6×. Add 100 μL/well Goat-Anti-C1q antibody diluted to 1:3000. Incubate for 60 minutes at 36-38° C. Wash with PBST 6×. Add 100 μL/well of donkey-anti-Goat-HRP antibody diluted 1:10000. Incubate for 60 minutes at 36-38° C. Wash with PBST 6×. Add 100 μL/well TMB substrate. Incubate for 10 minutes at 15-30° C. Read on ELISA plate reader at 450-540 nm.

[0220] Results: FIG. 6A shows that the caninized mAb designated can2H9VH4 IgGB ADCC (mut-1)/VL3, which has the genetic modification of D31A or the mAb designated can2H9 VH4 IgGB ADCC (mut-2)/VL3 which has the genetic modification N63A display considerable reduction in binding to C1q. On the other hand, the mAb designated can2H9 VH4 IgGB ADCC (1062)/VL3 which contains the combined D31A plus N63A genetic modifications lacks detectable binding to C1q.

[0221] In FIG. 6A, can2H9 VH4 IgGD/VL3 is a caninized antibody which contains the Fc from canine IgGD and can3B6 VH4/VL4 IgGB is a caninized antibody that does not bind to the coating antigen (PD-1 HIS), and caninized mAb designated can2H9 VH4/VL3 IgGB is an antibody that contains un-mutated IgGB Fc. FIG. 6B shows that the caninized mAb designated can2H9 VH4 IgGB ADCC (1059)/VL3 which contains the substitution P4A and the mAb designated can2H9 VH4 IgGB ADCC (1061)/VL3 which contains the substitution P95A display considerable reduction in binding to C1q, whereas the mAb designated can2H9 VH4 IgGB ADCC (1060)/VL3 which contains the substitution A93G display an enhancement in binding to C1q. On the other hand, the mAb designated can2H9 VH4 IgGB ADCC (1068)/VL3 which contains five substitutions (D31A, N63A, P4A, A93G, P95A), is completely lacking in binding to C1q. In FIG. 6B, the mAb designated can3B6 VH4/VL4 IgGB is a caninized antibody that does not bind to the coating antigen (PD-1 HIS), and caninized mAb designated can2H9 VH4/VL3 IgGB is an antibody that contains un-mutated IgGB Fc.

TABLE 4

MODIFIED cFc or NATIVE cFc WITH HINGE SEQUENCES			
#	N.	A.	Modified Fs
1*	✓		Modified Fc-clgGB
2*		✓	Modified Fc-clgGB
3*	✓		Modified Fc-clgGC
4*		✓	Modified Fc-clgGC
5 [#]	✓		cIgGD Fc with S of cIgGD hinge to P
6 [#]		✓	cIgGD Fc with S of cIgGD hinge to P
7	✓		cIgGD Fc with A hinge
8		✓	cIgGD Fc with A hinge
9	✓		cIgGD Fc with B hinge
10		✓	cIgGD Fc with B hinge

TABLE 4-continued

MODIFIED cFc or NATIVE cFc WITH HINGE SEQUENCES			
#	N.	A.	Modified Fs
11	✓		cIgGD Fc with C hinge
12		✓	cIgGD Fc with C hinge

*The substitutions are at P4, D31, N63, G64, T65, A93, and P95 of amino acid sequences SEQ ID NOs: 2 and 4; or at the nucleotides that encode those amino acids for nucleotide sequences SEQ ID NOs: 1 and 3.

^aSingle amino acid substitution as shown in Table 5 below in hinge region of IgGD.

TABLE 5

HINGE REGION SEQUENCES			
#	A.A.	Hinge	Sequence
109	✓	IgGA	FNECRCTDTPPCPVPEP
110	✓	IgGB	PKRENGRVPRPPDCPKCPAPEM
111	✓	IgGC	AKECECKCNCNNPCPGCGL
112	✓	IgGD ^a	PKESTCKCI <u>P</u> PCVPES

^aSingle amino acid substitution of a serine to a proline as in bold and underlined.

TABLE 6

CANINE PD-1/PD-L1 SEQUENCES									
#	N.	A.	PD-1	Description	#	N.	A.	PD-L1	Description
113	✓		✓	Full Length	119	✓		✓	Full Length
114		✓	✓	Full Length	120		✓	✓	Full Length
115	✓		✓	ECD	121	✓		✓	ECD
116		✓	✓	ECD	122		✓	✓	ECD
117	✓		✓	cECD-hIgG1	123	✓		✓	cECD-hIgG1
118		✓	✓	cECD-hIgG1	124		✓	✓	cECD-hIgG1
133	✓		✓	+signal seq.	135	✓		✓	+signal seq.
134		✓	✓	+signal seq.	136		✓	✓	+signal seq.
137		✓	✓	+signal seq.					

TABLE 7

NATIVE cFc SEQUENCES							
#	N.	A.		#	N.	A.	
125	✓		Fc-cIgGA	129	✓	Fc-cIgGB	
126		✓	Fc-clgGA	130		✓	Fc-clgGB
127	✓		Fc-clgGD	131	✓		Fc-clgGC
128		✓	Fc-cIgGD	132		✓	Fc-clgGC

TABLE 8

CDR AMINO ACID SEQUENCES		
#	A.A.	CDR
13	✓	VL CDR1 1B5, 3B6, 4D12, 7C9
14	✓	VL CDR1 2G9
15	✓	VL CDR1 2H9, 5G5
16	✓	VL CDR2 1B5, 7C9
17	✓	VL CDR2 2G9
18	✓	VL CDR2 2H9
19	✓	VL CDR2 3B6
20	✓	VL CDR2 4D12
21	✓	VL CDR2 5G5

TABLE 8-continued

CDR AMINO ACID SEQUENCES		
#	A.A.	CDR
22	✓	VL CDR3 1B5, 7C9
23	✓	VL CDR3 2G9
24	✓	VL CDR3 2H9
25	✓	VL CDR3 4D12
26	✓	VL CDR3 5G5
27	✓	VH CDR1 1B5, 3B6, 4D12
28	✓	VH CDR1 2G9
29	✓	VH CDR1 2H9, 5G5
30	✓	VH CDR1 7C9
31	✓	VH CDR2 1B5, 3B6, 7C9
32	✓	VH CDR2 2G9
33	✓	VH CDR2 2H9
34	✓	VH CDR2 4D12
35	✓	VH CDR2 5G5
36	✓	VH CDR3 1B5, 3B6, 4D12, 7C9
37	✓	VH CDR3 2G9
38	✓	VH CDR3 2H9
146	✓	VH CDR3 5G5

TABLE 9

INDIVIDUAL SUBSTITUTED CANINIZED HEAVY CHAINS			
#	N.	A.	
39	✓		3B6-VH3-CH1-hinge-FC-cIgGB Fc
40		✓	3B6-VH3-CH1-hinge-FC-cIgGB Fc
41	✓		3B6-VH3-CH1-hinge-FC-cIgGC Fc
42		✓	3B6-VH3-CH1-hinge-FC-cIgGC Fc
43	✓		2H9-VH4-CH1-hinge-FC-cIgGB Fc
44		✓	2H9-VH4-CH1-hinge-FC-cIgGB Fc
45	✓		2H9-VH4-CH1-hinge-FC-cIgGC Fc
46		✓	2H9-VH4-CH1-hinge-FC-cIgGC Fc
47	✓		2G9-VH6-CH1-hinge-FC-cIgGB Fc
48		✓	2G9-VH6-CH1-hinge-FC-cIgGB Fc
49	✓		2G9-VH6-CH1-hinge-FC-cIgGC Fc
50		✓	2G9-VH6-CH1-hinge-FC-cIgGC Fc
51	✓		7C9-VH3-CH1-hinge-FC-cIgGB Fc
52		✓	7C9-VH3-CH1-hinge-FC-cIgGB Fc
53	✓		7C9-VH3-CH1-hinge-FC-cIgGC Fc
54		✓	7C9-VH3-CH1-hinge-FC-cIgGC Fc
55	✓		1B5-VH3-CH1-hinge-FC-cIgGB Fc
56		✓	1B5-VH3-CH1-hinge-FC-cIgGB Fc
57	✓		1B5-VH3-CH1-hinge-FC-cIgGC Fc
58		✓	1B5-VH3-CH1-hinge-FC-cIgGC Fc
59	✓		5G5-VH3-CH1-hinge-FC-cIgGB Fc
60		✓	5G5-VH3-CH1-hinge-FC-cIgGB Fc
61	✓		5G5-VH3-CH1-hinge-FC-cIgGC Fc
62		✓	5G5-VH3-CH1-hinge-FC-cIgGC Fc
63	✓		4D12-VH3-CH1-hinge-FC-cIgGB Fc
64		✓	4D12-VH3-CH1-hinge-FC-cIgGB Fc
65	✓		4D12-VH3-CH1-hinge-FC-cIgGC Fc
66		✓	4D12-VH3-CH1-hinge-FC-cIgGC Fc

The potential specific substitutions are at P4, D31, N63, G64, T65, A93, and P95

TABLE 10

CORRELATION OF AMINO ACID RESIDUE POSITIONS OF NATIVE AND SUBSTITUTED cFc WITH THAT OF THE CORRESPONDING SUBSTITUTED CANINE HEAVY CHAINS [#]							
130/132	P4	D31	N63	G64	T65	A93	P95
2/4	4	31	63	64	65	93	95
40	239	266	298	299	300	328	330
42	237	264	296	297	298	326	328
44	244	271	303	304	305	333	335
46	242	269	301	302	303	331	333

TABLE 10-continued

CORRELATION OF AMINO ACID RESIDUE POSITIONS OF NATIVE AND SUBSTITUTED cFc WITH THAT OF THE CORRESPONDING SUBSTITUTED CANINE HEAVY CHAINS [#]							
130/132	P4	D31	N63	G64	T65	A93	P95
48	246	273	305	306	307	335	337
50	244	271	303	304	305	333	335
52	239	266	298	299	300	328	330
54	237	264	296	297	298	326	328
56	239	266	298	299	300	328	330
58	237	264	296	297	298	326	328
60	244	271	303	304	305	333	335
62	242	269	301	302	303	331	333
64	239	266	298	299	300	328	330
66	237	264	296	297	298	326	328

[#]First Column lists SEQ ID NOs.; remaining columns list corresponding amino acid positions. For the two native amino acid sequences (SEQ ID NOs. 130 and 132), the one letter code for the natural amino acid residues are also provided.

TABLE 11

INDIVIDUAL UNSUBSTITUTED CANINIZED HEAVY AND LIGHT CHAINS		
#	N.	A.
67	✓	3B6-VH3-CH1-hinge-FC-clgGA Fc
68	✓	3B6-VH3-CH1-hinge-FC-clgGA Fc
69	✓	3B6-VH3-CH1-hinge-FC-clgGD Fc
70	✓	3B6-VH3-CH1-hinge-FC-clgGD Fc
71	✓	3B6-VL3-CL-Kappa
72	✓	3B6-VL3-CL-Kappa
73	✓	2H9-VH4-CH1-hinge-FC-clgGA Fc
74	✓	2H9-VH4-CH1-hinge-FC-clgGA Fc
75	✓	2H9-VH4-CH1-hinge-FC-clgGD Fc
76	✓	2H9-VH4-CH1-hinge-FC-clgGD Fc
77	✓	2H9-VL3-CL-Kappa
78	✓	2H9-VL3-CL-Kappa
79	✓	2G9-VH6-CH1-hinge-FC-clgGA Fc
80	✓	2G9-VH6-CH1-hinge-FC-clgGA Fc
81	✓	2G9-VH6-CH1-hinge-FC-clgGD Fc
82	✓	2G9-VH6-CH1-hinge-FC-clgGD Fc
83	✓	2G9-VL3-CL-Kappa
84	✓	2G9-VL3-CL-Kappa
85	✓	7C9-VH3-CH1-hinge-FC-clgGA Fc
86	✓	7C9-VH3-CH1-hinge-FC-clgGA Fc
87	✓	7C9-VH3-CH1-hinge-FC-clgGD Fc
88	✓	7C9-VH3-CH1-hinge-FC-clgGD Fc
89	✓	7C9-VL3-CL-Kappa
90	✓	7C9-VL3-CL-Kappa
91	✓	1B5-VH3-CH1-hinge-FC-clgGA Fc
92	✓	1B5-VH3-CH1-hinge-FC-clgGA Fc
93	✓	1B5-VH3-CH1-hinge-FC-clgGD Fc
94	✓	1B5-VH3-CH1-hinge-FC-clgGD Fc
95	✓	1B5-VL3-CL-Kappa
96	✓	1B5-VL3-CL-Kappa
97	✓	5G5-VH3-CH1-hinge-FC-clgGA Fc
98	✓	5G5-VH3-CH1-hinge-FC-clgGA Fc
99	✓	5G5-VH3-CH1-hinge-FC-clgGD Fc
100	✓	5G5-VH3-CH1-hinge-FC-clgGD Fc
101	✓	SGS-VL3-CL-Kappa
102	✓	SGS-VL3-CL-Kappa
103	✓	4D12-VH3-CH1-hinge-FC-clgGA Fc
104	✓	4D12-VH3-CH1-hinge-FC-clgGA Fc
105	✓	4D12-VH3-CH1-hinge-FC-clgGD Fc
106	✓	4D12-VH3-CH1-hinge-FC-clgGD Fc
107	✓	4D12-VL3-CL-Kappa
108	✓	4D12-VL3-CL-Kappa

Example 5

Epitope Mapping of Anti-Canine Pd-1 Antibodies

Introduction

[0222] The interaction of antibodies with their cognate protein antigens is mediated through the binding of specific amino acids (paratopes) of the antibodies with specific amino acids (epitopes) of target antigens. An epitope is an antigenic determinant that causes a specific reaction by an immunoglobulin. It consists of a group of amino acids on the surface of the antigen.

[0223] A protein of interest may contain several epitopes that are recognized by different antibodies. The epitopes recognized by antibodies are classified as linear or conformational epitopes. Linear epitopes are formed by a stretch of continuous sequence of amino acids in a protein, while conformational epitopes are composed of amino acids that are discontinuous (e.g., far apart) in the primary amino acid sequence, but are brought together upon three-dimensional protein folding.

[0224] Epitope mapping refers to the process of identifying the amino acid sequences (i.e., epitopes) that are recognized by antibodies on their target antigens. Identification of epitopes recognized by monoclonal antibodies (mAbs) on target antigens has important applications. For example, it can aid in the development of new therapeutics, diagnostics, and vaccines. Epitope mapping can also aid in the selection of optimized therapeutic mAbs and help elucidate their mechanisms of action. Epitope information can also elucidate unique cancer epitopes and define the protective or pathogenic effects of vaccines.

[0225] Epitope mapping can be carried out using polyclonal or monoclonal antibodies and several methods are employed for epitope identification depending on the suspected nature of the epitope (i.e., linear versus conformational). Mapping linear epitopes is more straightforward and relatively easy to perform. For this purpose, commercial services for linear epitope mapping often employ peptide scanning. In this case, an overlapping set of short peptide sequences of the target protein are chemically synthesized and tested for their ability to bind antibodies of interest. The strategy is rapid, high-throughput, and relatively inexpensive to perform. On the other hand, mapping of discontinuous epitope is more technically challenging and requires more specialized techniques such as x-ray co-crystallography of a monoclonal antibody together with its target protein, Hydrogen-Deuterium (H/D) exchange, and/or Mass Spectroscopy coupled with enzymatic digestion.

Mapping of PD-1 Epitopes Using a ProImmune® MicroArray:

[0226] In order to identify the amino acids that form the epitopes for anti-PD1 mAbs, a total of 28 peptides that are 15 amino acids long and overlapping by 10 amino acids were chemically synthesized. This library of overlapping peptides was designed to cover the full length canine PD-1 protein. The determination of peptide-antibody binding was performed by attachment of antibody samples to the ProArray Ultra® peptide microarray, followed by incubation with a fluorescent-labelled secondary antibody. All peptides are synthesized separately, and then bound to the ProArray Ultra® slide surface alongside ProImmune® murine IgG

controls. This optimized process ensures that peptides are presented on the array in such a manner as to closely mimic the properties of the corresponding protein region, circumventing the inherent physiochemical variation of the free peptides themselves and making a compatible, combined peptide and protein array platform. The test analytes (peptides) are dispensed onto the ProArray Ultra® slide in discrete spots and appropriate gal-files enable exact alignment of the resulting array features back to the analyte deposited. ProArray Ultra® slides were blocked using a validated blocking buffer to reduce non-specific binding of the mAbs. They were then incubated with the mAb samples, followed by incubation with a specific fluorescent-labelled secondary antibody. After several washing steps, the ProArray Ultra® arrays were dried and scanned using a high-resolution fluorescence microarray scanning system. After scanning the fluorescent labelled ProArray Ultra® slides, the scanner recorded an image which was evaluated using image analysis software-enabling interpretation and quantification of the levels of fluorescent intensities associated with each fluorescent spot on the scanned microarray slide. The results of this experiment indicated some of the canine PD-1 peptides were recognized by some of the mAbs evaluated. The identity of the mAbs and the amino acid sequence recognized by these mAbs are listed in Table 12. This study indicates that mAb 2H9 recognizes an epitope located in the extracellular domain of canine PD-1 comprised of the amino acid sequence represented by SEQ ID NO: 138 and that mAb 1A1 recognizes an epitope comprising the amino acid sequence represented by SEQ ID NO: 138 and the overlapping amino acid sequence represented by the amino acid sequence represented by SEQ ID NO: 139.

Mapping of PD-1 Epitopes Using Mass Spectroscopy:

[0227] In order to identify potentially discontinuous epitopes recognized by anti-canine PD-1 a method based on chemical crosslinking and mass spectrometry detection was used (CovalX® Instrument Incorporated). The application of this technology to epitope mapping of canine PD-1 resulted in identification of at least portions of epitopes recognized by the indicated mAbs which are listed in Table 13. As can be seen from Table 13, mAb 3B6 recognizes at least a portion of an epitope located in the extracellular domain of canine PD-1 within the amino acid sequence represented by SEQ ID NO: 140 and that mAb 2G9 recognizes at least a portion of an epitope within the amino acid sequence represented by SEQ ID NO: 141. On the other hand, mAb 1E4 and mAb 1B5 recognize at least a portion of an epitope within the amino acid sequence represented by SEQ ID NO: 142 and acid sequence represented by SEQ ID NO: 143, respectively.

[0228] As depicted in FIG. 9A a determination performed by chemical cross-linking, High-Mass MALDI mass spectrometry and nLC-Orbitrap mass spectrometry shows that the epitope on canine PD-1 recognized by caninized antibody 2G9 comprises R₆₂, R₆₉, R₇₂, and R₇₅ of SEQ ID NO: 114. The analogous determination for the epitope on canine PD-1 recognized by caninized antibody 3B6 comprises R₇₅ and R₉₀ of SEQ ID NO: 114. Accordingly, R₇₅ appears to be a particularly important amino acid residue in one or more epitopes of canine PD-1. Interestingly, after performing these analyses, the amino acid sequence for the CDRs of 1A1 were found to be identical to that of 2G9. The consistency between the region on PD-1 that 2G9 binds with that

found for 1A1, which were obtained by these two very different methodologies, indicates that this region contains amino acid residues comprised by a PD-1 epitope that is recognized by the anti-canine PD-1 antibodies (see, Tables 12 and 13 below).

[0229] Moreover, the region of the amino acid sequence of PD-1 that is recognized by the blocking antibodies of the present invention tested is within the extracellular domain of canine PD-1. The region recognized is comprised by the following peptide (see, Tables 12 and 13 below).

(SEQ ID NO: 144)
NQTDKLAAFQEDRIEPGRDRRFRVM*RLPNGRDFHMSIVAARLND

[0230] Within this peptide, is a shorter peptide that is in bold. This shorter peptide was recognized with the ProImmune® MicroArray (see, Table 12).

(SEQ ID NO: 145)
DRIEPGRDRRFRVM*RLPNGR

[0231] Notably, R₆₂, R₆₉, R₇₂, and R₇₅ of SEQ ID NO: 114 are all comprised by both the longer peptide (SEQ ID NO: 144) and the shorter peptide (SEQ ID NO: 145), whereas R₉₀ of SEQ ID NO: 114 is in the longer peptide. These five arginine residues appear to be important amino acid residues in one or more epitopes of canine PD-1. As indicated in the Tables 6-8, the starred methionine residue (*) has also been reported as being a threonine residue.

TABLE 12

PD-1 EPITOPES RECOGNIZED BY ANTI-CANINE PD-1 MAABS USING PROIMMUNE ® MICROARRAY		
ANTIBODY	ANTIGEN PEPTIDE	SEQ ID NO:
2H9	GRDRRFRVM*RLPNGR	138
1A1 [#]	DRIEPGRDRRFRVM*R	139
1A1	GRDRRFRVM*RLPNGR	138

*This methionine residue has also been reported as being a threonine residue.
[#]The CDRs of 1A1 are identical to those of 2G9.

TABLE 13

PD-1 EPITOPES RECOGNIZED BY ANTI-CANINE PD-1 MAABS USING MASS SPECTROMETRY		
ANTIBODY	PEPTIDE ANTIGEN	SEQ ID NO:
3B6	RFRVM*RLPNGRDFHMSIVAARLND	140
2G9	LAAFQEDRIEPGRDRRFRVM*RLPNGR	141
1E4 [#]	EDRIEPGRDRRFRVM*RLPNGRDFHMSIVAAR	142
1B5	NQTDKLAAFQEDRIEPGRDRRFRVM*RLPNGR	143

*This methionine residue has also been reported as being a threonine residue.
[#]The CDRs of 1E4 are most closely related to those of 2G9.

[0232] All references cited herein are incorporated by reference to the same extent as if each individual publication, database entry (e.g., Genbank sequences or GeneID entries), patent application, or patent, was specifically and individually indicated to be incorporated by reference. This

-continued

SEQUENCE: 2

LGGXSVFIFF	PKPKDTLLIA	RTPEVTCVVV	XLDPEDPEVQ	ISWFDGKQM	QTAKTQPRE	60
QFXXXYRVVS	VLPIGHQDWL	KGKQFTCKVN	NKXLXSPIER	TISKARGQAH	QPSVYVLP	120
REELSKNTVS	LTCLIKDFFP	PDIDVEWQSN	GQQEPESKYR	TTPPQLDEDG	SYFLYSKLSV	180
DKSRWQRGDT	FICAVMHEAL	HNHYTQESLS	HSPGK			215

SEQ ID NO: 3 moltype = DNA length = 690

FEATURE Location/Qualifiers

misc_feature 1..690

 note = modified canine sequence

misc_difference 10..12

 note = n is a, c, g, or t

misc_difference 91..93

 note = n is a, c, g, or t

misc_difference 187..195

 note = n is a, c, g, or t

misc_difference 277..279

 note = n is a, c, g, or t

misc_difference 283..285

 note = n is a, c, g, or t

source 1..690

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 3

ctgggcggcn	nnagcgtgtt	tatttttccg	cgaaccga	aagatattct	ggtgaccg	60
cgcaccccca	ccgtgacctg	cgtggtggtg	nnctggatc	cggaaaaccc	ggaagtgcag	120
attagctggt	ttgtggatag	caaacagggtg	cagaccgcca	acaccagcc	gcgcgaagaa	180
cagagcnnnn	nnnnntatcg	cgtggtgagc	gtgctgccga	ttggccatca	ggattggctg	240
agcggcaaac	agtttaaatg	caaagtgaac	aacaaannnc	tgnnnagccc	gattgaagaa	300
attattagca	aaaccccggtg	ccaggcgcat	cagccgaacg	tgtatgtgct	gccgccgagc	360
cgcgatgaaa	tgagcaaaaa	caccgtgacc	ctgacctgcc	tggtgaaaga	tttttttccg	420
ccggaaattg	atgtggaatg	gcagagcaac	ggccagcagg	aaccggaaag	caaatatcgc	480
atgacccccg	cgcagctgga	tgaagatggc	agctattttc	tgtatagcaa	actgagcgtg	540
gataaaagcc	gctggcagcg	cggcgatacc	tttatttgcg	cggtgatgca	tgaagcgtg	600
cataaccatt	ataccagat	tagcctgagc	catagcccg	gcaaacataa	ccattatacc	660
cagattagcc	tgagccatag	cccgggcaaa				690

SEQ ID NO: 4 moltype = AA length = 215

FEATURE Location/Qualifiers

REGION 1..215

 note = modified canine sequence

VARIANT 4

 note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 31

 note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 63..65

 note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 93

 note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 95

 note = misc_feature - Xaa can be any naturally occurring amino acid

source 1..215

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 4

LGGXSVFIFF	PKPKDILVTA	RTPTVTCVVV	XLDPENPEVQ	ISWFDGSKQV	QTANTQPRE	60
QXXXXYRVVS	VLPIGHQDWL	SGKQFKCKVN	NKXLXSPIEE	IISKTPGQAH	QPNVYVLP	120
RDEMSKNTVT	LTCLVKDFFP	PEIDVEWQSN	GQQEPESKYR	MTPPQLDEDG	SYFLYSKLSV	180
DKSRWQRGDT	FICAVMHEAL	HNHYTQISLS	HSPGK			215

SEQ ID NO: 5 moltype = DNA length = 699

FEATURE Location/Qualifiers

misc_feature 1..699

 note = modified canine sequence

source 1..699

 mol_type = other DNA

 organism = synthetic construct

SEQUENCE: 5

ccgaaagaaa	gcacctgcaa	atgcattccg	ccgtgcccgg	tgccgaaag	cctgggccc	60
ccgagcgtgt	ttatttttcc	gccgaaaccg	aaagatatcc	tgccgattac	ccgcacccc	120
gaaattacct	gcgtggtgct	ggatctgggc	cgcggaagtc	cggaagtgca	gattagctgg	180

-continued

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tttgtggatg gcaaagaagt gcataccgcg aaaaccacgc cgcgcgaaca gcagtttaac 240
agcacctatc gcgtgggtgag cgtgctgccc attgaacatc aggattggct gaccggcaaa 300
gaatttaaat gcccgctgaa ccatattggc ctgccgagcc cgattgaacg caccattagc 360
aaagcgcgcg gccaggcgca tcagccgagc gtgtatgtgc tgccgcgagc cccgaaagaa 420
ctgagcagca gcgataccgt gaccctgacc tgccctgatta aagatttttt tccgcccggaa 480
attgatgtgg aatggcagag caacggccag ccggaaccgg aaagcaaata tcataccacc 540
gcgcgcgagc tggatgaaga tggcagctat tttctgtata gcaaactgag cgtggataaa 600
agccgctggc agcagggcga tacctttacc tgcgcggtga tgcatagaagc gctgcagaaac 660
cattataccg atctgagcct gagccatagc ccgggcaaaa 699

```

```

SEQ ID NO: 6          moltype = AA  length = 233
FEATURE              Location/Qualifiers
REGION              1..233
                    note = modified canine sequence
source              1..233
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 6
PKESTCKCIP PCVPESLGG PSVFIFPPKP KDILRITRTP EITCVVLDLG REDPEVQISW 60
FVDGKEVHTA KQPREQQFN STYRVSVSLP IEHQDWLTGK EFKCRVNHIG LPSPIERTIS 120
KARGQAHQPS VYVLPSPFKE LSSSDTVTLT CLIKDFFPPE IDVEWQSNQG PEPESKYHTT 180
APQLDEDEGSY FLYSKLSVDK SRWQQGDTFT CAVMHEALQN HYTDLSSLHS PGK 233

```

```

SEQ ID NO: 7          moltype = DNA  length = 699
FEATURE              Location/Qualifiers
misc_feature         1..699
                    note = modified canine sequence
source              1..699
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 7
tttaacgaat gccgctgcac cgataccccc cgtgcccggc tgccggaacc gctgggcccgc 60
ccgagcgtgt ttatttttcc gccgaaccgc aaagatatcc tgcgcattac ccgcaccccgc 120
gaaattacct gcgtggtgct ggaatctgggc cgcgaagatc cggaaagtga gattagctgg 180
tttgtggatg gcaaagaagt gcataccgcg aaaaccacgc cgcgcgaaca gcagtttaac 240
agcacctatc gcgtgggtgag cgtgctgccc attgaacatc aggattggct gaccggcaaa 300
gaatttaaat gcccgctgaa ccatattggc ctgccgagcc cgattgaacg caccattagc 360
aaagcgcgcg gccaggcgca tcagccgagc gtgtatgtgc tgccgcgagc cccgaaagaa 420
ctgagcagca gcgataccgt gaccctgacc tgccctgatta aagatttttt tccgcccggaa 480
attgatgtgg aatggcagag caacggccag ccggaaccgg aaagcaaata tcataccacc 540
gcgcgcgagc tggatgaaga tggcagctat tttctgtata gcaaactgag cgtggataaa 600
agccgctggc agcagggcga tacctttacc tgcgcggtga tgcatagaagc gctgcagaaac 660
cattataccg atctgagcct gagccatagc ccgggcaaaa 699

```

```

SEQ ID NO: 8          moltype = AA  length = 233
FEATURE              Location/Qualifiers
REGION              1..233
                    note = modified canine sequence
source              1..233
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 8
FNECRCTDTP PCVPEPLGG PSVFIFPPKP KDILRITRTP EITCVVLDLG REDPEVQISW 60
FVDGKEVHTA KQPREQQFN STYRVSVSLP IEHQDWLTGK EFKCRVNHIG LPSPIERTIS 120
KARGQAHQPS VYVLPSPFKE LSSSDTVTLT CLIKDFFPPE IDVEWQSNQG PEPESKYHTT 180
APQLDEDEGSY FLYSKLSVDK SRWQQGDTFT CAVMHEALQN HYTDLSSLHS PGK 233

```

```

SEQ ID NO: 9          moltype = DNA  length = 714
FEATURE              Location/Qualifiers
misc_feature         1..714
                    note = modified canine sequence
source              1..714
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 9
ccgaaacgcg aaaacggcgc cgtgccgcgc ccgcccggatt gcccgaaatg cccggcgccgc 60
gaaatgctgg gcggcccgcg cgtgtttatt tttccgcga aaccgaaaga tattctgcgc 120
attacccgca ccccggaatt tacctgcgtg gtgctggatc tgggcccgcga agatccggaa 180
gtgcagatta gctggtttgt ggaaggcaaa gaagtgcata ccgcgaaaac ccagccgcgc 240
gaacagcagt ttaacagcac ctatcgctg gtgagcgtgc tgccgattga acatcaggat 300
tggttgaccg gcaagaattt taaatgccgc gtgaaccata ttggcctgcc gagcccgcatt 360
gaacgcacca ttagcaaagc gcgcggccag gcgcatcagc cgagcgtgta tgtgctgccg 420
ccgagcccga aagaactgag cagcagcgat accgtgacct tgacctgect gattaagaat 480
ttttttccgc cggaaattga tgtggaatgg cagagcaacg gccagccgga accggaaagc 540
aaatatcata ccaccgcgcg gcagctggat gaagatggca gctattttct gtatagcaaa 600
ctgagcgtgg ataaaagccg ctggcagcag ggcgatacct ttacctgcgc ggtgatgcat 660

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gaagcgctgc agaaccatta taccgatctg agcctgagcc atagcccggg caaa 714

SEQ ID NO: 10 moltype = AA length = 238
FEATURE Location/Qualifiers
REGION 1..238
 note = modified canine sequence
source 1..238
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 10
PKRENGRVPR PPDCKPCPAP EMLGGPSVFI FPPKPKDILR ITRTPEITCV VLDLGREDPE 60
VQISWFVDGK EVHTAKTQPR EQQFNSTYRV VSVLPIEHQD WLTGKEPKCR VNHIGLPSPI 120
ERTISKARGQ AHQPSVYVLP PSPKELSSSD TVTLTCLIKD FFPPEIDVEW QSNQPEPES 180
KYHTTAPQLD EDGSYFLYSK LSVDKSRWQQ GDTFTCAVMH EALQNHYTDL SLSHSPGK 238

SEQ ID NO: 11 moltype = DNA length = 708
FEATURE Location/Qualifiers
misc_feature 1..708
 note = modified canine sequence
source 1..708
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 11
gccaaggagt gcgagtgcaa gtgcaactgc aacaactgcc cctgccccgg ctgcggcctg 60
ctgggcgggc ccagcgtgtt catcttcccc cccaagccca aggacatcct gagaatcacc 120
agaacccccg agatcacctg cgtggtgctg gacctgggca gagaggaccc cgagggtgcag 180
atcagctggt tcgtggacgg caaggaggtg cacaccgcca agaccagacc cagagagcag 240
cagttcaaca gcacctacag agtgggtgagc gtgctgcccc tcgagcacca ggactggctg 300
accggcaagg agttcaagtg cagagtgaac cacatcggcc tgcccagccc catcgagaga 360
accatcagca aggccagagg ccaggccccc cagcccagcg tgtacgtgct gccccccagc 420
cccaaggagc tgagcagcag gcacaccgtg accctgacct gcctgatcaa ggactttctt 480
ccccccgaga tcgacgtgga gtggcagagc aacggccagc ccgagcccca gagcaagtac 540
cacaccaccg cccccagct ggacgaggac ggcagctact tcctgtacag caagctgagc 600
ctggacaaga gcagatggga gcaggggcag accttcacct gcgcgctgat gcacgaggcc 660
ctgcagaacc actacaccga cctgagcctg agccacagcc ccggcaga 708

SEQ ID NO: 12 moltype = AA length = 236
FEATURE Location/Qualifiers
REGION 1..236
 note = modified canine sequence
source 1..236
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 12
AKECECKCNC NNPCPGCGL LGGPSVFIFP PKPKDILRIT RTPEITCVVL DLGREDPEVQ 60
ISWFDGKEV HTAKTQPREQ QFNSTYRVVS VLPPIEHQDWL TGKEFKCRVN HIGLPSPIER 120
TISKARGQAH QPSVYVLPSP PKELSSSDTV TLTCLIKDFP PPEIDVEWQS NGQPEPESKY 180
HTTAPQLDED GSYFLYSKLS VDKSRWQQGD TFTCAVMHEA LQNHYTDLSL SHSPGK 236

SEQ ID NO: 13 moltype = AA length = 17
FEATURE Location/Qualifiers
source 1..17
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 13
KSSQSLNLSV NQKNYLA 17

SEQ ID NO: 14 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 14
RSSQNIVHSN GNTYLE 16

SEQ ID NO: 15 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 15
HASQNINVWL S 11

SEQ ID NO: 16 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7

-continued

SEQUENCE: 16 FASTRVS	mol_type = protein organism = Mus musculus	7
SEQ ID NO: 17 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 17 KVSNRFS		7
SEQ ID NO: 18 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 18 KASHLHT		7
SEQ ID NO: 19 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 19 FASARVS		7
SEQ ID NO: 20 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 20 FASTRIS		7
SEQ ID NO: 21 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Mus musculus	
SEQUENCE: 21 KASNLHT		7
SEQ ID NO: 22 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 22 QQYFSTPLT		9
SEQ ID NO: 23 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 23 FQGSHPYPT		9
SEQ ID NO: 24 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 24 QQGQSWPLT		9
SEQ ID NO: 25 FEATURE source	moltype = AA length = 9 Location/Qualifiers 1..9 mol_type = protein organism = Mus musculus	
SEQUENCE: 25 QQYFSTPLT		9

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SEQ ID NO: 26	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 26		
QQGQSYPLT		9
SEQ ID NO: 27	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 27		
GYFTTYGMS		10
SEQ ID NO: 28	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 28		
GYTFTRYNMH		10
SEQ ID NO: 29	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 29		
GFNIKNTYMH		10
SEQ ID NO: 30	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 30		
GPSLTSYGVH		10
SEQ ID NO: 31	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 31		
WINIYSGIPT YADDFKG		17
SEQ ID NO: 32	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 32		
TIYPGYGDTN YNQKFKG		17
SEQ ID NO: 33	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 33		
RIAPANVDTK YAPKFQG		17
SEQ ID NO: 34	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 34		
WINIYSGMPT YADDFKG		17
SEQ ID NO: 35	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	

-continued

	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 35		
RIDPANVNTK YAPKFQG		17
SEQ ID NO: 36	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 36		
FDGPDY		6
SEQ ID NO: 37	moltype = AA length = 13	
FEATURE	Location/Qualifiers	
source	1..13	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 37		
EFADDYPIPP FDY		13
SEQ ID NO: 38	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Mus musculus	
SEQUENCE: 38		
IYYDYDGDID V		11
SEQ ID NO: 39	moltype = DNA length = 1350	
FEATURE	Location/Qualifiers	
misc_feature	1..1350	
	note = caninized mouse antibody	
misc_difference	715..717	
	note = n is a, c, g, or t	
misc_difference	796..798	
	note = n is a, c, g, or t	
misc_difference	892..900	
	note = n is a, c, g, or t	
misc_difference	982..984	
	note = n is a, c, g, or t	
misc_difference	988..990	
	note = n is a, c, g, or t	
source	1..1350	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 39		
gaggtgcagc tgggtgcagag cgggcgccgac ctgggtgaagc ccggcgccgag cgtgagactg		60
agctgcgtgg ccagcggcta cacttcacc acctacggca tgagctgggt gagacaggcc		120
cccgcaagg gccctgcagt gatgggctgg atcaacatct acagcggcat cccacacctac		180
gcccagcact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac		240
ctgcagctga acagcctgag agccgaggac accgcgtgt actactgcac cagattcgac		300
ggccccgac actggggcca gggcacccctg gtgaccgtga gcagcgccag caccaccgcc		360
cccagcgtgt tccccctggc ccccagctgc ggcagcacca gcggcagcac cgtggccctg		420
gcctgcctgg tgagcggcta ctccccgag cccgtgaccg tgagctggaa cagcggcagc		480
ctgaccagcg gcgtgcacac ctccccagc gtgctgcaga gcagcgccct gtacagcctg		540
agcagcatgg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg		600
gcccaccccc ccagcaagac caaggtggac aagcccgtgc ccaagagaga gaacggcaga		660
gtgcccagac cccccgactg ccccaagtgc cccgcccccg agatgctggg cggcnnnnagc		720
gtgttcactt tcccccccaa gcccaaggac accctgtgta tcgccagaac ccccaggtg		780
acctgcgtgg tggtnnnct ggacccccgag gaccccgagg tgagatcag ctggttcgtg		840
gacggcaagc agatgcagac cgccaagacc cagcccagag agggagcagt cnnnnnnnnn		900
tacagagtgg tgagcgtgct gcccatcggc caccaggact ggctgaaggg caagcagttc		960
acctgcaagg tgaacaacaa gnnnctgnnn agccccatcg agagaaccat cagcaaggcc		1020
agaggccagg cccaccagcc cagcgtgtac gtgctgcccc ccagcagaga ggagctgagc		1080
aagaacaccg tgagcctgac ctgcctgatc aaggacttct tccccccga catcgacgtg		1140
gagtggcaga gcaacggcca gcaggagccc gagagcaagt acagaaccac cccccccag		1200
ctggacgagg acggcagcta ctctcgtac agcaagctga gcgtggacaa gagcagatgg		1260
cagagaggcg acaccttcac ctgcgcctg atgcacgagg ccctgcacaa ccactacacc		1320
caggagagcc tgagccacag ccccgcaag		1350
SEQ ID NO: 40	moltype = AA length = 450	
FEATURE	Location/Qualifiers	
REGION	1..450	
	note = caninized mouse antibody	
VARIANT	239	

-continued

note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 266
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 298..300
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 328
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 330
 note = misc_feature - Xaa can be any naturally occurring amino acid
 source 1..450
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 40
 EVQLVQSGGD LVKPGGSVRL SCVASGYTFT TYGMSWVRQA PGKGLQWMGW INIYSGIPTY 60
 ADDFKGRFTF SLDTAKNTAY LQLNSLRAED TAVYYCTRPD GPDYWGQGT L VTVSSASTTA 120
 PSVFPPLAPSC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLYSL 180
 SSMVTVPSSR WPSETFTCNV AHPASKTKVD KPVPKRENGR VPRPPDCPKC PAPEMLGGXS 240
 VFIFPPKPKD TLLIARTPEV TCVVXLDPD DPEVQISWFV DGKQMOTAKT QPREEQFXXX 300
 YRVVSVLPIG HQDWLKGQF TCKVNNKXLX SPIERTISKA RGQAHQPSVY VLPPSREELS 360
 KNTVSLTCLI KDFPPDIDV EWQSNQQQEP ESKYRTTPPQ LDEGSGYFLY SKLSVDKSRW 420
 QRGDTFICAV MHEALHNHYT QESLSHSPGK 450
 SEQ ID NO: 41 moltype = DNA length = 1344
 FEATURE Location/Qualifiers
 misc_feature 1..1344
 note = caninized mouse antibody
 misc_difference 709..711
 note = n is a, c, g, or t
 misc_difference 790..792
 note = n is a, c, g, or t
 misc_difference 886..894
 note = n is a, c, g, or t
 misc_difference 976..978
 note = n is a, c, g, or t
 misc_difference 982..984
 note = n is a, c, g, or t
 source 1..1344
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 41
 gaggtgcagc tgggtgcagc cggtgcgagc ctggtgaagc cgggcggcag cgtgagactg 60
 agctgcgtgg ccagcggcta cacttcacc acctacggca tgagctgggt gagacaggcc 120
 ccggcaagg gcctgcagtg gatgggctgg atcaacatct acagcggcat cccacctac 180
 gccagcagct tcaaggcgag attcaccttc agcctggaca cgcgaagaa caccgcctac 240
 ctgcagctga acagcctgag agccgaggac accgcctgt actactgcac cagattcgac 300
 ggcccccagc actggggcca gggcacccctg gtgaccgtga gcagcggcag caccaccgcc 360
 cccagcgtgt tccccctggc cccagctgc ggcagccaga gcgcagcac cgtggccctg 420
 gcctgcctgg tgagcggcta catccccgag cccgtgaccg tgagctggaa cagcgtgagc 480
 ctgaccagcg gcgtgcacac ctccccagc gtgctgcaga gcagcggcct gtacagcctg 540
 agcagcatgg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
 gcccaccccg ccaccaacac caaggtggac aagcccgtgg ccaaggagtg cgagtgcagg 660
 tgcaactgca acaactgccc ctgcccggcg tgcggcctgc tgggcggcnn naggctgttc 720
 atcttcccc ccaagcccaa ggacatcctg gtgaccgcca gaacccccac cgtgacctgc 780
 gtggtggtgn nnctggaccc cgagaacccc gaggtgcaga tcagctggtt cgtggacagc 840
 aagcaggtgc agaccgcaa caccagccc agagaggagc agagcnnnnn nnnntacaga 900
 gtggtgagcg tgcctgccat cggccaccag gactggctga gcgcgaagca gttcaagtgc 960
 aaggtgaaca acaagnnnct gnnnagcccc atcgaggaga tcatcagcaa gacccccggc 1020
 caggccacac agcccaacgt gtacgtgctg cccccagca gagacgagat gagcaagaac 1080
 accgtgaccc tgacctgcct ggtgaaggac ttcttcccc cagagatcga cgtggagtgg 1140
 agagcaacg gccagcagga gcccgagagc aagtacagaa tgaccccccc ccagctggac 1200
 gaggacggca gctacttctt gtacagcaag ctgagcgtgg acaagagcag atggcagaga 1260
 ggcgacacct tcactctgcg cgtgatgcac gaggccctgc acaaccacta caccagatc 1320
 agcctgagcc acagcccgcg caag 1344
 SEQ ID NO: 42 moltype = AA length = 448
 FEATURE Location/Qualifiers
 REGION 1..448
 note = caninized mouse antibody
 VARIANT 237
 note = misc_feature - Xaa can be any naturally occurring amino acid

-continued

VARIANT 264
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 296..298
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 326
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 328
note = misc_feature - Xaa can be any naturally occurring amino acid

source 1..448
mol_type = protein
organism = synthetic construct

SEQUENCE: 42

EVQLVQSGGD	LVKPGGSRVL	SCVASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGIPTY	60
ADDFKGRFTF	SLDTAKNTAY	LQLNSLRAED	TAVYYCTFRD	GPDYWGQGTL	VTVSSASTTA	120
PSVFPLAPSC	GSQSGSTVAL	ACLVSGYIPE	PVTVSWNSVS	LTSGVHTFPS	VLQSSGLYSL	180
SSMVTVPSSR	WPSETFTCNV	AHPATNTKVD	KPVAKECECK	CNCNNPCPCG	CGLLGXSVF	240
IFPPKPKDIL	VTARTPTVTC	VVXLDPENP	EVQISWFDVS	KQVQTANTQP	REEQSXXXYR	300
VVSVLPIGHQ	DWLSGKQFKC	KVNNKXLKSP	IEEII SKTPG	QAHQPNVYVL	PPSRDEMSKN	360
TVTLTCLVKD	FFPPEIDVEW	QSNQQQEPES	KYRMTPPQLD	EDGSYFLYSK	LSVDKSRWQR	420
GDTFICAVMH	EALHNHYTQI	SLSHSPGK				448

SEQ ID NO: 43 moltype = DNA length = 1365

FEATURE Location/Qualifiers

misc_feature 1..1365
note = caninized mouse antibody

misc_difference 730..732
note = n is a, c, g, or t

misc_difference 811..813
note = n is a, c, g, or t

misc_difference 907..915
note = n is a, c, g, or t

misc_difference 997..999
note = n is a, c, g, or t

misc_difference 1003..1005
note = n is a, c, g, or t

source 1..1365
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 43

gaggtgcagc	tggtgcagag	cggcgccgac	ctggtgaagc	cggcgccgac	cgtgagactg	60
agctgcgtgg	ccagcggcgt	caacatcaag	aacacctaca	tgcactgggt	gagacaggcc	120
cccgccaagg	gcctgcagtg	gatcggcaga	atcgccccc	ccaacgtgga	caccaagtac	180
gcccccaagt	tccagggcaa	ggccaccatc	agcgccgaca	ccgccaagaa	caccgcctac	240
atgcagctga	acagcctgag	agccgaggac	accgcccgtg	actactgcgt	gctgatctac	300
tacgactacg	acggcgacat	cgacgtgtgg	ggccagggca	ccctgggtgac	cgtgagcagc	360
gccagcacca	ccgccccag	cgtgttcccc	ctggccccca	gctgcggcag	caccagcggc	420
agcaccgtgg	ccctggcctg	cctgggtgagc	ggctacttcc	ccgagcccgt	gaccgtgagc	480
tggaacagcg	gcagcctgac	cagcgccgtg	cacaccttcc	ccagcgtgct	gcagagcagc	540
ggcctgtaca	gcctgagcag	catggtgacc	gtgcccagca	gcagatggcc	cagcgagacc	600
ttcacctgca	acgtggccca	ccccgccagc	aagaccaagg	tggacaagcc	cgtgcccagg	660
agagagaacg	gcagagtggc	cagaccccc	gactgcccc	agtgcgccgc	ccccgagatg	720
ctgggcgccn	nnagcgtgtt	catcttcccc	ccccagcccc	aggacacct	gctgatcgcc	780
agaacccccg	aggtgacctg	cgtgggtggtg	nnnctggacc	ccgaggaccc	cgagggtcag	840
atcagctggt	tcgtggacgg	caagcagatg	cagaccgcca	agacccagcc	cagagaggag	900
cagttcnnnn	nnnnntacag	agtgggtgagc	gtgctgcccc	tcggccacca	ggactggctg	960
aagggcaagc	agttcacctg	caagggtgaac	aacaagnnnc	tgnnnagccc	catcgagaga	1020
accatcacga	agggccagag	ccaggccccac	cagccccagc	tgtacgtgct	gccccccagc	1080
agagaggagc	tgagcaagaa	caccgtgagc	ctgacctgcc	tgatcaagga	cttcttcccc	1140
cccacatcgc	acgtggagtg	gcagagcaac	ggccagcagg	agcccagagag	caagtacaga	1200
accaccccc	cccagctgga	caggagcggc	agctacttcc	tgtacagcaa	gctgagcgtg	1260
gacaagagca	gatggcagag	aggcgacacc	ttcatctcgc	ccgtgatgca	cgaggccctg	1320
cacaaccact	acacccagga	gagcctgagc	cacagccccg	gcaag		1365

SEQ ID NO: 44 moltype = AA length = 455

FEATURE Location/Qualifiers

REGION 1..455
note = caninized mouse antibody

VARIANT 244
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 271
note = misc_feature - Xaa can be any naturally occurring

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      amino acid
VARIANT      303..305
              note = misc_feature - Xaa can be any naturally occurring
              amino acid
VARIANT      333
              note = misc_feature - Xaa can be any naturally occurring
              amino acid
VARIANT      335
              note = misc_feature - Xaa can be any naturally occurring
              amino acid
source        1..455
              mol_type = protein
              organism = synthetic construct

SEQUENCE: 44
EVQLVQSGGD LVKPGGSRVL SCVASGFNIK NTYMHWVRQA PGKGLQWIGR IAPANVDTKY 60
APKFQGKATI SADTAKNTAY MQLNSLRAED TAVYYCVLIY YDYGDDIDVW GQGTSLVTSS 120
ASTTAPSVFP LAPSCGSTSG STVALACLVS GYFPEPVTVS WNSGSLTSGV HTFPSVLQSS 180
GLYSLSSMVT VPSSRWPSET FTCNVAHPAS KTKVDKPVPK RENGVRPRPP DCPKCPAPEM 240
LGGXSVFIFP PKPKDTLLIA RTPEVTCVVV XLDPEDPEVQ ISWFDVGKQM QTAKTPREE 300
QFXXXRVVVS VLPIGHQDWL KGKQFTCKVN NKXLXSPIER TISKARGQAH QPSVYVLPPS 360
REELSKNTVS LTCLIKDFFP PDIDVEWQSN GQQEPESKYR TTPQLDEDG SYFLYSKLSV 420
DKSRWQRGDT FICAVMHEAL HNHYTQESLS HSPGK 455

SEQ ID NO: 45      moltype = DNA length = 1359
FEATURE            Location/Qualifiers
misc_feature        1..1359
                    note = caninized mouse antibody
misc_difference      724..726
                    note = n is a, c, g, or t
misc_difference      805..807
                    note = n is a, c, g, or t
misc_difference      901..909
                    note = n is a, c, g, or t
misc_difference      991..993
                    note = n is a, c, g, or t
misc_difference      997..999
                    note = n is a, c, g, or t
source              1..1359
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 45
gaggtgcagc tggtgcagag cggtgcgcag ctggtgaagc ccggcgccag cgtgagactg 60
agctgcgtgg ccagcggtt caacactcaag aacacctaca tgcactgggt gagacaggcc 120
cccggaagg gcctgcagtg gatcggcaga atcgccccc ccaacgtgga caccaagtac 180
gcccccaagt tccagggcaa ggcaccatc agcgccgaca ccgccaagaa caccgcctac 240
atgcagctga acagcctgag agccgaggac accgcctgt actactgct gctgatctac 300
tacgactacg acggcgacat cgacgtgtgg ggccagggca ccctggtgac cgtgagcagc 360
gccagcacca ccgccccag cgtgttcccc ctggccccca gctgcggcag ccagagcggc 420
agcaccgtgg ccctggcctg cctggtgagc ggctacatcc ccgagccctg gaccgtgagc 480
tggaacacgc tgagcctgac cagcggtgtg cacaccttcc ccagcgtgct gcagagcagc 540
ggcctgtaca gcctgagcag catggtgacc gtgcccagca gcagatggcc cagcgagacc 600
ttcacctgca acgtggccca ccccgccacc aacaccaagg tggacaagcc cgtggccaag 660
gagtgcgagt gcaagtgcaa ctgcaacaac tgccccctgc ccggctgcgg cctgctgggc 720
ggcnnnagcg tgttcattt ccccccaag cccaaggaca tcctggtgac cgccagaacc 780
cccaccgtga cctgcgtggt ggtgnnctg gaccccgaga accccgaggt gcagatcagc 840
tggttcgtgg acagcaagca ggtgcagacc gccaacaccc agcccagaga ggagcagagc 900
nnnnnnmnt acagagtgtg gacgtgtgtg cccatcgccc accaggactg gctgagcggc 960
aagcagttca agtgcaaggt gaacaacaag nnnctgnnna gcccacatga ggagatcatc 1020
agcaagaccc ccggccaggc ccaccagccc aacgtgtacg tgctgcccc cagcagagac 1080
gagatgagca agaacaccgt gaccctgacc tgcctgttga aggacttctt cccccccgag 1140
atcgacgtgg agtggcagag caacggccag caggagcccg agagcaagta cagaatgacc 1200
ccccccagc tggacagagga cggcagctac ttcctgtaca gcaagctgag cgtggacaag 1260
agcagatggc agagaggcga caccttcac tgcgcctgga tgcacgaggc cctgcacaac 1320
cactacaccc agatcagcct gagccacagc cccggcaag 1359

SEQ ID NO: 46      moltype = AA length = 453
FEATURE            Location/Qualifiers
REGION              1..453
                    note = caninized mouse antibody
VARIANT            242
                    note = misc_feature - Xaa can be any naturally occurring
                    amino acid
VARIANT            269
                    note = misc_feature - Xaa can be any naturally occurring
                    amino acid
VARIANT            301..303

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-continued

note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 331
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 333
 note = misc_feature - Xaa can be any naturally occurring amino acid
 source 1..453
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 46
 EVQLVQSGGD LVKPGGSVRL SCVASGFNIK NTYMHWVRQA PGKGLQWIGR IAPANVDTKY 60
 APKFQQKATI SADTAKNTAY MQLNSLRAED TAVYYCVLIY YDYGDDIDVW GQGTSLTVSS 120
 ASTTAPSVFP LAPSCGSQSG STVALACLVS GYIPEPVTVS WNSVSLTSGV HTFPSVLQSS 180
 GLYSLSMMVT VPSSRWPSSET FTCNVAHPAT NTKVDKPVAK ECECKCNENN CPCPGCGLLG 240
 GKSVFIFPPK PKDILVTART PTVTCVVVXL DPENPEVQIS WFDVSKQVQT ANTQPREEQS 300
 XXXYRVSVL PIGHQDWLSG KQPKCKVNNK XLXSPIEEII SKTPGQAHQP NVYVLPSPRD 360
 EMSKNTVTLT CLVKDFFPPE IDVEWQSNQO QEPESKYRMT PPQLDEDGSY FLYSKLSVDK 420
 SRWRGRDTFI CAVMHEALHN HYTQISLSHS PGK 453

SEQ ID NO: 47 moltype = DNA length = 1371
 FEATURE Location/Qualifiers
 misc_feature 1..1371
 note = caninized mouse antibody
 misc_difference 736..738
 note = n is a, c, g, or t
 misc_difference 817..819
 note = n is a, c, g, or t
 misc_difference 913..921
 note = n is a, c, g, or t
 misc_difference 1003..1005
 note = n is a, c, g, or t
 misc_difference 1009..1011
 note = n is a, c, g, or t
 source 1..1371
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 47
 gaggtgcagc tgggtgcagc cggcgccgac ctggtgaagc cggcgccgac cgtgagactg 60
 agctgcgtgg ccagcggtga caccctcacc agatacaaca tgcactgggt gagacaggcc 120
 ccgggcaagg gcctgcagtg gatcgccacc atctaccccg gctacggcga caccagctac 180
 aaccagaagt tcaaggggcaa ggccaccctg agcgtggaca tcgccaagaa caccgcctac 240
 atgcagctga acagcctgag agccgaggac accgcctgtt acttctgcag cagagagttc 300
 gccgacgact accccatccc ccccttcgac tactggggcc agggcaccct ggtgaccgtg 360
 agcagcgcca gcaccaccgc ccccgccgtg tccccctgg ccccaagctg cggcagcacc 420
 agcggcgaca ccgtggccct ggccctgcct gtgagcggct acttccccga gcccgtagcc 480
 gtgagctgga acagcgccag cctgaccagc ggcgtgcaca ccttcccag cgtgctgcag 540
 agcagcgccc tgtacagct gagcagcatg gtgacgtgc ccagcagcag atggcccagc 600
 gagaccttca cctgcaacgt ggcccacccc gccagcaaga ccaagggtga caagcccgtg 660
 cccaagagag agaacggcag agtgcccaga ccccccgact gcccgaagtg ccccgcccc 720
 gagatgctgg gcggnnnnnn cgtgttcac tccccccca agcccaagga caccctgctg 780
 atcgccagaa cccccaggt gacctgcgtg gtggtgnnnc tggaccccga ggaccccag 840
 gtgcagatca gctgggtcgt ggacggcaag cagatgcaga ccgccaagac ccagcccaga 900
 gagagcagtg tcnnnnnnnn ntacagagtg gtgagcgtgc tgcccatcgg ccaccaggac 960
 tggctgaagg gcaagcagtt cacctgcaag gtgaacaaca agnnnctggn nagccccatc 1020
 gagagaacca tcagcaaggc cagaggccag gccaccagc ccagcgtgta cgtgctgccc 1080
 cccagcagag aggagctgag caagaacacc gtgagcctga cctgcctgat caaggacttc 1140
 tcccccccg acatcgacgt ggagtgccag agcaacggcc agcaggagcc cgagagcaag 1200
 tacagaacca cccccccca gctggacgag gacggcagct acttctgta cagcaagctg 1260
 agcgtggaca agagcagatg gcagagagcc gacaccttca tctgcgcgt gatgcaagag 1320
 gccctgcaca accactacac ccaggagagc ctgagccaca gccccggcaa g 1371

SEQ ID NO: 48 moltype = AA length = 457
 FEATURE Location/Qualifiers
 REGION 1..457
 note = caninized mouse antibody
 VARIANT 246
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 273
 note = misc_feature - Xaa can be any naturally occurring amino acid
 VARIANT 305..307
 note = misc_feature - Xaa can be any naturally occurring amino acid

-continued

VARIANT 335
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 337
note = misc_feature - Xaa can be any naturally occurring amino acid

source 1..457
mol_type = protein
organism = synthetic construct

SEQUENCE: 48

EVQLVQSGGD	LVPKGGSVRL	SCVASGYTFT	RYNMHWVRQA	PGKGLQWIGT	IYPGYGDTSY	60
NQKFKGKATL	SVDIAKNYAY	MLNLSLRAED	TAVYFCSREF	ADDYPIPPFD	YWGQGLVTV	120
SSASTTAPSV	FPLAPSCGST	SGSTVALACL	VSGYFPEPVT	VSWNSGLTS	GVHTFPSVLQ	180
SSGLYSLSSM	VTVPSSRWPS	ETFTCNVAHP	ASKTKVDKPV	PKRENGRVR	PPDCPKCPAP	240
EMLGKXSVFI	PPPKPKDTLL	IARTPEVTCV	VXLDPEDPE	VQISWFVDGK	QMOTAKTQPR	300
EQQFXXYRV	VSVLPIGHQD	WLKGKQFTCK	VNNKXLXSPI	ERTISKARGQ	AHQPSVYVLP	360
PSREELSKNT	VSLTCLIKDF	FPPDIDVEWQ	SNGQQEPESK	YRTTPPQLDE	DGSYFLYSKL	420
SVDKSRWQRG	DTFICAVMHE	ALHNHYTQES	LSHSPGK			457

SEQ ID NO: 49 moltype = DNA length = 1365

FEATURE Location/Qualifiers

misc_feature 1..1365
note = caninized mouse antibody

misc_difference 730..732
note = n is a, c, g, or t

misc_difference 811..813
note = n is a, c, g, or t

misc_difference 907..915
note = n is a, c, g, or t

misc_difference 997..999
note = n is a, c, g, or t

misc_difference 1003..1005
note = n is a, c, g, or t

source 1..1365
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 49

gaggtgcagc	tggtgcagag	cggcgccgac	ctggtgaagc	cggcgccgag	cgtgagactg	60
agctgcgtgg	ccagcggcga	caccttcacc	agatacaaca	tgactgggtg	gagacaggcc	120
ccggcgaagg	gcctgcagtg	gatcgccacc	atctaccccg	gctacggcga	caccagctac	180
aaccagaagt	tcaagggcga	ggccaccctg	agcgtggaca	tcgccaaaga	caccgcctac	240
atgcagctga	acagcctgag	agccgaggac	accgccgtgt	actctctcag	cagagagttc	300
gccgacgact	accccatccc	ccccttcgac	tactggggcc	agggcaccct	ggtgaccgtg	360
agcagcgcca	gcaccaccgc	ccccagcgtg	ttccccctgg	ccccagctg	cggcagccag	420
agcggcagca	ccgtggccct	ggcctgcctg	gtgagcggct	acatccccga	gcccgtagcc	480
gtgagctgga	acagcgtgag	cctgaccagc	ggcgtgcaca	ccttccccag	cgtgctgcag	540
agcagcggcc	tgtacagcct	gagcagcatg	gtgaccgtgc	ccagcagcag	atggcccagc	600
gagaccttca	cctgcaacgt	ggcccccccc	gccaccaaca	ccaaggtgga	caagcccgtg	660
gccaaaggagt	gcgagtgcga	gtgcaactgc	aacaactgcc	cctgccccgg	ctgcccgtg	720
ctgggcggcn	nnagcgtgtt	catcttcccc	cccaagccca	aggacatcct	ggtgaccgcc	780
agaacccccca	ccgtgacctg	cgtgggtggtg	nnnctggacc	ccgagaaccc	cagagtgca	840
atcagctggt	tcgtggacag	caagcaggtg	cagaccgcca	acaccagcc	cagagaggag	900
cagagcnnnn	nnnnntacag	agtgggtgagc	gtgctgccca	tcggccacca	ggactggctg	960
agcggcaagc	agtccaagtg	caaggtgaac	aacaagnnnc	tgnnnagccc	catcgaggag	1020
atcatcagca	agacccccgg	ccaggccccac	cagcccaacg	tgtacgtgct	gccccccagc	1080
agagacgaga	tgagcaagaa	caccgtgacc	ctgacctgcc	tggtgaagga	cttcttcccc	1140
cccagatcgc	acgtggagtg	gcagagcaac	ggccagcagg	agcccagag	caagtacaga	1200
atgaccccccc	cccagctgga	cgaggacggc	agctacttcc	tgtacagcaa	gctgagcgtg	1260
gacaagagca	gatggcagag	aggcgacacc	ttcatctcgc	ccgtgatgca	cagggccctg	1320
cacaaccact	acacccagat	cagcctgagc	cacagccccg	gcaag		1365

SEQ ID NO: 50 moltype = AA length = 455

FEATURE Location/Qualifiers

REGION 1..455
note = caninized mouse antibody

VARIANT 244
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 271
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 303..305
note = misc_feature - Xaa can be any naturally occurring amino acid

VARIANT 333
note = misc_feature - Xaa can be any naturally occurring

-continued

VARIANT amino acid
335
note = misc_feature - Xaa can be any naturally occurring amino acid

source 1..455
mol_type = protein
organism = synthetic construct

SEQUENCE: 50

EVQLVQSGGD	LVKPGGSVRL	SCVASGYTFT	RYNMHWVRQA	PGKGLQWIGT	IYPGYGDTSY	60
NQKFKGKATL	SVDIAKNTAY	MQLNSLRAED	TAVYFCSREF	ADDYPIPPFD	YWGQGLVTV	120
SSASTTAPSV	FPLAPSCGSQ	SGSTVALACL	VSGYIPEPVT	VSWNSVSLTS	GVHTFPSVLQ	180
SSGLYSLSSM	VTPSSRWPS	ETFTCNVAHP	ATNTKVDKPV	AKECECKCNC	NNCPCPGCGL	240
LGGXSVFIFP	PKPKDILVTA	RTPTVTCVVV	XLDPENPEVQ	ISWFVDSKQV	QTANTQPRE	300
QXXXXYRVVS	VLPIGHQDWL	SGKQFKCKVN	NKXLXSPIEE	IISKTPGQAH	QPNVYVLPSS	360
RDEMSKNTVT	LTCLVKDFFP	PEIDVEWQSN	GQQEPESKYR	MTPPQLDEDG	SYFLYSKLSV	420
DKSRWQRGDT	FICAVMHEAL	HNHYTQISLS	HSPGK			455

SEQ ID NO: 51 moltype = DNA length = 1350
FEATURE Location/Qualifiers
misc_feature 1..1350
note = caninized mouse antibody
misc_difference 715..717
note = n is a, c, g, or t
misc_difference 796..798
note = n is a, c, g, or t
misc_difference 892..900
note = n is a, c, g, or t
misc_difference 982..984
note = n is a, c, g, or t
misc_difference 988..990
note = n is a, c, g, or t
source 1..1350
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 51

gaggtgcagc	tggtgcagag	cggccccggc	ctggtgaagc	ccagccagag	cctgagcctg	60
acctgcgtgg	tgagcggtct	cagcctgacc	agctacggcg	tgactgggt	gagacaggcc	120
cccggaagg	gcctgcagtg	gatgggctgg	atcaacatct	acagcggcat	ccccacctac	180
gcccagcact	tcaagggcag	attcaccttc	agcctggaca	ccgccaagaa	caccgcctac	240
ctgcagctga	gcagcctgag	agccgaggac	accgccgtgt	actactgcgc	cagattcgac	300
ggccccgact	actggggcca	gggcaccctg	gtgaccgtga	gcagcgccag	caccaccgcc	360
cccagcgtgt	tccccctggc	cccagctgc	ggcagcacca	gcggcagcac	cgtggccctg	420
gcctgcctgg	tgagcggcta	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcggcct	gtacagcctg	540
agcagcatgg	tgaccgtgcc	cagcagcaga	tggcccagcg	agaccttcac	ctgcaacgtg	600
gcccccccg	ccagcaagac	caaggtggac	aagcccgtgc	ccaagagaga	gaacggcaga	660
gtgcccagac	cccccgactg	ccccaaagtc	cccgcctggc	agatgctggg	cggcnnnnagc	720
gtgttcacat	tcccccccaa	gcccgaaggac	accctgtgta	tcgccagaac	ccccgagggtg	780
acctgcgtgg	tggtgmnnt	ggaccccagc	gaccccagg	tgagatcag	ctggttcgtg	840
gacggcaagc	agatgcagac	gcgcaagacc	cagcccagag	aggagcagtg	cnnnnnnnnn	900
tacagagtg	tgagcgtgct	gcccacggcg	caccaggact	ggctgaagg	caagcagttc	960
acctgcaagg	tgaacaacaa	gnnnctgnnn	agccccatcg	agagaacat	cagcaaggcc	1020
agaggccagg	cccaccagcc	cagcgtgtac	gtgctgcccc	ccagcagaga	ggagctgagc	1080
aagaacaccg	tgagcctgac	ctgcctgatc	aaggacttct	tccccccga	catcgacgtg	1140
gagtggcaga	gcaacggcca	gcaggagccc	gagagcaagt	acagaaccac	cccccccag	1200
ctggacgagg	acggcagcta	cttcctgtac	agcaagctga	gcgtggacaa	gagcagatgg	1260
cagagaggcg	acaccttcac	ctgcgccgtg	atgcacgagg	ccctgcacaa	ccactacacc	1320
caggagagcc	tgagccacag	ccccggcaag				1350

SEQ ID NO: 52 moltype = AA length = 450
FEATURE Location/Qualifiers
REGION 1..450
note = caninized mouse antibody
VARIANT 239
note = misc_feature - Xaa can be any naturally occurring amino acid
VARIANT 266
note = misc_feature - Xaa can be any naturally occurring amino acid
VARIANT 298..300
note = misc_feature - Xaa can be any naturally occurring amino acid
VARIANT 328
note = misc_feature - Xaa can be any naturally occurring amino acid
VARIANT 330

-continued

```

note = misc_feature - Xaa can be any naturally occurring
amino acid
source 1..450
mol_type = protein
organism = synthetic construct

SEQUENCE: 52
EVQLVQSGPG LVKPSQSLSL TCVVSGFSLT SYGVHWVRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFTF SLDTAKNTAY LQLSSLRAED TAVYYCARFD GPDYWGQGT L VTVSSASTTA 120
PSVFPPLAPSC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLYSL 180
SSMVTVPSSR WPSETFTCNV AHPASKTKVD KPVPKRENGR VPRPPDCPKC PAPEMLGGXS 240
VFIFPPPKPKD TLLIARTPEV TCVVVXLDPE DPEVQISWFV DGKQMOTAKT QPREEQFXXX 300
YRVVSVLPPIG HQDWLKGKQF TCKVNNKXLX SPIERTISKA RGQAHQPSVY VLPPSREELS 360
KNTVSLTCLI KDFFPDIDV EWQSNQQQEP ESKYRTTPPQ LDEGSGYFLY SKLSVDKSRW 420
QRGDTFICAV MHEALHNHYT QESLSHSPGK 450

SEQ ID NO: 53 moltype = DNA length = 1344
FEATURE Location/Qualifiers
misc_feature 1..1344
note = caninized mouse antibody
misc_difference 709..711
note = n is a, c, g, or t
misc_difference 790..792
note = n is a, c, g, or t
misc_difference 886..894
note = n is a, c, g, or t
misc_difference 976..978
note = n is a, c, g, or t
misc_difference 982..984
note = n is a, c, g, or t
source 1..1344
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 53
gagggtgcagc tgggtgcagag cggcccccggc ctggtgaagc ccagccagag cctgagcctg 60
acctgcgtgg tgagcggcct cagcctgacc agctacggcg tgcactgggt gagacaggcc 120
cccgcaaggg gcctgcagtg gatgggctgg atcaacatct acagcggcat cccacacctac 180
gcccagcact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac 240
ctgcagctga gcagcctgag agccgaggac accgcctgt actactgcgc cagattcgac 300
ggccccgact actggggcca gggcacctct gtgaccgtga gcagcgccag caccaccgcc 360
cccagcgtgt tccccctggc cccagctgc ggcagccaga gcgcagcac cgtggccctg 420
gcctgcctgg tgagcggcct catccccgag cccgtgaccg tgagctggaa cagcgtgagc 480
ctgaccagcg gcgtgcacac ctccccagc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcatgg tgaccgtgcc cagcagcaga tggcccagcg agacottcac ctgcaacgtg 600
gcccaccccg ccaccaacac caaggtggac aagcccgctg ccaaggagtg cgagtgcaag 660
tgcaactgca acaactgccc ctgccccggc tgcggcctgc tggcgccgcn nagcgtgttc 720
atcttcccc ccaagcccaa ggacatcctg gtgaccgcca gaacccccc cgtgacctgc 780
gtggtggtgn nnctggaccc cgagaacccc gaggtgcaga tcagctggtt cgtggacagc 840
aagcaggtgc agaccgcaa cagccagccc agagaggagc agagcnnnnn nnnntacaga 900
gtggtgagcg tgcgtgccat cggccaccag gactggctga gcgcgaagca gttcaagtgc 960
aagggtgaaca acaagnnnct gnnnagcccc atcgaggaga tcatcagcaa gacccccggc 1020
caggcccacc agcccaacgt gtacgtgctg cccccagca gagacgagat gagcaagaac 1080
accgtgaccc tgacctgctt ggtgaaggac ttcttcccc cagagatcga cgtggagtgg 1140
cagagcaacg gccagcagga gcccgagagc aagtacagaa tgaccccccc ccagctggac 1200
gaggacggca gctacttctt gtacagcaag ctgagcgtgg acaagagcag atggcagaga 1260
ggcgacacct tcactctgdc gtgatgcac gagggcctgc acaaccacta caccagatc 1320
agcctgagcc acagccccgg caag 1344

SEQ ID NO: 54 moltype = AA length = 448
FEATURE Location/Qualifiers
REGION 1..448
note = caninized mouse antibody
VARIANT 237
note = misc_feature - Xaa can be any naturally occurring
amino acid
VARIANT 264
note = misc_feature - Xaa can be any naturally occurring
amino acid
VARIANT 296..298
note = misc_feature - Xaa can be any naturally occurring
amino acid
VARIANT 326
note = misc_feature - Xaa can be any naturally occurring
amino acid
VARIANT 328
note = misc_feature - Xaa can be any naturally occurring
amino acid

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source                1..448
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 54
EVQLVQSGPG LVKPSQSLSL TCVVSGFSLT SYGVHWRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFTF SLDTAKNTAY LQLSSLRAED TAVYYCARFD GPDYWGQGTL VTVSSASTTA 120
PSVFLAPASC GSQSGSTVAL ACLVSGYIPE PVTVSWNSVS LTSGVHTFPS VLQSSGLYSL 180
SSMVTVPSSR WPSETFTCNV AHPATNTKVD KPVAKCECK CNCNNPCPG CGLLGXSVF 240
IFPPKPKDIL VTARTPTVTC VVVXLDPENP EVQISWFVDS KQVQTANTQP RREQSXXXYR 300
VVSVLPIGHQ DWLSGKQFKC KVNKKXLKSP IEEIISKTPG QAHQPNVYVL PPSRDEMSKN 360
TVTLTCLVKD FFPPEIDVEW QSNQQQEPES KYRMTPPQLD EDGSYFLYSK LSVDKSRWQR 420
GDTFICAVMH EALHNHYTQI SLSHSPGK 448

SEQ ID NO: 55          moltype = DNA length = 1350
FEATURE               Location/Qualifiers
misc_feature           1..1350
                      note = caninized mouse antiodody
misc_difference        715..717
                      note = n is a, c, g, or t
misc_difference        796..798
                      note = n is a, c, g, or t
misc_difference        892..900
                      note = n is a, c, g, or t
misc_difference        982..984
                      note = n is a, c, g, or t
misc_difference        988..990
                      note = n is a, c, g, or t
source                1..1350
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 55
gagggtgcagc tgggtgcagag cggcgccgag gtgaagaagc cggcgccag cgtgaaggtg 60
agctgcaagg ccagcggcta cacttcacc acctacgga tgagctgggt gagacaggcc 120
cccgcaagg gacctgagtg gatgggctgg atcaacatct acagcggeat cccacctac 180
gcccagcact tcaagggcag attcgccctg agcctggaca ccagcaccag caccgcctac 240
atggagctga acagcctgag agccgaggac accgcccgtg actactgcgc cagattcgac 300
ggccccgact actggggcca gggcaccctg gtgaccgtga gcagcgccag caccacgcc 360
cccagcgtgt tccccctggc cccagctgc ggcagcacca ggcgcagcac cgtggccctg 420
gcctgccttg tgagcggcta ctccccgag cccgtgaccg tgagctggaa cagcggcagc 480
ctgaccagcg gcgtgcacac ctccccagc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcatgg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gcccaccgcc ccagcaagac caaggtggac aagcccgtgc ccaagagaga gaacggcaga 660
gtgcccagac ccccgactg ccccaagtgc cccgcccccg agatgctggg cggnnnnagc 720
gtgttcatct tcccccccaa gcccaggac accctgtgta tcgccagaac ccccgaggtg 780
acctgcgtgg tgggtgnnct ggaccccgag gaccccgagg tgcagatcag ctggttcgtg 840
gacggcaagc agatgcagac gcccaagacc cagcccagag aggagcagtt cnnnnnnnnn 900
tacagagtgg tgagcgtgct gcccacggc caccaggact ggctgaaggg caagcagttc 960
acctgcaagg tgaacaacaa gnnnctgnnn agccccatcg agagaacct cagcaaggcc 1020
agaggccagg cccaccagcc cagcgtgtac gtgctgcccc ccagcagaga ggagctgagc 1080
aagaacaccg tgagcctgac ctgcctgata aaggacttct tccccccga catcgacgtg 1140
gagtggcaga gcaacggcca gcaggagccc gagagcaagt acagaaccac cccccccag 1200
ctggacgagg acggcagcta ctctctgtac agcaagctga gcgtggacaa gagcagatgg 1260
cagagaggcg acaccttcat ctgcgcctg atgcacgagg ccctgcacaa ccactacacc 1320
caggagagcc tgagccacag ccccgcaag 1350

SEQ ID NO: 56          moltype = AA length = 450
FEATURE               Location/Qualifiers
REGION                1..450
                      note = caninized mouse antibody
VARIANT               239
                      note = misc_feature - Xaa can be any naturally occurring
                      amino acid
VARIANT               266
                      note = misc_feature - Xaa can be any naturally occurring
                      amino acid
VARIANT               298..300
                      note = misc_feature - Xaa can be any naturally occurring
                      amino acid
VARIANT               328
                      note = misc_feature - Xaa can be any naturally occurring
                      amino acid
VARIANT               330
                      note = misc_feature - Xaa can be any naturally occurring
                      amino acid
source                1..450
                      mol_type = protein

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                                organism = synthetic construct
SEQUENCE: 56
EVQLVQSGAE VKKPGASVKV SCKASGYTFT TYGMSWVRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFAL SLDTSTSTAY MELNSLRAED TAVYYCARFD GPDYWGQGTI VTVSSASTTA 120
PSVFFPLAPSC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLYSL 180
SSMVTVPSSR WPSETFTCNV AHPASKTKVD KVPVKRENGR VPRPPDCPKC PAPEMLGGXS 240
VFIFPPKPKD TLLIARTPEV TCVVVLDPE DPEVQISWV DKGQMQTAKT QPREEQFXXX 300
YRVVSVLPIG HQDWLKGKQF TCKVNNKXLX SPIERTISKA RGQAHQPSVY VLPPSREELS 360
KNTVSLTCLI KDFPPPDIDV EWQSNQQQEP ESKYRTTPPQ LDEDGSYFLY SKLSVDKSRW 420
QRGDTFICAV MHEALHNHYT QESLSHSPGK 450

SEQ ID NO: 57                moltype = DNA length = 1344
FEATURE                      Location/Qualifiers
misc_feature                  1..1344
                                note = caninized mouse antibody
misc_difference               709..711
                                note = n is a, c, g, or t
misc_difference               790..792
                                note = n is a, c, g, or t
misc_difference               886..894
                                note = n is a, c, g, or t
misc_difference               976..978
                                note = n is a, c, g, or t
misc_difference               982..984
                                note = n is a, c, g, or t
source                        1..1344
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 57
gaggtgcagc tggtgcagag cggtgcagc gtgaagaagc ccgtgcagc cgtgaaggtg 60
agctgcaagg ccagcggtga cacttcacc acctacggca tgagctgggt gagacaggcc 120
cccgcaagg gcctgcagtg gatgggcttg atcaacatct acagcggcat cccacacctac 180
gccgacgact tcaagggcag attcgccctg agcctggaca ccagcaccag caccgcctac 240
atggagctga acagcctgag agccgaggac accgcccgtg actactgcgc cagattcgac 300
ggccccgact actggggcca gggcaccctg gtgaccgtga gcagcggcag caccaccgcc 360
cccagcgtgt tccccctggc ccccgactgc ggcagccaga gcggcagcac cgtggccctg 420
gcctgcctgg tgagcgggta catccccgag cccgtgaccg tgagctggaa cagcgtgagc 480
ctgaccagcg gcgtgcacac cttccccagc gtgctgcaga gcagcggcct gtacagcctg 540
agcagcatgg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gcccaacccc ccaccaaacac caaggtggac aagcccgtgg ccaaggagtg cagtgcaag 660
tgcaactgca acaactgccc ctgccccggc tgcggcctgc tgggcggcnn naggcgtgtc 720
atcttcccc ccaagcccaa ggacatcctg gtgaccgcca gaacccccac cgtgacctgc 780
gtggtggtgn nntggagacc cgagaacccc gaggtgcaga tcagctggtt cgtggacagc 840
aagcaggtgc agaccgcaa caccagccc agagaggagc agagcnnnnn nnnntacaga 900
gtggtgagcg tgctgcccc cggccaccag gactggctga gcggcaagca gttcaagtgc 960
aaggtgaaca acaagnnct gnnnagcccc atcgaggaga tcatcagcaa gacccccggc 1020
caggccccac agcccaacgt gtacgtgctg cccccagca gagacgagat gagcaagaac 1080
accgtgacct tgacctgctt ggtgaaggac ttcttcccc cagagatcga cgtggagtg 1140
cagagcaacg gccagcagga gcccgagagc aagtacagaa tgaccccccc ccagctggac 1200
gaggacggca gctacttctt gtacagcaag ctgagcgtgg acaagagcag atggcagaga 1260
ggcgacacct tcactctgcg cgtgatgcac gaggccctgc acaaccacta caccagatc 1320
agcctgagcc acagccccgg caag 1344

SEQ ID NO: 58                moltype = AA length = 448
FEATURE                      Location/Qualifiers
REGION                        1..448
                                note = caninized mouse antibody
VARIANT                       237
                                note = misc_feature - Xaa can be any naturally occurring
                                amino acid
VARIANT                       264
                                note = misc_feature - Xaa can be any naturally occurring
                                amino acid
VARIANT                       296..298
                                note = misc_feature - Xaa can be any naturally occurring
                                amino acid
VARIANT                       326
                                note = misc_feature - Xaa can be any naturally occurring
                                amino acid
VARIANT                       328
                                note = misc_feature - Xaa can be any naturally occurring
                                amino acid
source                        1..448
                                mol_type = protein
                                organism = synthetic construct
SEQUENCE: 58

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EVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGIPTY	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCARFD	GPDYWGQGT	VTVSSASTTA	120
PSVFPPLAPSC	GSQSGSTVAL	ACLVSGYIPE	PVTVSWNSVS	LTSGVHTFPS	VLQSSGLYSL	180
SSMVTVPSSR	WPSETFTCNV	AHPATNTKVD	KPVAKECECK	CNCNNCPCPG	CGLLGXSVF	240
IFPPKPKDIL	VTARTPTVTC	VVVXLDPENP	EVQISWFDVS	KQVQTANTQP	REEQSXXXYR	300
VVSVLPIGHQ	DWLSGKQFKC	KVNNKXLXSP	IEEIIISKTPG	QAHQPNVYVL	PPSRDEMSKN	360
TVTLTCLVKD	FFPPEIDVW	QSNQQEPES	KYRMTPPQLD	EDGSYFLYSK	LSVDKSRWQR	420
GDTFICAVMH	EALHNHYTQI	SLSHSPGK				448

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SEQ ID NO: 59      moltype = DNA length = 1365
FEATURE           Location/Qualifiers
misc_feature       1..1365
                  note = caninized mouse antibody
misc_difference    730..732
                  note = n is a, c, g, or t
misc_difference    811..813
                  note = n is a, c, g, or t
misc_difference    907..915
                  note = n is a, c, g, or t
misc_difference    997..999
                  note = n is a, c, g, or t
misc_difference    1003..1005
                  note = n is a, c, g, or t
source            1..1365
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 59
gaggtgcagc tggtgcagag cgtggccgag ctggtgaagc cggcgccag cgtgaaggtg 60
agctgcaccg tgagcggctt caacatcaag aacacctaca tgcactgggt gagacaggcc 120
cccgcaaggg gcctgcagtg gatcggcaga atcgaccccg ccaacgtgaa caccaagtac 180
gcccccaagt tccagggcag agccaccatc accgcccaga ccagaccaa caccgcctac 240
atgcagctga gcagcctgag agccgaggac accgcccgtg actactgcgc cagaatctac 300
tacgactacg acgcgacat cgacgtgtgg ggccagggca ccctgggtgac cgtgagcagc 360
gccagcacca cgcgccccag cgtgttcccc ctggcccccga gctgcggcag caccagcggc 420
agcaccgtgg ccctggcctg cctgggtgagc ggctacttcc ccgagcccgt gaccgtgagc 480
tggaacacgg gcagcctgac cagcggcggt cacccttccc ccagcgtgct gcagagcagc 540
ggcctgtaca gcctgagcag catggtgacc gtgccagcga gcagatggcc cagcagagacc 600
ttcacctgca acgtggccca ccccgccagc aagaccaagg tggacaagcc cgtgcccagg 660
agagagaacg gcagagtgcc cagacccccc gactgcccga agtgcccgcg ccccgagatg 720
ctgggcccgn nnagcgtgtt catcttcccc cccaagccca aggacacct gctgatcgcc 780
agaacccccg aggtgacctg cgtggtggtg nnnctggacc cggaggaccg cgaggtgcag 840
atcagctggt tcgtggacgg caagcagatg cagaccgcca agaccagcc cagagaggag 900
cagttcnnnn nnnntacag atggtgagc gtgctgcccc tcggccacca ggactggctg 960
aagggcaagc agttcacctg caaggtgaac aacaagnnnc tgnnnagccc catcgagaga 1020
accatcagca agggcagagg ccaggccccc cagcccagcg tgtacgtgct gccccccagc 1080
agagaggagc tgagcaagaa caccgtgagc ctgacctgcc tgatcaagga cttcttcccc 1140
cccgacatcg acgtggagtg gcagagcaac gggcagcagg agcccagag caagtacaga 1200
accaccccc ccagctgga cgaggacggc agctacttcc tgtacagcaa gctgagcgtg 1260
gacaagagca gatggcagag aggcagacac ttcactctgc cgtgatgca cgaggccctg 1320
cacaaccact acaccagga gagcctgagc cacagccccc gcaag 1365

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SEQ ID NO: 60      moltype = AA length = 455
FEATURE           Location/Qualifiers
REGION           1..455
                  note = caninized mouse antibody
VARIANT          244
                  note = misc_feature - Xaa can be any naturally occurring
                  amino acid
VARIANT          271
                  note = misc_feature - Xaa can be any naturally occurring
                  amino acid
VARIANT          303..305
                  note = misc_feature - Xaa can be any naturally occurring
                  amino acid
VARIANT          333
                  note = misc_feature - Xaa can be any naturally occurring
                  amino acid
VARIANT          335
                  note = misc_feature - Xaa can be any naturally occurring
                  amino acid
source           1..455
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 60
EVQLVQSVAE LVKPGASVKV SCTVSGFNIK NTYMHWVRQA PGKGLQWIGR IDPANVNTKY 60
APKFQGRATI TADTSTNTAY MQLSSLRAED TAVYYCARIY YDYGDIWVW GQGLTVTVSS 120

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ASTTAPSVFP	LAPSCGSTSG	STVALACLVS	GYFPEPVTVS	WNSGSLTSGV	HTFPSVLQSS	180
GLYSLSSMVT	VPSSRWPSET	FTCNVAHPAS	GTKVDKPVVK	RENGRVPRPP	DCPKCPAPEM	240
LGGXSVFIFP	PKPKDTLLIA	RTPEVTCVVV	XLDPEDPEVQ	ISWFDGKQM	QTAKTQPREE	300
QFXXXYRVVS	VLPIGHQDWL	KGKQFTCKVN	NKXLXSPIER	TISKARGQAH	QPSVYVLPSS	360
REELSKNTVS	LTCLIKDFFP	PDIDVEWQSN	GQQEPESKYR	TPPQLDEDG	SYFLYSKLSV	420
DKSRWQRGDT	FICAVMHEAL	HNYHTQESLS	HSPGK			455

SEQ ID NO: 61 moltype = DNA length = 1359
 FEATURE Location/Qualifiers
 misc_feature 1..1359
 note = caninized mouse antibody
 misc_difference 724..726
 note = n is a, c, g, or t
 misc_difference 805..807
 note = n is a, c, g, or t
 misc_difference 901..909
 note = n is a, c, g, or t
 misc_difference 991..993
 note = n is a, c, g, or t
 misc_difference 997..999
 note = n is a, c, g, or t
 source 1..1359
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 61

gaggtgcagc	tggtgcagag	cgtggccgag	ctggtgaagc	ccggcgccag	cgtgaaggtg	60
agctgcaccg	tgagcggctt	caacatcaag	aacacctaca	tgcactgggt	gagacaggcc	120
cccgcaagg	gcctgcagtg	gatcggcaga	atcgaccccg	ccaacgtgaa	caccaagtac	180
gcccccaagt	tccagggcag	agccaccatc	accgcccaga	ccagcaccaa	caccgcctac	240
atgcagctga	gcagcctgag	agccgaggac	accgcccgtg	actactgcgc	cagaatctac	300
tacgactacg	acggcgacat	cgacgtgtgg	ggccagggca	ccctggtgac	cgtgagcagc	360
gccagcacca	ccgccccccg	cgtgttcccc	ctggccccca	gctgcccgcg	caccagcggc	420
agcacccgtg	ccctggcctg	cctggtgagc	ggctacttcc	ccgagcccgt	gaccgtgagc	480
tggaacagcg	gcagcctgac	cagcggcgctg	cacaccttcc	ccagcgtgct	gcagagcagc	540
ggcctgtaca	gcctgagcag	catggtgacc	gtgcccagca	gcagatggcc	cagcgagacc	600
ttcacctgca	acgtggccca	ccccgccagc	aagaccaagg	tggaacaagg	cgtggccaag	660
gagtgcgagt	gcaagtgcaa	ctgcaacaac	tgcccctgcc	ccggtcgccg	cctgctgggc	720
ggcnnnnagc	tggtcatctt	cccccccaag	cccaaggaca	tcctgggtgac	cgccagaacc	780
cccaccgtga	cctgctgtgt	ggtgnnnctg	gaccccgaga	accccgaggt	gcagatcagc	840
tggttctgtg	acagcaagca	ggtgcagacc	gccaacaccc	agcccagaga	ggagcagagc	900
nnnnnnnnnt	acagagtggt	gagcgtgctg	cccatcggcc	accaggactg	gctgagcggc	960
aagcagttca	agtgcaaggt	gaacaacaag	nnnctgnnna	gccccatcga	ggagatcatc	1020
agcaagaccc	ccggccaggc	ccaccagccc	aacgtgtacg	tgtgcccccc	cagcagagac	1080
gagatgagca	agaacaccgt	gaccctgacc	tgccctgtga	aggacttctt	cccccccgag	1140
atcgacgtgg	agtggcagag	caacggccag	caggagcccg	agagcaagta	cagaatgacc	1200
cccccccagc	tggaagagga	cggcagctac	ttcctgtaca	gcaagctgag	cgtggacaag	1260
agcagatggc	agagaggcga	caacctcatc	tgccgcgtga	tgccagaggg	cctgcacaac	1320
cactacaccc	agatcagcct	gagccacagc	cccgcaag			1359

SEQ ID NO: 62 moltype = AA length = 453
 FEATURE Location/Qualifiers
 REGION 1..453
 note = caninized mouse antibody
 VARIANT 242
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 269
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 301..303
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 331
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 333
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 source 1..453
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 62

EVQLVQSVAE	LVKPGASVKV	SCTVSGFNK	NTYMHWVRQA	PGKGLQWIGR	IDPANVNTKY	60
APKFQGRATI	TADTSTNTAY	QLSSSLRAED	TAVYYCARIY	YDYDGDIDVW	GQGTLTVTSS	120
ASTTAPSVFP	LAPSCGSTSG	STVALACLVS	GYFPEPVTVS	WNSGSLTSGV	HTFPSVLQSS	180
GLYSLSSMVT	VPSSRWPSET	FTCNVAHPAS	GTKVDKPVAK	ECECKNCNN	CPCPGCGLLG	240

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GXSVPFIFPPK	PKDILVTART	PTVTCVVVXL	DPENPEVQIS	WFVDSKQVQT	ANTQPREEQS	300
XXXYRVVSVL	PIGHQDWLSG	KQFKCKVNNK	XLXSPIEIII	SKTPGQAHQP	NVYVLPSPRD	360
EMSKNTVTLT	CLVKDFFPPE	IDVEWQSNQG	QEPESKYRMT	PPQLDEDGSY	FLYSKLSVDK	420
SRWQRGDTFI	CAVMHEALHN	HYTQISLSHS	PGK			453

SEQ ID NO: 63 moltype = DNA length = 1350
FEATURE Location/Qualifiers
misc_feature 1..1350
 note = caninized mouse antibody
misc_difference 715..717
 note = n is a, c, g, or t
misc_difference 796..798
 note = n is a, c, g, or t
misc_difference 892..900
 note = n is a, c, g, or t
misc_difference 982..984
 note = n is a, c, g, or t
misc_difference 988..990
 note = n is a, c, g, or t
source 1..1350
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 63

gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc	cggcgccgag	cgtgaaggtg	60
agctgcaagg	ccagcggcta	caccttcacc	acctacggca	tgagctgggt	gagacaggcc	120
cccggaagg	gcctgcagtg	gatgggcttg	atcaacatct	acagcggcat	gcccacctac	180
gccgacgact	tcaagggcag	attcgccctg	agcctggaca	ccagcaccag	caccgcctac	240
atggagctga	acagcctgag	agccgaggag	accgccgtgt	actactgcac	cagattcgac	300
ggccccgact	actggggcca	gggcacccctg	gtgaccgtga	gcagcgccag	caccaccgcc	360
cccagcgtgt	tccccctggc	ccccagctgc	ggcagcacca	gcggcagcac	cgtggccctg	420
gctgcctgg	tgagcggcta	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcggcct	gtacagcctg	540
agcagcatgg	tgaccgtgcc	cagcagcaga	tggcccagcg	agaccttcac	ctgcaacgtg	600
gcccaccccg	ccagcaagac	caaggtggac	aagcccgctg	ccaagagaga	gaacggcaga	660
gtgcccagac	cccccgactg	cccccaagtgc	cccgcctccg	agatgctggg	cggcnnnnagc	720
gtgttcacat	tcccccccaa	gcccgaaggac	accctgctga	tcgccagaac	ccccgaggtg	780
acctgcgtgg	tggtgnnnct	ggaccccgag	gaccccgagg	tgagatcag	ctgggttcgtg	840
gacggcaagc	agatgcagac	cgccaagacc	cagcccagag	aggagcagtg	cnnnnnnnnn	900
tacagagtgg	tgagcgtgct	gcccacggcg	caccaggact	ggctgaaggg	caagcagttc	960
acctgcaagg	tgaacaacaa	gnnnctgnnn	agccccatcg	agagaacat	cagcaaggcc	1020
agaggccagg	cccaccagcc	cagcgtgtac	gtgctgcccc	ccagcagaga	ggagctgagc	1080
aagaacaccg	tgagcctgac	ctgcctgatc	aaggacttct	tcccccccca	catcgacgtg	1140
gagtggcaga	gcaacggcca	gcaggagccc	gagagcaagt	acagaaccac	cccccccag	1200
ctggacgagg	acggcagcta	cttccctgtac	agcaagctga	gcgtggacaa	gagcagatgg	1260
cagagaggcg	acaccttcac	ctgcgcctg	atgcacgagg	ccctgcacaa	ccactacacc	1320
caggagagcc	tgagccacag	ccccggcaag				1350

SEQ ID NO: 64 moltype = AA length = 450
FEATURE Location/Qualifiers
REGION 1..450
 note = caninized mouse antibody
VARIANT 239
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
VARIANT 266
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
VARIANT 298..300
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
VARIANT 328
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
VARIANT 330
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
source 1..450
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 64

EVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGMPTY	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCTRPD	GPDYWGQGTL	VTSSASTTA	120
PSVFLPLAPC	GSTSGSTVAL	ACLVSGYFPE	PVTVSWNSGS	LTSGVHTFPS	VLQSSGLYSL	180
SSMVTVPSSR	WPSETFTCNV	AHPASKTKVD	KPVPKRENGR	VPRPPDCPKC	PAPEMLGGXS	240
VFIFPPKPKD	TLLIARTPEV	TCVVVXLDPE	DPEVQISWFV	DGKQMOTAKT	QPREEQFXXX	300
YRVVSVLPIG	HQDWLKGKQF	TCKVNNKXLX	SPIERTISKA	RGQAHQPSVY	VLPPSREELS	360

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KNTVSLTCLI KDFPPPDIDV EWQSNQQQEP ESKYRTTPPQ LDEGGSYFLY SKLSVDKSRW 420
 QRGDTFICAV MHEALHNHYT QESLSHSPGK 450

SEQ ID NO: 65 moltype = DNA length = 1344
 FEATURE Location/Qualifiers
 misc_feature 1..1344
 note = caninized mouse antibody
 misc_difference 709..711
 note = n is a, c, g, or t
 misc_difference 790..792
 note = n is a, c, g, or t
 misc_difference 886..894
 note = n is a, c, g, or t
 misc_difference 976..978
 note = n is a, c, g, or t
 misc_difference 982..984
 note = n is a, c, g, or t
 source 1..1344
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 65

gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc	cggcgccgag	cgtgaaggtg	60
agctgcaagg	ccagcgggta	caccttcacc	acctacggca	tgagctgggt	gagacaggcc	120
cccgcaagg	gcctgcagtg	gatgggctgg	atcaacatct	acagcggcat	gcccacctac	180
gccgacgact	tcaaggggcag	attcgccctg	agcctggaca	ccagcaccag	caccgcctac	240
atggagctga	acagcctgag	agccgaggac	accgcccgtg	actactgcac	cagattcgac	300
ggccccgact	actggggcca	gggcaccctg	gtgaccgtga	gcagcggccg	caccaccgcc	360
cccagcgtgt	tccccctggc	ccccagctgc	ggcagcacca	gcggcagcac	cgtggccctg	420
gcctgccttg	tgagcgggta	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcggcct	gtacagcctg	540
agcagcatgg	tgacgtgccc	cagcagcaga	tgggccagcg	agaccttcac	ctgcaacgtg	600
gcccccccg	ccagcaagac	caaggtggac	aagcccgtgg	ccaaggagtg	cagtgcaag	660
tgcaactgca	acaactgccc	ctgccccggc	tgcggcctcg	tgggcggcnn	nagcgtgttc	720
atcttcccc	ccaagcccaa	ggacatcctg	gtgaccgcca	gaacccccac	cgtgacctgc	780
gtggtggtgn	nnctggaccc	cgagaacccc	gaggtgcaga	tcagctgggt	cgtggacagc	840
aagcaggtgc	agaccgcca	caccagccc	agagaggagc	agagcnnnnn	nnntacaga	900
gtggtgagcg	tgctgcccc	cggccaccag	gactggctga	gcggcaagca	gttcaagtgc	960
aagtggaaca	acaagnnnct	gnnnagcccc	atcgaggaga	tcatcagcaa	gacccccggc	1020
caggcccacc	agcccaacgt	gtacgtgctg	ccccccagca	gagacgagat	gagcaagaac	1080
accgtgaccc	tgacctgccc	ggtgaaggac	ttcttcccc	ccgagatcga	cgtggagtgg	1140
cagagcaacg	gccagcagga	gcccagagagc	aagtacagaa	tgaccccccc	ccagctggac	1200
gaggacggca	gctacttccc	gtacagcaag	ctgagcgtgg	acaagagcag	atggcagaga	1260
ggcgacacct	tcatotgcgc	cgtgatgcac	gaggccctgc	acaaccacta	caccagatc	1320
agcctgagcc	acagccccgg	caag				1344

SEQ ID NO: 66 moltype = AA length = 448
 FEATURE Location/Qualifiers
 REGION 1..448
 note = caninized mouse antibody
 VARIANT 237
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 264
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 296..298
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 326
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 VARIANT 328
 note = misc_feature - Xaa can be any naturally occurring
 amino acid
 source 1..448
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 66

EVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INISGMPTT	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCTRFD	GPDYWGQGL	VTVSSASTTA	120
PSVFLAPSC	GSTSGSTVAL	ACLVSGYFPE	PVTVSWNSGS	LTSGVHTFPS	VLQSSGLYSL	180
SSMVTVPSSR	WPSETFTCNV	AHPASKTKVD	KPVAKECECK	CNCNCPCPG	CGLLGXSVF	240
IFPPKPKDIL	VTARTPTVTC	VVVXLDPENP	EVQISWFVDS	KQVQTANTQP	RERQSSXXYR	300
VVSVLPIGHQ	DWLSGKQFKC	KVNNKXLP	IEEIIKSTPG	QAHQPNVYVL	PPSRDEMSKN	360
TVTLTCLVKD	FPFPEIDVEW	QSNQQQEPES	KYRMTPPQLD	EDGSYFLYSK	LSVDKSRWQR	420
GDTFICAVMH	EALHNHYTQI	SLSHSPGK				448

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SEQ ID NO: 67      moltype = DNA length = 1338
FEATURE           Location/Qualifiers
misc_feature       1..1338
                   note = caninized mouse antibody
source            1..1338
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 67
gaggtgcagc tgggtgcagag cggcgccgac ctggtgaagc cggcgccgac cgtgagactg 60
agctgcgtgg ccagcggtga cacttcacc acctacggca tgagctgggt gagacaggcc 120
cccgcaagg gcctgcagtg gatgggctgg atcaacatct acagcgccat cccacctac 180
gccgacgact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac 240
ctgcagctga acagcctgag agccgaggac accgccgtgt actactgcac cagattcgac 300
ggccccgact actggggcca gggcaccctg gtgaccgtga gcagcgccag caccaccgcc 360
ccccgctgt tccccctggc cccagctgc ggcagcacca gcgcagcac cgtggccctg 420
gcctgcctgg tgagcggtga cttccccgag cccgtgaccg tgagctggaa cagcgccagc 480
ctgaccagcg gcgtgcacac cttccccagc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcatgg tgaccgtgac cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gtgcaccccg ccagcaaac caaggtggac aagcccgtgt tcaacgagtg cagatgcacc 660
gacaccccc cctgccccgt gcccgagccc ctgggcccgc ccagcgtgct gatcttcccc 720
cccaagccca aggacatcct gagaatcacc agaaccccc aggtgacctg cgtggtgctg 780
gacctgggca gagaggaccc cgaggtgcag atcagctggt tcgtggacgg caaggaggtg 840
cacaccgcca agacccagcc cagagagcag cagttcaacg gcacctacag agtggtgagc 900
gtgctgcccc tcgagcacca ggactggctg accggcaagg agttcaagtg cagagtgaac 960
cacatcgacc tgcaccagcc catcgagaga acctacagca aggccagagg cagagccacc 1020
aagccccagc tgtacgtgct gccccccagc cccaaggagc tgagcagcag cgacaccgtg 1080
agcatcacct gctgatcaaa ggacttctac ccccccgaca tcgacgtgga gtggcagagc 1140
aacggccagc agagagccga gagaaagcac agaattgaccc cccccagct ggacgaggac 1200
ggcagctact tcctgtacac caagctgagc gtggacaaga gcagatggca gcaggggcag 1260
cccttcacct cgccgctgat gcacgagacc ctgcagaacc actacaccga cctgagcctg 1320
agccacagcc cgggcaag 1338

SEQ ID NO: 68      moltype = AA length = 446
FEATURE           Location/Qualifiers
REGION           1..446
                   note = caninized mouse antibody
source            1..446
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 68
EVQLVQSGGD LVKPGGSVRL SCVASGYTFT TYGMSWVRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFTF SLD TAKNTAY LQLNSLRAED TAVYYCTREFD GPDYWGQGT LVTSSASTTA 120
PSVFPPLPSC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLHSL 180
SSMVTVPSSR WPSETFTCNV VHPASNTKVD KPVFNECRCT DTPPCPVPEP LGGPSVLIFP 240
PKPKDILRIT RTPVETCVVL DLGREDEPEVQ ISWFDVGKEV HTAKTQGREQ QFNGTYRVVS 300
VLPIEHQDWL TGKEFKCRVN HIDLPSPIER TISKARGRAH KPSVYVLPSS PKELSSSDTV 360
SITCLIKDFY PPDIDVEVQS NGQQEPERKH RMTPPQLDED GSYFLYSKLS VDKSRWQQGD 420
PPTCAVMHET LQNHYYTDL LSHSPGK 446

SEQ ID NO: 69      moltype = DNA length = 1338
FEATURE           Location/Qualifiers
misc_feature       1..1338
                   note = caninized mouse antibody
source            1..1338
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 69
gaggtgcagc tgggtgcagag cggcgccgac ctggtgaagc cggcgccgac cgtgagactg 60
agctgcgtgg ccagcggtga cacttcacc acctacggca tgagctgggt gagacaggcc 120
cccgcaagg gcctgcagtg gatgggctgg atcaacatct acagcgccat cccacctac 180
gccgacgact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac 240
ctgcagctga acagcctgag agccgaggac accgccgtgt actactgcac cagattcgac 300
ggccccgact actggggcca gggcaccctg gtgaccgtga gcagcgccag caccaccgcc 360
ccccgctgt tccccctggc cccagctgc ggcagcacca gcgcagcac cgtggccctg 420
gcctgcctgg tgagcggtga cttccccgag cccgtgaccg tgagctggaa cagcgccagc 480
ctgaccagcg gcgtgcacac cttccccagc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcacgg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gtgcaccccg ccagcaaac caaggtggac aagcccgtgc ccaaggagag cactgcaag 660
tgcatcagcc cctgccccgt gcccgagagc ctgggcccgc ccagcgtgtt catcttcccc 720
cccaagccca aggacatcct gagaatcacc agaaccccc agatcacctg cgtggtgctg 780
gacctgggca gagaggaccc cgaggtgcag atcagctggt tcgtggacgg caaggaggtg 840
cacaccgcca agacccagcc cagagagcag cagttcaaca gcacctacag agtggtgagc 900
gtgctgcccc tcgagcacca ggactggctg accggcaagg agttcaagtg cagagtgaac 960
cacatcgccc tgcaccagcc catcgagaga acctacagca aggccagagg ccaggccacc 1020
cagcccgagc tgtacgtgct gccccccagc cccaaggagc tgagcagcag cgacaccgtg 1080

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accctgacct gcctgatcaa ggaattcttc ccccccgaga tcgacgtgga gtggcagagc 1140
aacggccagc ccgagcccga gagcaagtag cacaccaccg cccccagct ggacgaggac 1200
ggcagctact tcctgtacag caagctgagc gtggacaaga gcagatggca gcagggcgac 1260
accttcacct gcgccgtgat gcacgaggcc ctgcagaacc actacaccga cctgagcctg 1320
agccacagcc ccggcaag                                     1338

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SEQ ID NO: 70      moltype = AA length = 446
FEATURE           Location/Qualifiers
REGION            1..446
                  note = caninized mouse antibody
source            1..446
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 70
EVQLVQSGGD LVKPGGSVRL SCVASGYTFT TYGMSWVRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFTF SLDTAKNTAY LQLNSLRAED TAVYYCTREFD GPDYWGQGTL VTVSSASTTA 120
PSVFPPLAPSC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLYSL 180
SSTVTVPSSR WPSETFTCNV VHPASNTKVD KVPVKESTCK CISPCVPFES LGGPSVFIFP 240
PKPKDILRIT RTPETTCVVL DLGREDPEVQ ISWFDGKEV HTAKTQPREQ QFNSTYRVVS 300
VLPIDHEDWL TGKEFKCRVN HIGLPSPIER TISKARGQAH QPSVYVLPPS PKELSSSDTV 360
TLTCLIKDFP PPEIDVEWQS NGQPEPESKY HTTAPQLDED GSYFLYSKLS VDKSRWQQGD 420
TFTCAVMHEA LQNHYTDLNL SHSPGK                                     446

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SEQ ID NO: 71      moltype = DNA length = 672
FEATURE           Location/Qualifiers
misc_feature       1..672
                  note = caninized mouse antibody
source            1..672
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 71
gacatcgtag tgaccagac cccctgagc ctgagcgtag gcccgcgga gcccgccagc 60
atgagctgca agagcagcca gagcctgctg aacagcgtag accagaagaa ctacctggcc 120
tggtacagac agaagcccgg ccagagcccc caggtgctgg tgaacttcgc cagcgccaga 180
gtgagcggcg tgcccagacg attcatcgcc agcggcagcg gcaccgactt caccctgaga 240
atcagcagag tggaggccga cgacctgggc gtgtactact gccagcagta ctacagcacc 300
cccctgacct tcggccaggg caccagagtg gagctgaaga gaaacgacgc ccagcccgcc 360
gtgtacctgt tccagccagg ccccgaccag ctgcacaccg gcagcgccag cgtgggtgtg 420
ctgctgaaca gcttctaccc caaggacatc aacgtgaagt ggaaggtgga cggcgtgatc 480
caggacaccg gcatccagga gagcgtgacc gagcaggaca gcaaggacag cactacagc 540
ctgagcagca cccctgaccat gagcagcacc gagtacctga gccacgagct gtacagctgc 600
gagatcaccg acaagagcct gccagcacc ctgatcaaga gcttcacagag aagcgagtg 660
cagagagtgg ac                                     672

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```

SEQ ID NO: 72      moltype = AA length = 224
FEATURE           Location/Qualifiers
REGION            1..224
                  note = caninized mouse antibody
source            1..224
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 72
DIVMTQTPLS LSVSPGEPAS MSCKSSQSLN NSVNQKNYLA WYRQKPGQSP QVLVYFASAR 60
VSGVPDRFIG SSGTDFTLR ISRVADDLG VYQCQYFST PLTFGQGTGL ELKRNDAPQA 120
VYLFQSPSPDQ LHGSASVVC LLNSFYPKDI NVKWKVDGVI QDTGIQESVT EQDSKDSTYS 180
LSSTLTMSST EYLSHELYSC EITHKSLPST LIKSPQRSEC QRVD                                     224

```

```

SEQ ID NO: 73      moltype = DNA length = 1353
FEATURE           Location/Qualifiers
misc_feature       1..1353
                  note = caninized mouse antibody
source            1..1353
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 73
gaggtgcagc tgggtgcagag cggcgccgac ctggtgaagc ccggcgccag cgtgagactg 60
agctgcgtgg ccagcggtct caacatcaag aacacctaca tgcactgggt gagacaggcc 120
cccgcaaggg gctcgagtg gatcgccaga atcgcccccg ccaacgtgga caccagtag 180
gcccccaagt tccagggcaa ggccaccatc agcgccgaca ccgccaagaa caccgcctac 240
atgacgtgta acagcctgag agccgaggac accgcccgtg actactgcgt gctgatctac 300
tacgactacg ccggcgacat cgacgtgtgg ggccagggca cctgggtgac cgtgagcagc 360
gccagcacca accgcccag cgtgttcccc ctggcccccga gctgcggcag caccagcgcc 420
agcaccgtgg cctgggctg cctgggtgagc ggctacttcc ccgagccgtg gaccgtgagc 480
tggaacagcg gcagcctgac cagcgcgctg cacaccttcc ccagcgtgct gcagagcagc 540
ggcctgcaca gctcgagcag catggtgacc gtgcccagca gcagatggcc cagcgagacc 600
ttcacctgca acgtggtgca ccccgccagc aacaccaagg tggacaagcc cgtgttcaac 660

```

-continued

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gagtgcagat gcaccgacac cccccctgc cccgtgccg agccctggg cgccccagc 720
gtgctgatct tccccccaa gcccaaggac atcctgagaa tcaccagaac ccccgagggtg 780
acctgcgtgg tgctggacct gggcagagag gaccccgagg tgcagatcag ctggttcgtg 840
gacggcaagg aggtgcacac cgccaagacc cagagcagag agcagcagtt caacggcacc 900
tacagagtgg tgagcgtgct gcccatcgag caccaggact ggctgaccgg caaggagtgc 960
aagtgcagag tgaaccacat cgacctgccc agcccatcg agagaacct cagcaaggcc 1020
agaggcagag cccacaagcc cagcgtgtac gtgctgcccc ccagcccaa ggagctgagc 1080
agcagcgaca ccgtgagcat cacctgcctg atcaaggact tctaccccc cgacatcgac 1140
gtggagtggc agagcaacgg ccagcaggag cccgagagaa agcacagaat gacccccccc 1200
cagctggacg aggcaggcag ctacttctg tacagcaagc tgagcgtgga caagagcaga 1260
tggcagcagg gcgacccctt cacctgcgcc gtgatgcacg agaccctgca gaaccactac 1320
accgacctga gcctgagcca cagccccggc aag 1353

```

```

SEQ ID NO: 74      moltype = AA length = 451
FEATURE           Location/Qualifiers
REGION            1..451
                  note = caninized mouse antibody
source            1..451
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 74
EVQLVQSGGD LVKPGGSVRL SCVASGFNIK NTYMHWVRQA PGKGLQWIGR IAPANVDTKY 60
APKFQGGKATI SADTAKNTAY MQLNSLRAED TAVYYCVLIY YDYGDDIDVW GQGTLTIVSS 120
ASTTAPSVFP LAPSCGSTSG STVALACLVS GYFPEPVTVS WNSGSLTSGV HTPPSVLQSS 180
GLHSLSSMVT VPSSRWPSSET FTCNVVHPAS NTKVDKPVFN ECRCTDTPPC PVPEPLGGPS 240
VLIFPPKPKD ILRITRTPEV TCVVLDLGRE DPEVQISWFV DGKEVHTAKT QSREQQFNST 300
YRVVSVLPLE HQDWLTGKEF KCRVNHIDLP SPIERTISKA RGRAHKPSVY VLPSPKELS 360
SSDTSVITCL IKDFYPPDID VEWQSNQQQE PERKHRMTPP QLDEDEGSYFL YSKLSVDKSR 420
WQQGDPFTCA VMHETLQNHY TDLSLSHSPG K 451

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```

SEQ ID NO: 75      moltype = DNA length = 1353
FEATURE           Location/Qualifiers
misc_feature       1..1353
                  note = caninized mouse antibody
source            1..1353
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 75
gagggtgcagc tggtgcagag cgggcgccgac ctggtgaagc ccggcgccgag cgtgagactg 60
agctgcgtgg ccagcggtct caacatcaag aacacctaca tgcactgggt gagacaggcc 120
ccggcaagg gcctgcagt gatcggcaga atcgcccccg ccaacgtgga caccaagtac 180
gcccccaagt tccagggcaa ggcaccatc agcgccgaca ccgccaagaa caccgcctac 240
atgcagctga acagcctgag agccgaggac accgccgtgt actactgctg gctgatctac 300
tacgactacg acggcgacat cgacgtgtgg ggccagggca ccctgggtgac cgtgagcagc 360
gcgcagacca ccgcccccg cgtgttcccc ctggccccca gctgcggcag caccagcggc 420
agcaccgtgg ccttgccctg cctgggtgagc ggctacttcc ccgagcccg gaccgtgagc 480
tggaacacgg gcagcctgac cagcgcgctg cacaccttcc ccagcgtgct gcagagcagc 540
ggcctgtaca gcctgagcag caccgtgacc gtgcccagca gcagatggcc cagcgagacc 600
ttcacctgca acgtgtgtga ccccgccagc aacaccaagg tggacaagcc cgtgcccagg 660
gagagcacct gcaagtgcac cagccccctgc cccgtgcccg agagcctggg cgccccagc 720
gtgttcactt tccccccaa gcccaaggac atcctgagaa tcaccagaac ccccgagatc 780
acctgcgtgg tgctggacct gggcagagag gaccccgagg tgcagatcag ctggttcgtg 840
gacggcaagg aggtgcacac cgccaagacc cagcccagag agcagcagtt caacagcacc 900
tacagagtgg tgagcgtgct gcccatcgag caccaggact ggctgaccgg caaggagtgc 960
aagtgcagag tgaaccacat cgccctgccc agcccatcg agagaacct cagcaaggcc 1020
agaggccagg cccaccagcc cagcgtgtac gtgctgcccc ccagcccaa ggagctgagc 1080
agcagcgaca ccgtgacct gacctgcctg atcaaggact tcttcccccc cgagatcgac 1140
gtggagtggc agagcaacgg ccagcccgag cccgagagca agtaccacac caccgcccc 1200
cagctggacg aggcaggcag ctacttctg tacagcaagc tgagcgtgga caagagcaga 1260
tggcagcagg gcgacacctt cacctgcgcc gtgatgcacg aggccttgcga gaaccactac 1320
accgacctga gcctgagcca cagccccggc aag 1353

```

```

SEQ ID NO: 76      moltype = AA length = 451
FEATURE           Location/Qualifiers
REGION            1..451
                  note = caninized mouse antibody
source            1..451
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 76
EVQLVQSGGD LVKPGGSVRL SCVASGFNIK NTYMHWVRQA PGKGLQWIGR IAPANVDTKY 60
APKFQGGKATI SADTAKNTAY MQLNSLRAED TAVYYCVLIY YDYGDDIDVW GQGTLTIVSS 120
ASTTAPSVFP LAPSCGSTSG STVALACLVS GYFPEPVTVS WNSGSLTSGV HTPPSVLQSS 180
GLYSLSSVT VPSSRWPSSET FTCNVVHPAS NTKVDKPVFN ESTCKCISPC PVPEPLGGPS 240
VLIFPPKPKD ILRITRTPEI TCVVLDLGRE DPEVQISWFV DGKEVHTAKT QPREQQFNST 300
YRVVSVLPLE HQDWLTGKEF KCRVNHIGLP SPIERTISKA RGAHQPSVY VLPSPKELS 360

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SDTDTLTLCL IKDFPPEID VEWQSNQOPE PESKYHTTAP QLDEDSYFL YSKLSVDKSR 420
WQGGDTFTCA VMHEALQNHY TDLSLSHSPG K 451

```

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SEQ ID NO: 77      moltype = DNA length = 654
FEATURE           Location/Qualifiers
misc_feature       1..654
                   note = caninized mouse antibody
source            1..654
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 77
gacatcgtga tgacccagac ccccttgagc ctgagcgtga gcctgggcca gcccgcagc 60
atcagctgcc acgccagcca gaacatcaac gtgtggctga gctgggtacag acagaagccc 120
ggccagatcc cccagctgct gatctacaag gccagccacc tgcacacgg cgtggccgac 180
agattcagcg gcagcggcag cggcaccgac ttcaccctga gaatcagcag agtggaggcc 240
gacgacgccc gcgtgtacta ctgccagcag gccagagct ggccctgac cttcgccag 300
ggcaccgaag tggagatcaa gagaaacgac gccagcccc cctgtacct gttccagccc 360
agcccccacc agctgcacac cggcagcgcc agcgtggtgt gcctgtgtaa cagctttcac 420
cccaaggaca tcaacgtgaa gtggaagggt gacggcgtga tccaggacac cgccatccag 480
gagagcgtga ccgcaaggac agcacctaca gcctgagcag caccctgacc 540
atgagcagca ccgagtcact gagccacgag ctgtacagct gcgagatcac ccacaagagc 600
ctgccagca cctgatcaaa gagcttccag agaagcgagt gccagagagt ggac 654

```

```

SEQ ID NO: 78      moltype = AA length = 218
FEATURE           Location/Qualifiers
REGION           1..218
                   note = caninized mouse antibody
source            1..218
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 78
DIVMTQTPLS LSVSLGEPAS ISCHASQNIN VWLSWYRQKP GQIPQLLIYK ASHLHTGVDP 60
RFSGSGSGTD FTLRISRVEA DDAGVYVCQQ QGSWPLTFGQ GTKVEIKRND AQPAVYLFQP 120
SPDQLHTGSA SVVCLLSFY PKDINVKWKV DGVIQDTGIQ ESVTEQDSKD STYLSSTLT 180
MSSTEYLSHE LYSCEITHKS LPSTLIKSFQ RSECQRVD 218

```

```

SEQ ID NO: 79      moltype = DNA length = 1359
FEATURE           Location/Qualifiers
misc_feature       1..1359
                   note = caninized mouse antibody
source            1..1359
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 79
gaggtgcagc tgggtgcagag cggcgccgac ctggtgaagc ccggcggcag cgtgagactg 60
agctgcgtgg ccagcggcta cacttcacc agatacaaca tgcactgggt gagacaggcc 120
cccggaagg gcctgcagtg gatcgccacc atctaccccg gctacggcga caccagctac 180
aaccagaagt tcaagggcaa ggcacccttg agcgtggaca tcgccaagaa caccgcctac 240
atgcagctga acagcctgag agccgaggac accgcgtgt actctgcag cagagagttc 300
gccgacgact accccatccc ccccttcgac tactggggcc agggcaccct ggtgaccgtg 360
agcagcgcca gcaccaccgc ccccgagctg tccccctgg ccccgagctg cggcagcacc 420
agcgccagca ccgtggccct ggcctgcctg gtgagcggct acttcccga gcccgtagcc 480
gtgagctgga acagcggcag cctgaccagc ggcgtgcaca ccttcccag cgtgctgcag 540
agcagcggcc tgcacagcct gagcagcatg gtgaccgtgc ccagcagcag atggcccagc 600
gagaccttca cctgcaacgt ggtgcacccc gccagcaaca ccaaggtgga caagccctg 660
tcaacagagt gcagatgcac cgacaccccc cctgccccg tgcccagacc cctgggcggc 720
cccagcgtgc tgaatttccc ccccaagccc aaggacatcc tgagaatcac cagaaccccc 780
gaggtgacct gcgtggtgct ggacctgggc agagaggacc ccgaggtgca gatcagctgg 840
ttcgtggacg gcaaggaggt gcacaccgcc aagaccaga gcagagagca gcagttcaac 900
ggcaccctac gagtggtagc cgtgctgccc atcgagcacc aggactggct gaccggcaag 960
gagttcaagt gcagagtgaa ccacatcgac ctgccagacc ccatcgagag aaccatcagc 1020
aaggccagag gcagagccca caagcccagc gtgtacgtgc tgccccccag ccccaaggag 1080
ctgagcagca gcgacaccgt gagcaccacc tgcctgatca aggacttcta cccccccgac 1140
atcgacgtgg agtggcagag caacggccag caggagcccg agagaaagca cagaatgacc 1200
ccccccagc tggacgagga cggcagctac ttctgtaca gcaagctgag cgtggacaag 1260
agcagatggc agcagggcga ccccttcacc tgcgcgtga tgcacgagac cctgcagaac 1320
cactacaccg acctgagcct gagccacagc cccggcaag 1359

```

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SEQ ID NO: 80      moltype = AA length = 453
FEATURE           Location/Qualifiers
REGION           1..453
                   note = caninized mouse antibody
source            1..453
                   mol_type = protein
                   organism = synthetic construct

```

```

SEQUENCE: 80

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EVQLVQSGGD	LVKPGGSVRL	SCVASGYTFT	RYNMHWVRQA	PGKGLQWIGT	IYPGYGDTSY	60
NQKFKGKATL	SVDIAKNTAY	MQLNSLRAED	TAVYFCSREF	ADDYPIPPFD	YWQGGLVTV	120
SSASTTAPSV	FPLAPSCGST	SGSTVALACL	VSGYFPEPVT	VSWNSGSLTS	GVHTFPSVLQ	180
SSGLHSLSSM	VTVPSSRWPS	ETFTCNVVHP	ASNTKVDKPV	FNECRCTDTP	PCVPPEPLGG	240
PSVLIFPPKP	KDILRITRTP	EVTCTVLDLG	REDPEVQISW	FVDGKEVHTA	KTQSRQQQFN	300
GTYYRVSVLP	IEHQDWLTGK	EFKCRVNHID	LPSPIERTIS	KARGRAHKPS	VYVLPPSPKE	360
LSSSDTVSIT	CLIKDFYPPD	IDVEWQSNQO	QEPERKHRMT	PPQLDEDEGSY	FLYSKLSVDK	420
SRWQQGDPFT	CAVMHETLQN	HYTDLSLSHS	PGK			453

SEQ ID NO: 81 moltype = DNA length = 1359
 FEATURE Location/Qualifiers
 misc_feature 1..1359
 note = caninized mouse antibody
 source 1..1359
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 81

gaggtgcagc	tggtgcagag	cggcgccgac	ctggtgaagc	cggcgccgag	cgtgagactg	60
agctgcgtgg	ccagcggtga	caccttcacc	agatacaaca	tgcactgggt	gagacaggcc	120
cccggaaggg	gcctgcagtg	gacgcggcacc	atctaccccg	gctacggcga	caccagctac	180
aaccagaagt	tcaagggcaa	ggccaccctg	agcgtggaca	tcgccaagaa	caccgcctac	240
atgcagctga	acagcctgag	agccgaggag	accgccgtgt	acttctgcag	cagagagttc	300
gccgacgact	accccatccc	cccttcgcag	tactggggcc	agggcacccc	ggtgaccgtg	360
agcagcgcca	gcaccaccgc	ccccagcgtg	ttccccctgg	ccccagactg	cggcagcacc	420
agcggcagca	ccgtggccct	ggcctgcctg	gtgagcggct	acttccccga	gcccgtagcc	480
gtgagctgga	acagcggcag	cctgaccagc	ggcgtgcaca	ccttccccag	cgtgctgcag	540
agcagcgccc	tgtacagcct	gagcagcacc	gtgaccgtgc	ccagcagcag	atggcccagc	600
gagaccttca	cctgcaacgt	ggtgcacccc	gccagcaaca	ccaaggtgga	caagcccgtg	660
cccaaggaga	gcacctgcaa	gtgcacacgc	ccctgccccg	tgcccgagag	cctgggcccgc	720
cccagcgtgt	tcattctccc	ccccaaagccc	aaggacatcc	tgagaatcac	cagaaccccc	780
gagattcacct	cgctggtgct	ggacctgggc	agagaggacc	ccgaggtgca	gacagctggg	840
ttcgtggacg	gcaaggaggt	gcacaccgcc	aagaccccag	ccagagagca	gcagttcaac	900
agcacctaca	gagtggtgag	cgtgctgccc	atcgagcacc	aggactggct	gaccggcaag	960
gagttcaagt	gcagagtgaa	ccacatcgcc	ctgcccagcc	ccatcgagag	aaccatcagc	1020
aaggccagag	gccaggccca	ccagcccagc	gtgtacgtgc	tgccccccag	ccccaaaggag	1080
ctgagcagca	gcgacaccgt	gaccctgacc	tgcctgatca	aggactctct	cccccccag	1140
atcgacgtgg	agtggcagag	caacggccag	cccagcccgc	agagcaagta	ccacaccacc	1200
gccccccagc	tggacagaga	cggcagctac	ttcctgtaca	gcaagctgag	cgtggacaag	1260
agcagatggc	agcagggcga	caccttcacc	tgcgcctgta	tgcacgaggc	cctgcagaa	1320
cactacaccg	acctgagcct	gagccacagc	cccggcaag			1359

SEQ ID NO: 82 moltype = AA length = 453
 FEATURE Location/Qualifiers
 REGION 1..453
 note = caninized mouse antibody
 source 1..453
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 82

EVQLVQSGGD	LVKPGGSVRL	SCVASGYTFT	RYNMHWVRQA	PGKGLQWIGT	IYPGYGDTSY	60
NQKFKGKATL	SVDIAKNTAY	MQLNSLRAED	TAVYFCSREF	ADDYPIPPFD	YWQGGLVTV	120
SSASTTAPSV	FPLAPSCGST	SGSTVALACL	VSGYFPEPVT	VSWNSGSLTS	GVHTFPSVLQ	180
SSGLYLSST	VTVPSSRWPS	ETFTCNVVHP	ASNTKVDKPV	PKESTCKCIS	PCVPPEPLGG	240
PSVFIFFPKP	KDILRITRTP	EITCVVLDLG	REDPEVQISW	FVDGKEVHTA	KTQPREQQFN	300
STYYRVSVLP	IEHQDWLTGK	EFKCRVNHIG	LPSPIERTIS	KARGQAHPQS	VYVLPPSPKE	360
LSSSDTVTLT	CLIKDFFPPE	IDVEWQSNQO	PEPESKYHTT	PPQLDEDEGSY	FLYSKLSVDK	420
SRWQQGDTFT	CAVMHEALQN	HYTDLSLSHS	PGK			453

SEQ ID NO: 83 moltype = DNA length = 666
 FEATURE Location/Qualifiers
 misc_feature 1..666
 note = caninized mouse antibody
 source 1..666
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 83

gacatcgtga	tgaccacagc	ccccctgagc	ctgcccgtga	gcctgggcga	gcccggccagc	60
atcagctgca	gaagcagcca	gaacatcgtg	cacagcaacg	gcaacaccta	cctggagtg	120
tacagacaga	agcccgccca	gagcccccag	ctgctgatct	acaaggtgag	caacagattc	180
agcggcgtgc	ccgacagatt	cagcggcagc	ggcagcggca	ccgacttcac	cctgagaatc	240
agcagagtgg	aggccgacga	cccgccgtg	tactactgct	tccagggcag	ccacgtgccc	300
tacaccttcg	gccaggccac	caaggtggag	atcaagagag	acgcccagcc	cgcctgtgac	360
ctgttcacgc	ccagccccga	ccagctgcac	accggcagcg	ccagcgtggg	gtgcctgctg	420
aacagcttct	accccaagga	catcaactgt	aagtgggaag	tggacggcgt	gatccaggac	480
accgcctacc	aggagagcgt	gaccgagcag	gacagcaagg	acagcaccta	cagcctgagc	540
agcaccctga	ccatgagcag	caccgagtag	ctgagccacg	agctgtacag	ctgcgagatc	600

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accacacaaga gcctgcccag caccctgata aagagcttcc agagaagcga gtgccagaga 660
gtggac 666

```

```

SEQ ID NO: 84      moltype = AA length = 222
FEATURE           Location/Qualifiers
REGION            1..222
                  note = caninized mouse antibody
source            1..222
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 84
DIVMTQTPLS LPVSLGEPAS ISCRSSQNIV HSENGNTYLEW YRQKPGQSPQ LLIYKVSNR 60
SGVPDRFSGS GSGTDFTLRI SRVEADDAGV YYCFQGSHPV YTFGQGTKVE IKRDAQPAVY 120
LFQPSPDQLH TGSASVVCLL NSFYPKDINV KWKVDGVIQD TGIQESVTEQ DSKDSTYSLS 180
STLTMSSTEY LSHELYSCIE THKSLPSTLI KSFQRSECQR VD 222

```

```

SEQ ID NO: 85      moltype = DNA length = 1338
FEATURE           Location/Qualifiers
misc_feature       1..1338
                  note = caninized mouse antibody
source            1..1338
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 85
gaggtgcagc tggtgcagag cggccccggc ctggtgaagc ccagccagag cctgagcctg 60
acctgcgtgg tgagcggctt cagcctgacc agctacggcg tgcactgggt gagacaggcc 120
cccgcaagg gcctgcagtg gatgggctgg atcaacatct acagcgccat cccacacctac 180
gccgacgact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac 240
ctgcagctga gcagcctgag agccgaggac accgcccgtg actactgcgc cagattcgac 300
ggccccgact actggggcca gggcacccctg gtgaccgtga gcagcgccag caccaccgcc 360
cccagcgtgt tcccctggc cccagctgc ggcagcacca gcgcagcac cgtggccctg 420
gcctgcctgg tgagcggcta cttcccagc cccgtgaccg tgagctggaa cagcggcagc 480
ctgaccagcg gcgtgcacac cttcccagc gtgctgcaga gcagcgccct gcacagcctg 540
agcagcatgg tgaccgtgac cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gtgcaccccg ccagcaaac caaggtggac aagcccgtgt tcaacgagtg cagatgcacc 660
gacaccccc cctgcccctg gcccgagccc ctgggcccgc ccagcgtgct gatcttcccc 720
cccaagcccc agggacatct gagaatcacc agaaccccc aggtgacctg cgtggtgctg 780
gacctgggca gagaggaccc cgaggtgcag atcagctggt tcgtggacgg caaggagggtg 840
cacaccgcca agaccagag cagagagcag cagttcaacg gcacctacag agtggtgagc 900
gtgctgcccc tcgagcacca ggaactggctg accggcaagg agttcaagtg cagagtgaac 960
cacatcgacc tgcccagccc catcgagaga accatcagca aggccagagg cagagccacc 1020
aagcccagcg tgtacgtgct gccccccagc cccaaggagc tgagcagcag cgacaccgtg 1080
agcatcacct gctgatcaa ggaacttcac cccccgaca tcgacgtgga gtggcagagc 1140
aacggccagc agggagcccg gagaaagcac agaattgacc cccccagct ggacgaggac 1200
ggcagctact tcctgtacag caagctgagc gtggacaaga gcagatggca gcaggggcagc 1260
cccttcacct gcgcctgat gacagagacc ctgcagaacc actacaccga cctgagcctg 1320
agccacagcc ccggcaag 1338

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SEQ ID NO: 86      moltype = AA length = 446
FEATURE           Location/Qualifiers
REGION            1..446
                  note = caninized mouse antibody
source            1..446
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 86
EVQLVQSGPG LVKPSQSLSL TCVVSGFSLT SYGVHWRQA PGKGLQWMGW INIYSGIPT 60
ADDFKGRFTF SLDTAKNITAY LQLSSLRAED TAVYYCARFD GPDYWGQGT VVSSASTTA 120
PSVFPPLAPSC GSTSGSTVAL ACLVSGYPPE PVTVSWNSGS LTSGVHTFPS VLQSSGLHSL 180
SSMVTVPSSR WPSETFTCNV VHPASNTKVD KPVFNECRCT DTPPCPVPEP LGGPSVLIFP 240
PKPKDILRIT RTPEVTCVVL DLGREDPEVQ ISWFDGKEV HTAKTQSRQ QFNGTYRVVS 300
VLPIEQDWL TGKEFKCRVN HIDLPSPIER TISKARGRAH KPSVYVLPSS PKELSSSDTV 360
SITCLIKDFY PPDIDVEWQS NGQQEPERKH RMTPPQLDED GSYFLYSKLS VDKSRWQQGD 420
PFTCAVMHET LQNHYTDLNL SHSPGK 446

```

```

SEQ ID NO: 87      moltype = DNA length = 1338
FEATURE           Location/Qualifiers
misc_feature       1..1338
                  note = caninized mouse antibody
source            1..1338
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 87
gaggtgcagc tggtgcagag cggccccggc ctggtgaagc ccagccagag cctgagcctg 60
acctgcgtgg tgagcggctt cagcctgacc agctacggcg tgcactgggt gagacaggcc 120
cccgcaagg gcctgcagtg gatgggctgg atcaacatct acagcgccat cccacacctac 180

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gccgacgact tcaagggcag attcaccttc agcctggaca ccgccaagaa caccgcctac 240
ctgcagctga gcagcctgag agccgaggac accgcctgt actactgcgc cagattcgac 300
ggccccgact actggggcca gggcacctct gtgaccgtga gcagcgccag caccaccgcc 360
cccagcgtgt tccccctggc ccccagctgc ggcagcacca gcggcagcac cgtggccctg 420
gcctgcctgg tgagcggteta cttccccgag cccgtgaccg tgagctggaa cagcggcagc 480
ctgaccagcg gcgtgcacac cttccccagc gtgctgcaga gcagcgccct gtacagcctg 540
agcagcaccg tgaccgtgcc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gtgcaccccg ccagcaacac caaggtggac aagcccgctg ccaaggagag cacctgcaag 660
tgcatcagcc cctgccccct gcccagagag ctggggcgcc ccagcgtgt catcttcccc 720
cccaagccca aggacatcct gagaatcacc agaaccctcg agatcacctg cgtggtgctg 780
gacctgggca gagaggcccc cgaggtgcag atcagctggt tcgtggacgg caaggaggtg 840
cacaccgcca agaccagcc cagagagcag cagttcaaca gcacctacag agtggtgagc 900
gtgctgcccc tcgagcacca ggaactggctg accggcaagg agttcaagt cagagtgaac 960
cacatcgccc tgcccagccc catcgagaga accatcagca aggccagagg ccaggccccc 1020
cagccccagc tgtagctgct gccccccagc cccaaggagc tgagcagcag cgacaccgtg 1080
accctgacct gctgatcaaa ggactttctt ccccccgaga tcgacgtgga gtggcagagc 1140
aacggccagc ccagggccga gaggccccg caccaccagg cccccagct ggacgaggac 1200
ggcagctact tctgtacag caagctgagc gtggacaaga gcagatggca gcaggggcag 1260
accttcacct gcgcgctgat gcacgaggcc ctgcagaacc actacaccga cctgagcctg 1320
agccacagcc ccggcaag

```

```

SEQ ID NO: 88      moltype = AA length = 446
FEATURE           Location/Qualifiers
REGION            1..446
                  note = caninized mouse antibody
source            1..446
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 88
EVQLVQSGPG LVKPSQSLSL TCVVSGFSLT SYGVHVRQA PGKGLQWMGW INIYSGIPTY 60
ADDFKGRFTF SLDTAKNTAY LQLSSLRAED TAVYYCARFD GPDYWGQGTL VTVSSASTTA 120
PSVFPPLPASC GSTSGSTVAL ACLVSGYFPE PVTVSWNSGS LTSGVHTFPS VLQSSGLYSL 180
SSTVTVPSSR WPSFTFTCNV VHPASNTKVD KVPKKESTCK CISPCVPES LGGPSVFIFP 240
PKPKDILRIT RTPBITCVVL DLGREDEEVQ ISWFDGKEV HTAKTQPREQ QFNSTYRVVS 300
VLPPIEQDWL TGKEFKCRVN HIGLPSPIER TISKARGQAH QPSVYVLPSS PKELSSSDTV 360
TLTCLIKDFP PPEIDVEVQS NGQPEPESKY HTTAPQLDED GSYFLYSKLS VDKSRWQQGD 420
TFTCAVMHEA LQNHYTDLNL SHSPGK

```

```

SEQ ID NO: 89      moltype = DNA length = 672
FEATURE           Location/Qualifiers
misc_feature       1..672
                  note = caninized mouse antibody
source            1..672
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 89
gacatcgtga tgaccagac cccctgagc ctgagcgtga gcccggcgga gcccgcacg 60
atgagctgca agagcagcca gagcctgctg aacagcgtga accagaagaa ctacctggcc 120
tggtacagac agaagcccg cagagcccc caggtgctgg tgtacttcgc cagcaccaga 180
gtgagcggcg tgcccagacg attcatcgcc agcggcagcg gcaccgactt caccctgaga 240
atcagcagag tggaggccga cgacctgggc gtgtactact gccagcagta cttcagcacc 300
cccctgacct tcggccaggg caccgaagctg gagctgaaga gaaacgacgc ccagcccgcc 360
gtgtacctgt tccagcccg ccccgaccag ctgcacaccg gcagcgccag cgtggtgtgc 420
ctgtgaaca gcttctaccc caaggacatc aacgtgaagt ggaaggtgga cggcgtgatc 480
caggacaccg gcattccagg gagcgtgacc gagcaggaca gcaaggacag cactacagc 540
ctgagcagca cccctgacct gagcagcacc gagtacctga gccacgagct gtacagctgc 600
gagatcacc acaagagcct gcccagcacc ctgatcaaga gcttcacag aagcgagtgc 660
cagagagtgg ac

```

```

SEQ ID NO: 90      moltype = AA length = 224
FEATURE           Location/Qualifiers
REGION            1..224
                  note = caninized mouse antibody
source            1..224
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 90
DIVMTQTPLS LSVSPGEPAS MSCKSSQSLN NSVNQKNYLA WYRQKPGQSP QVLVYFASTR 60
VSGVPDRFIG SGSGTDFTLR ISRVEADDLG VYKQYFST PLTFGQGTGL ELKRNDQPA 120
VYLFQSPDPQ LHTGSASVVC LLNSFYPKDI NVKWKVDGVI QDTGIQESVT EQDSKSDSTYS 180
LSSTLTMSST EYLSHELYSC EITHKSLPST LKSPQRSEC QRVD

```

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SEQ ID NO: 91      moltype = DNA length = 1338
FEATURE           Location/Qualifiers
misc_feature       1..1338
                  note = caninized mouse antibody

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-continued

source 1..1338
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 91

gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc	cggcgccag	cgtgaagggtg	60
agctgcaagg	ccagcggcta	caccttcacc	acctacggca	tgagctgggt	gagacaggcc	120
cccggcaagg	gcctgcagt	gatgggctgg	atcaacatct	acagcggcat	ccccacctac	180
gcccagcact	tcaagggcag	attcgccctg	agcctggaca	ccagcaccag	caccgcctac	240
atggagctga	acagcctgag	agccgaggac	accgcccgtg	actactgcgc	cagattcgac	300
ggccccgact	actggggcca	gggcaccctg	gtgaccgtga	gcagcgccag	caccaccgcc	360
cccagcgtgt	tccccctggc	ccccagctgc	ggcagcacca	gcggcagcac	cgtggccctg	420
gcctgcctgg	tgagcggcta	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcggcct	gcacagcctg	540
agcagcatgg	tgaccgtgcc	cagcagcaga	tggcccagcg	agaccttcac	ctgcaacgtg	600
gtgcaccccc	ccagcaaacac	caaggtggac	aagcccgtgc	tcaacgagtg	cagatgcacc	660
gacaccccc	cctgcccgtg	gcccagagccc	ctgggcggcc	ccagcgtgct	gatcttcccc	720
cccaagccca	aggacatcct	gagaatcacc	agaacccccg	aggtgacctg	cgtgggtgctg	780
gacctgggca	gagaggaccc	cgaggtgcag	atcagctggt	tcgtggacgg	caaggaggtg	840
cacaccgcca	agacccagag	caagagcagc	cagttcaacg	gcacctacag	agtgggtgagc	900
gtgctgcccc	tcgagcacca	ggactggctg	accggcaagg	agttcaagtg	cagagtgaac	960
cacatcgacc	tgcccagccc	catcgagaga	accatcagca	aggccagagg	cagagccccc	1020
aagcccagcg	tgtacgtgct	gccccccagc	cccaaggagc	tgagcagcag	cgacaccgtg	1080
agcatcacct	gcctgatcaa	ggactttctc	ccccccgaca	tcgacgtgga	gtggcagagc	1140
aacggccagc	aggagcccca	gagaaagcac	agaatgaccc	ccccccagct	ggacgaggac	1200
ggcagctact	tcctgtacag	caagctgagc	gtggacaaga	gcagatggca	gcaggggcgac	1260
cccttcacct	gcgcccgtgat	gcacgagacc	ctgcagaacc	actacaccga	cctgagcctg	1320
agccacagcc	ccggcaag					1338

SEQ ID NO: 92 moltype = AA length = 446
FEATURE Location/Qualifiers
REGION 1..446
note = caninized mouse antibody

source 1..446
mol_type = protein
organism = synthetic construct

SEQUENCE: 92

EVQLVQSGAE	VKKPGASVKV	SCKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGIPTY	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCARFD	GPDYWGQGTL	VTSSASTTA	120
PSVFPPLAPSC	GSTSGSTVAL	ACLVSIFYFPE	PVTVSWNSGS	LTSGVHTFPS	VLQSSGLHSL	180
SSMVTVPSSR	WPSSTFTCNV	VHPASNTKVD	KPVFNECRCT	DTPPCPVPEP	LGGPSVLIFP	240
PKPKDILRIT	RTPEVTCVVL	DLGREDPEVQ	ISWFVDGKEV	HTAKTQSREQ	QFNGTYRVVS	300
VLPIEHQDWL	TGKEFKCRVN	HIDLPSPIER	TISKARGRAH	KPSVYVLPFS	PKELSSSDTV	360
STCLIKDFY	PPDIDVWEQS	NGQQEPERKH	RMTPPQLDED	GSYFLYSKLS	VDKSRWQQGD	420
PFTCAVMHET	LQNHYTDLSL	SHSPGK				446

SEQ ID NO: 93 moltype = DNA length = 1338
FEATURE Location/Qualifiers
misc_feature 1..1338
note = caninized mouse antibody

source 1..1338
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 93

gaggtgcagc	tggtgcagag	cggcgccgag	gtgaagaagc	cggcgccag	cgtgaagggtg	60
agctgcaagg	ccagcggcta	caccttcacc	acctacggca	tgagctgggt	gagacaggcc	120
cccggcaagg	gcctgcagt	gatgggctgg	atcaacatct	acagcggcat	ccccacctac	180
gcccagcact	tcaagggcag	attcgccctg	agcctggaca	ccagcaccag	caccgcctac	240
atggagctga	acagcctgag	agccgaggac	accgcccgtg	actactgcgc	cagattcgac	300
ggccccgact	actggggcca	gggcaccctg	gtgaccgtga	gcagcgccag	caccaccgcc	360
cccagcgtgt	tccccctggc	ccccagctgc	ggcagcacca	gcggcagcac	cgtggccctg	420
gcctgcctgg	tgagcggcta	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcggcct	gtacagcctg	540
agcagcaccc	tgaccgtgcc	cagcagcaga	tggcccagcg	agaccttcac	ctgcaacgtg	600
gtgcaccccc	ccagcaaacac	caaggtggac	aagcccgtgc	ccaaggagag	cacctgcaag	660
tgcatcaccc	cctgcccctg	gcccagagagc	ctgggcggcc	ccagcgtggt	catcttcccc	720
cccaagccca	aggacatcct	gagaatcacc	agaacccccg	agatcacctg	cgtgggtgctg	780
gacctgggca	gagaggaccc	cgaggtgcag	atcagctggt	tcgtggacgg	caaggaggtg	840
cacaccgcca	agacccagcg	cagagagcag	cagttcaaca	gcacctacag	agtgggtgagc	900
gtgctgcccc	tcgagcacca	ggactggctg	accggcaagg	agttcaagtg	cagagtgaac	960
cacatcgccc	tgcccagccc	catcgagaga	accatcagca	aggccagagg	ccaggccccc	1020
cagcccagcg	tgtacgtgct	gccccccagc	cccaaggagc	tgagcagcag	cgacaccgtg	1080
accctgacct	gcctgatcaa	ggactttctc	ccccccgaga	tcgacgtgga	gtggcagagc	1140
aacggccagc	ccgagcccca	gagcaagtac	cacaccaccg	ccccccagct	ggacgaggac	1200
ggcagctact	tcctgtacag	caagctgagc	gtggacaaga	gcagatggca	gcaggggcgac	1260
accttcacct	gcgcccgtgat	gcacgaggcc	ctgcagaacc	actacaccga	cctgagcctg	1320
agccacagcc	ccggcaag					1338

-continued

SEQ ID NO: 94 moltype = AA length = 446
 FEATURE Location/Qualifiers
 REGION 1..446
 note = caninized mouse antibody
 source 1..446
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 94

EVQLVQSGAE	VKKPGASVKV	CKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGIPTY	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCARPD	GPDYWGQGT	LVSSASTTA	120
PSVFPLAPSC	GSTSGSTVAL	ACLVSGYFPE	PVTVSWNSGS	LTSGVHTFPS	VLQSSGLYSL	180
SSTVTVPSSR	WPSETFTCNV	VHPASNTKVD	KVPVKESTCK	CISPCVPES	LGGPSVFIFP	240
PKPKDILRIT	RTPEITCVVL	DLGREDPEVQ	ISWFDGKEV	HTAKTQPREQ	QFNSTYRVVS	300
VLPIEHQDWL	TGKEFKCRVN	HIGLPSPIER	TISKARGQAH	QPSVYVLPPS	PKELSSSDTV	360
TLTCLLKDF	PPEDVVEVQS	NGQPEPESKY	HTTAPQLDED	GSYFLYSKLS	VDKSRWQQGD	420
TFTCAVMHEA	LQNHYTDL	SL	SHSPGK			446

SEQ ID NO: 95 moltype = DNA length = 672
 FEATURE Location/Qualifiers
 misc_feature 1..672
 note = caninized mouse antibody
 source 1..672
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 95

gacatcgtag	tgacccagac	ccccctgagc	ctgagcgtga	gccccggcga	gccccccagc	60
atgagctgca	agagcagcca	gagcctgctg	aacagcgtga	accagaagaa	ctacctggcc	120
tggtagacag	agaagcccg	ccagagcccc	caggtgctgg	tgtacttcgc	cagcaccaga	180
gtgagcggcg	tgccccagag	attcctcgcc	agcggcagcg	gcaccgactt	caccctgaga	240
atcagcagag	tggaggccga	cgacctgggc	gtgtactact	gccagcagta	cttcagcacc	300
ccccctgacct	tcggccaggg	caccaagctg	gagctgaaga	gaaacgacgc	ccagccccgc	360
gtgtacctgt	tccagcccg	ccccgaccag	ctgcacaccg	gcagcgccag	cgtggtgtgc	420
ctgctgaaca	gcttctaccc	caaggacatc	aacgtgaagt	ggaaggtgga	cggcgtgatc	480
caggacaccg	gcattccagga	gagcgtgacc	gagcaggaca	gcaaggacag	cacctacagc	540
ctgagcagca	ccctgaccat	gagcagcacc	gagtagctga	gccacgagct	gtacagctgc	600
gagatcacc	acaagagcct	gccccagcacc	ctgatcaaga	gcttccagag	aagcagtgcc	660
cagagagtgg	ac					672

SEQ ID NO: 96 moltype = AA length = 224
 FEATURE Location/Qualifiers
 REGION 1..224
 note = caninized mouse antibody
 source 1..224
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 96

DIVMTQTPLS	LSVSPGEPAS	MSCKSSQSLL	NSVNQKNYLA	WYRQKPGQSP	QVLVYFASTR	60
VSGVDRFIG	SGSGTDFTLR	ISRVEADDLG	VYYCQYFST	PLTFGQGTKL	ELKRNDQPA	120
VYLFQSPDQ	LHTGSASVVC	LLNSFYPKDI	NVKKVVDGVI	QDTGIQESVT	EQDSKDSTYS	180
LSSTLTMSST	EYLSHELYSC	EITHKSLPST	LIKSPQRSEC	QRVD		224

SEQ ID NO: 97 moltype = DNA length = 1353
 FEATURE Location/Qualifiers
 misc_feature 1..1353
 note = caninized mouse antibody
 source 1..1353
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 97

gaggtgcagc	tggtgcagag	cgtggccgag	ctggtgaagc	ccggcgccag	cgtgaaggtg	60
agctgcaccg	tgagcggctt	caacatcaag	aacacctaca	tgcactgggt	gagacaggcc	120
cccgcaagg	gcctgcagtg	gatcggcaga	atcgaccccc	ccaacgtgaa	caccaagtac	180
gcccccaagt	tccagggcag	agccaccatc	accgcccaga	ccagcaccac	caccgcctac	240
atgcagctga	gcagcctgag	agccgaggac	accgcccgtg	actactgcgc	cagaatcttc	300
tacgactacg	acggcgacat	cgacgtgtgg	ggccagggca	ccctgggtgac	cgtgagcagc	360
gccagcacca	ccgccccag	cgtgttcccc	ctggccccca	gctgcggcag	caccagcggc	420
agcaccgtgg	ccctggcctg	cctgggtgagc	ggctacttcc	ccgagcccgt	gaccgtgagc	480
tggaacagcg	gcagcctgac	cagcggcgtg	cacaccttcc	ccagcgtgct	gcagagcagc	540
ggcctgcaca	ccctgagcag	catggtgacc	gtgcccagca	gcagatggcc	cagcagagacc	600
ttcacctgca	acgtggtgca	ccccgccagc	aacaccaagg	tggacaagcc	cgtgttcaac	660
gagtgagatg	gcacgcagac	ccccccctgc	cccgctgccc	agccccctgg	cggccccagc	720
gtgctgatct	tcccccccaa	gccccaggac	atcctgagaa	tcaccagaac	ccccgagggt	780
acctgcgtgg	tgctggacct	gggcagagag	gacccccagg	tgagatcagc	ctgggtcgtg	840
gacggcaagg	aggtgcacac	cgccaagacc	cagagcagag	agcagcagtt	caacggcacc	900
tacagagtgg	tgagcgtgct	gccccatcag	caccaggact	ggctgaccgg	caaggagttc	960

-continued

aagtgcagag	tgaaccacat	cgacctgccc	agcccatcg	agagaacct	cagcaaggcc	1020
agaggcagag	cccacaagcc	cagcgtgtac	gtgctgcccc	ccagcccaaa	ggagctgagc	1080
agcagcgaca	ccgtgagcat	cacctgcctg	atcaaggact	tctaccccc	cgacatcgac	1140
gtggagtggc	agagcaacgg	ccagcaggag	cccagagaaa	agcacagaat	gacccccccc	1200
cagctggacg	aggacggcag	ctacttctg	tacagcaagc	tgagcgtgga	caagagcaga	1260
tggcagcagg	gcgacccctt	cacctgcgcc	gtgatgcacg	agaccctgca	gaaccactac	1320
accgacctga	gcctgagcca	cagccccggc	aag			1353

SEQ ID NO: 98 moltype = AA length = 451
 FEATURE Location/Qualifiers
 REGION 1..451
 note = caninized mouse antibody
 source 1..451
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 98

EVQLVQSVAE	LVKPGASVKV	SCTVSGFNIK	NTYMHWVRQA	PGKGLQWIGR	IDPANVNTKY	60
APKFQGRATI	TADTSTNTAY	QLSSSLRAED	TAVYYCARIF	YDYDGDIDVW	GQGTSLTVSS	120
ASTTAPSVFP	LAPSCGSTSG	STVALACLVS	GYFPEPVTVS	WNSGSLTSGV	HTFPSVLQSS	180
GLHSLSSMVT	VPSSRWPSSET	FTCNVVPAS	NTKVDKPVFN	ECRCTDTPPC	PVPEPLGGPS	240
VLIFPPPKPD	ILRITRTPEV	TCVVDLGLRE	DPEVQISWFV	DGKEVHTAKT	QSRQQFNGT	300
YRVVSVLPPIE	HQDWLTGKEF	KCRVNHIDL	SPIERTISKA	RGRAHKPSVY	VLPPSPKELS	360
SSDTSITCL	IKDFYPPDID	VEWQSNQQE	PERKHRMTPP	QLDEDGSYFL	YSKLSVDKSR	420
WQQGDPFTCA	VMHETLQNH	TDLSSLHSPG	K			451

SEQ ID NO: 99 moltype = DNA length = 1353
 FEATURE Location/Qualifiers
 misc_feature 1..1353
 note = caninized mouse antibody
 source 1..1353
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 99

gaggtgcagc	tggtgcagag	cgtggccgag	ctgggtgaagc	ccggcgccag	cgtgaagggtg	60
agctgcaccg	tgagcggctt	caacatcaag	aacacctaca	tgcactgggt	gagacaggcc	120
cccgcaagg	gcctgcagtg	gatcggcaga	atcgaccccc	ccaacgtgaa	caccaagtac	180
gcccccaagt	tccagggcag	agccaccatc	accgcccaga	ccagcaccaa	caccgcctac	240
atgcagctga	gcagcctgag	agccgaggac	accgcccgtg	actactgcgc	cagaatcttc	300
tacgactacg	acggcgacat	cgacgtgtgg	ggccagggca	ccctgggtgac	cgtgagcagc	360
gccagcacca	ccgccccag	cgtgttcccc	ctggccccca	gctgcggcag	caccagcggc	420
agcacccgtg	ccctggcctg	cctgggtgagc	ggctacttcc	ccgagcccgt	gaccgtgagc	480
tggaaacagc	gcagcctgag	cagcggcggtg	cacaccttcc	ccagcgtgct	gcagagcagc	540
ggcctgtaca	gcctgagcag	caccgtgacc	gtgcccagca	gcagatggcc	cagcgagacc	600
ttcacctgca	acgtgggtgca	ccccgccagc	aacaccaagg	tggacaagcc	cgtgccccag	660
gagagcacct	gcaagtgcac	cagccccctg	cccgtgcccc	agagcctggg	cggccccagc	720
gtgttcactc	tcccccccaa	gcccaaggac	atcctgagaa	tcaccagaa	ccccgagatc	780
acctgcgtgg	tgctggacct	gggcagagag	gaccccgagg	tgcagatcag	ctgggtcgtg	840
cagcgcaagg	aggtgcacac	cgccaagacc	cagcccagag	agcagcagtt	caacagcacc	900
tacagagtgg	tgagcgtgtg	gcccacgcag	caccaggact	ggctgaccgg	caaggagtgc	960
aagtgcagag	tgaaccacat	cggcctgccc	agcccatcg	agagaacct	cagcaaggcc	1020
agaggccagg	cccaccagcc	cagcgtgtac	gtgctgcccc	ccagcccaaa	ggagctgagc	1080
agcagcgaca	ccgtgacctt	gacctgcctg	atcaaggact	tcttcccccc	cgagatcgac	1140
gtggagtggc	agagcaacgg	ccagcccag	cccagagaca	agtaccacac	caccgcccc	1200
cagctggacg	aggacggcag	ctacttctg	tacagcaagc	tgagcgtgga	caagagcaga	1260
tggcagcagg	gcgacacctt	cacctgcgcc	gtgatgcacg	aggccctgca	gaaccactac	1320
accgacctga	gcctgagcca	cagccccggc	aag			1353

SEQ ID NO: 100 moltype = AA length = 451
 FEATURE Location/Qualifiers
 REGION 1..451
 note = caninized mouse antibody
 source 1..451
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 100

EVQLVQSVAE	LVKPGASVKV	SCTVSGFNIK	NTYMHWVRQA	PGKGLQWIGR	IDPANVNTKY	60
APKFQGRATI	TADTSTNTAY	QLSSSLRAED	TAVYYCARIF	YDYDGDIDVW	GQGTSLTVSS	120
ASTTAPSVFP	LAPSCGSTSG	STVALACLVS	GYFPEPVTVS	WNSGSLTSGV	HTFPSVLQSS	180
GLYSLSSVT	VPSSRWPSSET	FTCNVVPAS	NTKVDKPVFK	ESTCKCISPC	PVPESLGGPS	240
VFIFPPPKPD	ILRITRTPEI	TCVVDLGLRE	DPEVQISWFV	DGKEVHTAKT	QPREQQFNST	300
YRVVSVLPPIE	HQDWLTGKEF	KCRVNHIGLP	SPIERTISKA	RGAHQPSVY	VLPPSPKELS	360
SSDTSITCL	IKDFYPPDID	VEWQSNQQE	PESKYHTTAP	QLDEDGSYFL	YSKLSVDKSR	420
WQQGDTFTCA	VMHEALQNH	TDLSSLHSPG	K			451

SEQ ID NO: 101 moltype = DNA length = 654
 FEATURE Location/Qualifiers

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misc_feature      1..654
                  note = caninized mouse antibody
source            1..654
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 101
gacatcgtga tgacccagac cccctgagc ctgagcgtga gcctggcgga gcccggcagc 60
atcagctgcc acgccaagcca gaacatcaac gtgtggctga gctggtagac acagaagccc 120
ggccagatcc cccagctgct gatctacaag gccagcaacc tgcacaccgg cgtgcccagc 180
agattcagcg gcagcggcag cggcaccgac ttcaccctga gaatcagcag agtggaggcc 240
gacgacgccc gcgtgtacta ctgccagcag gccagagcgt acccctgac ctccggccag 300
ggcaccgaag tggagatcaa gagaaacgac gccagccccg ccgtgtacct gttccagccc 360
agccccgacc agctgcacac cggcagcgcc agcgtggtgt gcctgctgaa cagcttctac 420
cccaaggaca tcaacgtgaa gtggaagggt gacggcgtga tccaggacac cggcattccag 480
gagagcgtga ccgagcagga cagcaaggac agcacctaca gcctgagcag caccctgacc 540
atgagcagca ccgagtacct gagccacgag ctgtacagct gcgagatcac ccacaagagc 600
ctgcccgaca cctgatcaa gagcttccag agaagcgagt gccagagagt ggac 654

SEQ ID NO: 102      moltype = AA length = 218
FEATURE            Location/Qualifiers
REGION            1..218
                  note = caninized mouse antibody
source            1..218
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 102
DIVMTQTPLS LSVSLGEPAS ISCHASQNIN VWLSWYRQKP GQIPQLLIYK ASNLHTGVDP 60
RFSGSGSGTD FTLRISRVEA DDAGVYYCQQ GQSYPLTFGQ GTKVEIKRND AQPAYVLFQP 120
SPDQLHTGSA SVVCLLNSFY PKDINVKWKV DGVIQDTGIQ ESVTEQDSKD STYLSSTLT 180
MSSTEYLSHE LYSCEITHKS LPSTLIKSFQ RSECQRVD 218

SEQ ID NO: 103      moltype = DNA length = 1338
FEATURE            Location/Qualifiers
misc_feature      1..1338
                  note = caninized mouse antibody
source            1..1338
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 103
gaggtgcagc ccgtgcagag cggcgcccag gtgaagaagc cggcgcccag cgtgaagggtg 60
agctgcaagg ccagcggcta caccctcacc acctacggca tgagctgggt gagacaggcc 120
cccgcaaggc gcctgcagtg gatgggctgg atcaacatct acagcggcat gccacacctac 180
gccgacgact tcaagggcag attcgccctg agcctggaca ccagcaccag caccgcctac 240
atggagctga acagcctgag agccgaggac accgcctgt actactgcac cagattcgac 300
ggccccgact actggggcca gggcaccctg gtgaccgtga gcagcggcag caccaccgcc 360
cccagcgtgt tccccctgtg cccagctgc gccagcaccg gcgcagcac cgtggccctg 420
gcctgcctgg tgagcggcta cttccccgag cccgtgaccg tgagctggaa cagcggcagc 480
ctgaccagcg gcctgcacac cttccccagc gtgctgcaga gcagcggcct gcacagcctg 540
agcagcatgg tgaccctgccc cagcagcaga tggcccagcg agaccttcac ctgcaacgtg 600
gtgcaccccc ccagcaaacac caaggtggac aagcccctgt tcaacgagtg cagatgcacc 660
gacaccccc cctgccccctg gcccgagccc ctgggcgggc ccagcgtgct gatcttcccc 720
cccaagccca aggacatcgt gagaatcacc agaaccccc aggtgacctg cgtggtgctg 780
gacctgggca gagaggaccc cgaggtgcag atcagctggt tcgtggacgg caaggagggtg 840
cacaccgcca agacccagag cagagagcag cagttcaacg gcacctacag agtggtgagc 900
gtgctgcccc tcgagcacca ggactggctg accggcaagg agttcaagtg cagagtgaac 960
cacatcgacc tgcccagccc catcgagaga acctacagca agggccagagg cagagccacc 1020
aagcccagcg tgtacgtgct gccccccagc cccaaggagc tgagcagcag cgacaccgtg 1080
agcatcacct gcctgatcaa ggacttctac ccccccgaca tcgacgtgga gtggcagagc 1140
aacggccagc agggagccga gagaaagcac agaattgaccc cccccagct ggacgaggac 1200
ggcagctact tcctgtacag caagctgagc gtggacaaga gcagatggca gcaggggcagc 1260
cccttcacct gcgcctgatg gcacgagacc ctgcagaacc actacaccga cctgagcctg 1320
agccacagcc ccggcaag 1338

SEQ ID NO: 104      moltype = AA length = 446
FEATURE            Location/Qualifiers
REGION            1..446
                  note = caninized mouse antibody
source            1..446
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 104
EVQLVQSGAE VKKPGASVKV SCKASGYTFT TYGMSWVRQA PGKGLQWMGW INIYSGMPTY 60
ADDFKGRFAL SLDTSTSTAY MELNSLRAED TAVYYCTFRD GPDYWGQGLT VTSSASTTA 120
PSVFPLAPSC GSTSGSTVAL ACLVSGYFPE PVTVSWSNGS LTSGVHTFPS VLQSSGLHSL 180
SSMVTVPSSR WPSSETFTCNV VHPASNTKVD KPVFNECRCT DTPPCVPEP LGGPSVLIFP 240
PKPKDILRIT RTPEVTCVVL DLGREDPEVQ ISWFVDGKEV HTAKTQSQREQ QFNQTYRVVS 300

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VLPIEHQDWL	TGKEFKCRVN	HIDLPSPIER	TISKARGRAH	KPSVYVLPPS	PKELSSSDTV	360
SITCLIKDFY	PPDIDVEWQS	NGQEPERKH	RMTPPQLDED	GSYFLYSKLS	VDKSRWQQGD	420
PFTCAVMHET	LQNHYTDLNL	SHSPGK				446

SEQ ID NO: 105 moltype = DNA length = 1338
 FEATURE Location/Qualifiers
 misc_feature 1..1338
 note = caninized mouse antibody
 source 1..1338
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 105

gaggtgcagc	tggtgcagag	cggtgcagag	gtgaagaagc	cggtgcagag	cgtgaagggtg	60
agctgcaagg	ccagcggtga	caccttcacc	acctacggca	tgagctgggt	gagacaggcc	120
cccgcaagg	gcctgcagtg	gatgggctgg	atcaacatct	acagcgccat	gcccacctac	180
gcccagcact	tcaagggcag	attcgccctg	agcctggaca	ccagcaccag	caccgcctac	240
atggagctga	acagcctgag	agccgaggag	accgccgtgt	actactgcac	cagattcgac	300
ggccccgact	actggggcca	gggcaccctg	gtgaccctga	gcagcgccag	caccaccgcc	360
cccagcgtgt	tcccctggc	cccagctgc	ggcagcacc	gcggcagcac	cgtggccctg	420
gcctgcctgg	tgagcggtga	cttccccgag	cccgtgaccg	tgagctggaa	cagcggcagc	480
ctgaccagcg	gcgtgcacac	cttccccagc	gtgctgcaga	gcagcgccct	gtacagcctg	540
agcagcaccg	tgaccgctgc	cagcagcaga	tggtccagcg	agaccttcac	ctgcaacgtg	600
gtgcaacccg	ccagcaaac	caaggtggac	aagcccgctg	ccaaggagag	cacctgcaag	660
tgcatcagcc	cctgccccgt	gcccagagag	ctgggcggcc	ccagcgtgtt	catcttcccc	720
cccaagccca	aggacatcct	gagaatcacc	agaacccccg	agatcacctg	cgtggtgctg	780
gacctgggca	gagaggaccc	cgaggtgcag	atcagctggg	tcgtggacgg	caaggagggtg	840
cacaccgcc	agaccagcc	cagagagcag	cagttcaaca	gcacctacag	agtgggtgagc	900
gtgctgcccc	tcagcagcca	ggactggctg	accggcaagg	agttcaagtg	cagagtgaac	960
cacatcgccc	tgccccagcc	cacagagaga	accatcagca	aggccagagg	ccaggccccc	1020
cagccccagc	tgtagctgct	gccccccagc	cccaaggagc	tgagcagcag	cgacaccctg	1080
accctgacct	gcctgatcaa	ggacttcttc	ccccccgaga	tcgacgtgga	gtggcagagc	1140
aacggccagc	cgagcccgga	gagcaagtag	cacaccaccc	ccccccagct	ggacgaggac	1200
ggcagctact	tcctgtacag	caagctgagc	gtggacaaga	gcagatggca	gcaggggcgac	1260
accttcacct	cgcccgctgat	gcacgaggcc	ctgcagaacc	actacaccga	cctgagcctg	1320
agccacagcc	ccggcaag					1338

SEQ ID NO: 106 moltype = AA length = 446
 FEATURE Location/Qualifiers
 REGION 1..446
 note = caninized mouse antibody
 source 1..446
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 106

EVQLVQSGAE	VKKPGASVKV	SKKASGYTFT	TYGMSWVRQA	PGKGLQWMGW	INIYSGMPTY	60
ADDFKGRFAL	SLDTSTSTAY	MELNSLRAED	TAVYYCTRFD	GPDYWGQGT	VTVSSASTTA	120
PSVFPPLAPSC	GSTSGSTVAL	ACLVSGYFPE	PVTVSWNSGS	LTSGVHTFPS	VLQSSGLYSL	180
SSTVTVPSSR	WPSTFTTCNV	VHPASNTKVD	KVPKESTCK	CISPCVPES	LGGPSVFIFP	240
PKPKDILRIT	RTPEITCVVL	DLGREDEEVQ	ISWFDGKEV	HTAKTQPREQ	QFNSTYRVVS	300
VLPIEHQDWL	TGKEFKCRVN	HIDLPSPIER	TISKARGQAH	QPSVYVLPPS	PKELSSSDTV	360
TLTCLIKDFY	PPDIDVEWQS	NGQPEPESKY	HTTAPQLDED	GSYFLYSKLS	VDKSRWQQGD	420
TFTCAVMHEA	LQNHYTDLNL	SHSPGK				446

SEQ ID NO: 107 moltype = DNA length = 672
 FEATURE Location/Qualifiers
 misc_feature 1..672
 note = caninized mouse antibody
 source 1..672
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 107

gacatcgtga	tgaccacagc	ccccctgagc	ctgagcgtga	gccccggcga	gccccccagc	60
atgagctgca	agagcagcca	gagcctgctg	aacagcgtga	accagaagaa	ctacctggcc	120
tggtacagac	agaagcccg	ccagagcccc	caggtgctgg	tgtacttcgc	cagcaccaga	180
atcagcgccg	tgcccgacag	attcatcgcc	agcgccagcg	gcacgcactt	cacctgaga	240
atcagcagag	tggaaggccga	cgacctgggc	gtgtactact	gccagcagta	cttcagcacc	300
ccccgacct	tcggccaggg	caccaagctg	gagctgaaga	gaaacgcagc	ccagcccgcc	360
gtgtacctgt	tccagcccg	ccccgaccag	ctgcacaccg	gcagcgccag	cgtgggtgtg	420
ctgctgaaca	gcttctaccc	caaggacatc	aacgtgaagt	ggaagggtga	cggcgtgatc	480
caggacaccg	gcattccagga	gagcgtgacc	gagcaggaca	gcaaggacag	cacctacagc	540
ctgagcagca	ccctgacct	gagcagcacc	gagtagctga	gccacgagct	gtacagctgc	600
gagatcacc	acaagagcct	gcccagcacc	ctgatcaaga	gcttccagag	aagcgagtg	660
cagagagtg	ac					672

SEQ ID NO: 108 moltype = AA length = 224
 FEATURE Location/Qualifiers

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REGION 1..224
note = caninized mouse antibody

source 1..224
mol_type = protein
organism = synthetic construct

SEQUENCE: 108
DIVMTQTPLS LSVSPGEPAS MSCKSSQSLL NSVNQKNYLA WYRQKPGQSP QVLVYFASTR 60
ISGVDPDRFIG SGSGTDFTLR ISRVEADDLG VYYCQQYFST PLTFGQGTKL ELKRNDQAQPA 120
VYLFQPSPDQ LHTGSASVVC LLNSFYPKDI NVKWKVDGVI QDTGIQESVT EQDSKDSTYS 180
LSSTLTMSST EYLSHELYSC EITHKSLPST LIKSPQRSEC QRVD 224

SEQ ID NO: 109 moltype = AA length = 17
FEATURE Location/Qualifiers
source 1..17
mol_type = protein
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 109
FNECRCTDTP PCPVPEP 17

SEQ ID NO: 110 moltype = AA length = 22
FEATURE Location/Qualifiers
source 1..22
mol_type = protein
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 110
PKRENGRVPR PPDCKPCPAP EM 22

SEQ ID NO: 111 moltype = AA length = 20
FEATURE Location/Qualifiers
source 1..20
mol_type = protein
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 111
AKECECKCNC NNCPGCGCL 20

SEQ ID NO: 112 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
note = modified canine
source 1..17
mol_type = protein
organism = synthetic construct

SEQUENCE: 112
PKESTCKCIP PCVPES 17

SEQ ID NO: 113 moltype = DNA length = 795
FEATURE Location/Qualifiers
source 1..795
mol_type = unassigned DNA
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 113
ctagactccc ctgacaggcc ctggagcccg ctcaccttct ccccggcgca gctcacggtg 60
caggaggagg agaacgccac gttcacctgc agcctggccg acatccccga cagcttcgtg 120
ctcaactggt accgcctgag ccccgcaac cagacggaca agctggccgc cttccaggag 180
gaccgcatcg agccgggccc ggacaggcgc ttccgcgtca tgcgggtgcc caacgggccc 240
gacttcacaca tgagcatcgt cgtgctgcgc ctcaacgaca gcggcatcta cctgtgcccg 300
gccatctacc tgccccccaa cacacagatc aacgagagtc cccgcgcaga gctctccgtg 360
acggagagaa ccctggagcc ccccacacag agccccagcc ccccacccag actcagcgcc 420
cagttgcagg ggctggctcat cggcgctcac agcgtgctgg tgggtgtcct gctactgctg 480
ctgctgacct gggctcctggc cgtgtctctc cccagggcca cccgaggtgc ctgtgtgtgc 540
gggagcgagg acgagcctct gaaggaggcc cccgatgcag cgcccgctct caccctggac 600
tacggggagc tggacttcca gtggcgagag aagacgcccg agccccgggc gccctgtgcc 660
ccggagcaga ccgagtatgc caccatcgct ttcccgggca ggccggcgct cccggggccc 720
agggcctcgg ccagcagcct gcaggagacc cagcctccga gccccgagga cggaccggcc 780
ctgtggcccc tctga 795

SEQ ID NO: 114 moltype = AA length = 264
FEATURE Location/Qualifiers
source 1..264
mol_type = protein
organism = Canis lupus
note = Canis familiaris

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SEQUENCE: 114
 LDSPDRPWSP LTFSPAQLTV QEGENATFTC SLADIPDSFV LNWYRLSPRN QTDKLAAPQE 60
 DRIEPGRDRR FRVMRLPNGR DFHMSIVAAR LNDSGIYLCG AIYLPPTQI NESPRAELSV 120
 TERTLEPPTQ SPSPPPRLSG QLQGLVIGVT SVLVGVLLLL LLTWVLA AVF PRATRGACVC 180
 GSEDEPLKEG PDAAPVFTLD YGELDFQWRE KTPEPPAPCA PEQTEYATIV FPGRPASPGR 240
 RASASSLQGA QPPSPEDGPG LWPL 264

SEQ ID NO: 115 moltype = DNA length = 438
 FEATURE Location/Qualifiers
 source 1..438
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 115
 ctggattccc cgcacagacc ctggagccct ctcaccttct cccctgcccc gctgaccgtc 60
 caggaaggcg agaagtcacc cttcacctgc agcctcgccg acatccccga cagcttcgtg 120
 ctgaactggt acagactgag ccccagggaac cagaccgaca agctggccgc ttccaggag 180
 gacaggatcg aaccggcgag ggacaggagg tttaggggtca tgaggctgcc caacggcagg 240
 gacttcaca tgtccatcgt ggccgcccaga ctgaacgact ccggcatcta cctgtgcggc 300
 gctatctacc tgccccccaa caccagatc aacgagagcc ccagggccga actgagcgtg 360
 acagagagaa ccctggaacc tccccccag agcccttccc ctctcctag actgagcgga 420
 cagctgcagg gcctggtg 438

SEQ ID NO: 116 moltype = AA length = 146
 FEATURE Location/Qualifiers
 source 1..146
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 116
 LDSPDRPWSP LTFSPAQLTV QEGENATFTC SLADIPDSFV LNWYRLSPRN QTDKLAAPQE 60
 DRIEPGRDRR FRVMRLPNGR DFHMSIVAAR LNDSGIYLCG AIYLPPTQI NESPRAELSV 120
 TERTLEPPTQ SPSPPPRLSG QLQGLV 146

SEQ ID NO: 117 moltype = DNA length = 1128
 FEATURE Location/Qualifiers
 misc_feature 1..1128
 note = canine and human sequence
 source 1..1128
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 117
 ctggattccc cgcacagacc ctggagccct ctcaccttct cccctgcccc gctgaccgtc 60
 caggaaggcg agaagtcacc cttcacctgc agcctcgccg acatccccga cagcttcgtg 120
 ctgaactggt acagactgag ccccagggaac cagaccgaca agctggccgc ttccaggag 180
 gacaggatcg aaccggcgag ggacaggagg tttaggggtca tgaggctgcc caacggcagg 240
 gacttcaca tgtccatcgt ggccgcccaga ctgaacgact ccggcatcta cctgtgcggc 300
 gctatctacc tgccccccaa caccagatc aacgagagcc ccagggccga actgagcgtg 360
 acagagagaa ccctggaacc tccccccag agcccttccc ctctcctag actgagcgga 420
 cagctgcagg gcctggtggg taccgacaaa actcacacat gcccaccgtg cccagcacct 480
 gaactcctgg ggggaccgtc agtcttctc ttccccccaa aacccaagga caccctcatg 540
 atctcccga cccctgaggt cacatgcgtg gtggtggacg tgagccacga agaccctgag 600
 gtcaagtcca actggtacgt ggacggcgtg gaggtgcata atgccaagac aaagccggcg 660
 gaggagcagt acaacagcac gtaccgtgtg gtcacgctcc tcaccgtcct gcaccaggac 720
 tggctgaatg gcaaggagta caagtgcagg gtctccaaca agccctccc agccccatc 780
 gagaaaacca tctccaaagc caaagggcag ccccgagaa caccaggtga caccctgcc 840
 ccatccgggg atgagctgac caagaaccag gtcagcctga cctgcctggt caaaggcttc 900
 tatcccagcg acatcgccgt ggagtgggag agcaatgggc agccggagaa caactacaag 960
 accacgcctc ccgtgctgga ctccgacggc tctcttctcc tctacagcaa gctcaccgtg 1020
 gacaagagca ggtggcagca ggggaacgtc ttctcatgct ccgtgatgca tgaggctctg 1080
 cacaaccact acacgcagaa gagcctctcc ctgtctccgg gtaaatga 1128

SEQ ID NO: 118 moltype = AA length = 375
 FEATURE Location/Qualifiers
 REGION 1..375
 note = canine and human sequence
 source 1..375
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 118
 LDSPDRPWSP LTFSPAQLTV QEGENATFTC SLADIPDSFV LNWYRLSPRN QTDKLAAPQE 60
 DRIEPGRDRR FRVMRLPNGR DFHMSIVAAR LNDSGIYLCG AIYLPPTQI NESPRAELSV 120
 TERTLEPPTQ SPSPPPRLSG QLQGLVGTDK THTCPPCAP ELLGGPSVFL FPPKPKDTLM 180
 ISRTPEVTCV VVDVSHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD 240
 WLNKEYKKK VSNKALPAPI EKTISKAKGQ PREPQVYTLPSRDELTKNQ VSLTCLVKGF 300
 YPSDIAVEWE SNGQPENNYK TTPPVLDSDG SFPLYSKLTV DKSRRWQQNV FSCVMHEAL 360

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HNHYTQKSLS LSPGK 375

SEQ ID NO: 119 moltype = DNA length = 816
 FEATURE Location/Qualifiers
 source 1..816
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 119

tttacgatca	cagtttctaa	ggacctgtat	gtggtagagt	atggtggcaa	tgtgacaatg	60
gaatgcaaat	tcccggtgga	aaaacagtta	aacttgtttg	cactaatcgt	ctactgggaa	120
atggaggata	aaaaaattat	acaatttgtg	aatggaaagg	aagacctgaa	agttcagcac	180
agcagctaca	gccagagggc	tcagctattg	aaggaccagc	tcttcttggg	gaaggctgcg	240
cttcagatca	cagatgtgag	attgcaggat	gcaggggttt	actgctgctt	gacgggctat	300
ggcgggtgctg	actacaagcg	gattactttg	aaagttcatg	ccccgtaccg	caacatcagc	360
caaagaattt	ctgtgggatcc	gtcacctct	gaacatgaac	taatgtgtca	ggctgagggt	420
taccctgagg	ctgaagtcac	ctggacaagc	agtgaaccac	gagtcctgag	tggcaaaacc	480
accatcacta	attccaatag	ggaagagaag	cttttcaatg	tgaccagcac	gctgaacatc	540
aatgcaacag	ctaatagagat	tttctactgc	acttttcaaa	gatcaggtcc	tgaggaaaac	600
aatactgccg	agttggtcac	cccagaacga	ctgcccgttc	cagcaagtga	gaggactcat	660
ttcatgatcc	tgggaccttt	cctgttgctt	cttggtgtag	tcctggcagt	cactttctgt	720
ctaaaaaaac	atggggagaat	gatggatgtg	gaaaaatgtt	gcacccgaga	taggaactca	780
aagaaacgaa	atgatataca	attgaagag	acataa			816

SEQ ID NO: 120 moltype = AA length = 271
 FEATURE Location/Qualifiers
 source 1..271
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 120

FTITVSKDLY	VVEYGGNVTM	ECKFPVEKQL	NLFALIVWE	MEDKKIIQFV	NGKEDLKVQH	60
SSYSQRAQLL	KDQLFLGKAA	LQITDVRLQD	AGVYCCLIGY	GGADYKRITL	KVHAPYRNIS	120
QRISVDPVTS	EHLMCQAEQ	YPEAEVIWTS	SDHRVLSGKT	TITNSNREEK	LFNVTSTLNI	180
NATANEIFYC	TFQRSQPEEN	NTAELVIPER	LPVPASERTH	FMILGPFLLL	LGVVLAVTFC	240
LKKHGRMDIV	EKCCTDRNS	KKRNDIQFEE	T			271

SEQ ID NO: 121 moltype = DNA length = 660
 FEATURE Location/Qualifiers
 source 1..660
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 121

tttaccatca	ccgtgtccaa	ggacctgtac	gtggtcgagt	acggcggcaa	tgtgaccatg	60
gagtgcgaagt	tcccgttgga	gaagcagctg	aacctgttcg	ccctcatcgt	gtactgggag	120
atggaggaca	agaagatcat	ccagttcgtg	aacggcaagg	aggacctgaa	ggtgcagcac	180
tccagctact	cccagagagc	ccagctgctg	aaggaccagc	tgttcctggg	caaggccgcc	240
ctgcagatca	ccgacgtgag	actgcaggac	gccggcgtgt	attgctgcct	gacgggctac	300
ggaggcgccg	actacaagag	gatcacccctg	aagggtgcatg	cacctacag	gaacatcagc	360
cagaggatca	gcgtcgatcc	cgtgaccagc	gagcacgagc	tgatgtgcca	agccgagggc	420
tatcccgagg	ccgaagtgat	ctggaccagc	agcgaccaca	gggtcctgag	cggcaagacc	480
accatcacca	acagcaacag	ggaggagaag	ctgttcaacg	tgaccagcac	cctcaacatc	540
aacgccaccg	ccaacgagat	cttctactgc	accttcacaga	ggagcggccc	cgaagagaaac	600
aacaccgccg	agctggtgat	ccccgagaga	ctgcctgtgc	ctgccagcga	gaggaccac	660

SEQ ID NO: 122 moltype = AA length = 220
 FEATURE Location/Qualifiers
 source 1..220
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 122

FTITVSKDLY	VVEYGGNVTM	ECKFPVEKQL	NLFALIVWE	MEDKKIIQFV	NGKEDLKVQH	60
SSYSQRAQLL	KDQLFLGKAA	LQITDVRLQD	AGVYCCLIGY	GGADYKRITL	KVHAPYRNIS	120
QRISVDPVTS	EHLMCQAEQ	YPEAEVIWTS	SDHRVLSGKT	TITNSNREEK	LFNVTSTLNI	180
NATANEIFYC	TFQRSQPEEN	NTAELVIPER	LPVPASERTH			220

SEQ ID NO: 123 moltype = DNA length = 1350
 FEATURE Location/Qualifiers
 source 1..1350
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 123

tttaccatca	ccgtgtccaa	ggacctgtac	gtggtcgagt	acggcggcaa	tgtgaccatg	60
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gagtgcgaagt tccccgtgga gaagcagctg aacctgttcg ccctcatcgt gtactgggag 120
atggaggagaca agaagatcat ccagttcgtg aacggcaagg aggaacctgaa ggtgcagcac 180
tccagctact cccagagagc ccagctgctg aaggaccagc tggtcctggg caaggccgcc 240
ctgcagatca ccgacgtgag actgcaggac gccggcggtg attgctgctt gatcggctac 300
ggaggcgccg actacaagag gatcacctg aagggtgcatg caccctacag gaacatcagc 360
cagaggatca gcgtcgatcc cgtgaccagc gagcacgagc tgatgtgcca agccgagggc 420
tatcccgagg ccgaagtgat ttggaccagc agcgaccaca gggtcctgag cggcaagacc 480
accatcacca acagcaacag ggaggagaag ctgttcaacg tgaccagcac cctcaacatc 540
aacgccaccg ccaacgagat cttctactgc acctccaga ggagcgcccc cgaagagaa 600
aacaccgccc agctggtgat ccccgagaga ctgctgtgct ctgccagcga gaggaccac 660
ggtaccgaca aaactcacac atgccaccg tgcaccagc ctgaactcct ggggggaccg 720
tcagtcttcc tcttcccccc aaaacccaag gacacctca tgatctcccc gaccctgag 780
gtcacatgcy tgggtggtgga cgtgagccac gaagaccctg aggtcaagtt caactggtac 840
gtggacggcg tggaggtgca taatgccaa acaaagccgc gggaggagca gtacaacagc 900
acgtaccgtg tggtcagcgt cctcaccgtc ctgaccagc actggtgtaa tggcaaggag 960
tacaagtga aggtctccaa caaagccctc ccagcccca tcgagaaaac catctccaaa 1020
gccaagggc agcccccaga accacaggtg tacaccctgc cccatcccc ggatgagctg 1080
accaagaacc aggtcagcct gacctgctg gtcaaaggct tctatccag cgacatcgcc 1140
gtggagtggt agagcaatgg gcagccggag aacaactaca agaccagcc tccgtgctg 1200
gactccgacg gctccttctt cctctacagc aagctcaccg tggacaagag cagggtggcag 1260
caggggaacg tcttctcatg ctccgtgatg catgaggctc tgcacaacca ctacacgcag 1320
aagagcctct cctgtctccc gggtaaatga 1350

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SEQ ID NO: 124 moltype = AA length = 449
FEATURE Location/Qualifiers
source 1..449
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 124
FTITVSKDLY VVEYGGNVMT ECKFPVEKQL NLFALIVWE MEDKKIIQFV NGKEDLKVQH 60
SSYSQRAQLL KDQLFLGKAA LQITDVRLQD AGVYCLLIGY GGADYKRITL KVHAPYRNIS 120
QRISVDPVTS EHELMCQAEQ YPEAEVIWTS SDHRLVSGKT TITNSNREEK LFNVTSTLNI 180
NATANEIFYC TFQSGPREN NTAELVIPER LPVPASERTH GTDKTHTCPP CPAPELLGGP 240
SVFLFPKPKP DTLMSRTP ETCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSRDEL 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSQSVM HEALHNYHTQ KSLSLSPGK 449

SEQ ID NO: 125 moltype = DNA length = 648
FEATURE Location/Qualifiers
source 1..648
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 125
ctgggcgggc ccagcgtgct gatcttcccc cccaagccca aggaatcctt gagaatcacc 60
agaacccccg aggtgacctg cgtggtgctg gacctgggca gagaggaccg cgagggtgag 120
atcagctggt tcgtggacgg caaggaggtg cacaccgcca agaccagag cagagagcag 180
cagttcaacg gcacctacag agtggtgagc gtgctgcccc tcgagcacca ggactggctg 240
accggcaagg agttcaagtg cagagtgaac cacatcgacc tgcccagccc catcgagaga 300
accatcagca aggccagagg cagagcccac aagcccagcg tgtacgtgct gccccccagc 360
cccaaggagc tgagcagcag cgacaccgtg agcatcacct gcctgatcaa ggacttctac 420
ccccccgaca tcgacgtgga gtggcagagc aacggccagc aggaagccga gagaaagcac 480
agaatgaccc cccccagct ggacgaggac ggcagctact tctgtacag caagctgagc 540
gtggacaaga gcagatggca gcaggcgac ccttcacat gcgcctgat gcacgagacc 600
ctgcagaacc actacaccga cctgagcctg agccacagcc ccggcaag 648

SEQ ID NO: 126 moltype = AA length = 216
FEATURE Location/Qualifiers
source 1..216
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 126
LGGPSVLIFP PKPKDILRIT RTPEVTCVVL DLGREDPEVQ ISWFDVGKEV HTAKTQSREQ 60
QFNGTYRVVS VLPIDHQDWL TGKEFKCRVN HIDLPSPIER TISKARGRAH KPSVYVLPPS 120
PKELSSSDTV SITCLIKDFY PPIDVWEQWS NGQQEPERKH RMTPPQLDED GSYFLYSKLS 180
VDKSRWQQGD PFTCAVMHET LQNHVTDLSL SHSPGK 216

SEQ ID NO: 127 moltype = DNA length = 648
FEATURE Location/Qualifiers
source 1..648
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

-continued

SEQUENCE: 127
ctgggcgggc ccagcgtgtt catcttcccc cccaagccca aggacatcct gagaatcacc 60
agaacccccg agatcacctg cgtggtgctg gacctgggca gagaggaccc cgagggtgcag 120
atcagctggt tcgtggacag caaggaggtg cacaccgcca agaccagacc cagagagcag 180
cagttcaaca gcacctacag agtggtagag gtgctgcccc tcgagcacca ggactggctg 240
accggcaagg agttcaagtg cagagtgaac cacatcggcc tgccccagccc catcgagaga 300
accatcagca agggcagagg ccaggccccc cagcccagcg tgtacgtgct gccccccagc 360
cccaaggagc tgagcagcag cgacaccgtg accctgacct gcctgatcaa ggactttctt 420
ccccccgaga tcgacgtgga gtggcagagc aacggccagc ccgagcccca gagcaagtac 480
cacaccaccg cccccagct ggacgaggac ggcagctact tcctgtacag caagctgagc 540
gtggacaaga gcagatggca gcaggcgagc accttcacct gcgccgtgat gcacgagggc 600
ctgcagaacc actacaccga cctgagcctg agccacagcc ccggcaag 648

SEQ ID NO: 128 moltype = AA length = 216
FEATURE Location/Qualifiers
source 1..216
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 128
LGGPSVFIFP PKPKDILRIT RTPEITCVVL DLGREDPEVQ ISWFVDGKEV HTAKTQPREQ 60
QFNSTYRVVS VLPIDHQDWL TGKEFKCRVN HIGLPSPIER TISKARGQAH QPSVYVLPPS 120
PKELSSSDTV TLTCLIKDFP PPEIDVEWQS NGQPEPESKY HTTAPQLDED GSYFLYSKLS 180
VDKSRWQQGD TFTCAVMHEA LQNHYTDL SL SHSPGK 216

SEQ ID NO: 129 moltype = DNA length = 645
FEATURE Location/Qualifiers
source 1..645
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 129
ctgggcgggc ccagcgtgtt catcttcccc cccaagccca aggacacctt gctgatcgcc 60
agaacccccg aggtgacctg cgtggtggtg gacctggacc ccgaggaccc cgagggtgcag 120
atcagctggt tcgtggacag caagcagatg cagaccgcca agaccagacc cagagaggag 180
cagttcaacg gcacctacag agtggtagag gtgctgcccc tcggccacca ggactggctg 240
aaggcgcaagc agttcacctg caaggtgaac aacaaggccc tgccccagccc catcgagaga 300
accatcagca agggcagagg ccaggccccc cagcccagcg tgtacgtgct gccccccagc 360
agagaggagc tgagcaagaa caccgtgagc ctgacctgcc tgatcaagga cttcttcccc 420
cccgacatcg acgtggagtg gcagagcaac ggccagcagg agcccagag caagtacaga 480
accaccccc cccagctgga cgaggacggc agctacttcc tgtacagcaa gctgagcgtg 540
gacaagagca gatggcagag aggcgacacc ttcactctcg ccgtgatgca cgaggccctg 600
cacaaccact acaccagga gagcctgagc cacagccccc gcaag 645

SEQ ID NO: 130 moltype = AA length = 215
FEATURE Location/Qualifiers
source 1..215
 mol_type = protein
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 130
LGGPSVFIFP PKPKDTLLIA RTPEVTCVVV DLDPEDPEVQ ISWFVDGKQM QTAKTQPREE 60
QFNSTYRVVS VLPIDHQDWL KGKQFTCKVN NKALPSPIER TISKARGQAH QPSVYVLPPS 120
REELSKNTVS LTCLIKDFFP PDIDVEWQSN GQPEPESKYR TTPQLDEDG SYFLYSKLSV 180
DKSRWQRGDT FICAVMHEAL HNHYTQESLS HSPGK 215

SEQ ID NO: 131 moltype = DNA length = 645
FEATURE Location/Qualifiers
source 1..645
 mol_type = unassigned DNA
 organism = Canis lupus
 note = Canis familiaris

SEQUENCE: 131
ctgggcgggc ccagcgtgtt catcttcccc cccaagccca aggacatcct ggtgaccgcc 60
agaacccccca ccgtgacctg cgtggtggtg gacctggacc ccgagaaccc cgagggtgcag 120
atcagctggt tcgtggacag caagcaggtg cagaccgcca acaccagacc cagagaggag 180
cagagcaacg gcacctacag agtggtagag gtgctgcccc tcggccacca ggactggctg 240
agcggcaagc agttcaagtg caaggtgaac aacaaggccc tgccccagccc catcgaggag 300
atcatcagca agaccccccg ccaggccccc cagcccaacg tgtacgtgct gccccccagc 360
agagacgaga tgagcaagaa caccgtgacc ctgacctgcc tgggtgaagga cttcttcccc 420
cccgagatcg accgtggagtg gcagagcaac ggccagcagg agcccagag caagtacaga 480
atgaccccc cccagctgga cgaggacggc agctacttcc tgtacagcaa gctgagcgtg 540
gacaagagca gatggcagag aggcgacacc ttcactctcg ccgtgatgca cgaggccctg 600
cacaaccact acaccagat cagcctgagc cacagccccc gcaag 645

SEQ ID NO: 132 moltype = AA length = 215

-continued

FEATURE
source Location/Qualifiers
1..215
mol_type = protein
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 132
LGGPSVFIFF PKPKDILVTA RTPTVTCVVV DLDPENPEVQ ISWPFVDSKQV QTANTQPRE 60
QSNGTYRVVS VLPIGHQDWL SGKQFKCKVN NKALPSPIEE IISKTPGQAH QPNVYVLPSS 120
RDEMSKNTVT LTCLVKDFFP PEIDVEWQSN GQQEPESKYR MTPPQLDEDG SYFLYSKLSV 180
DKSRWQRGDT FICAVMHEAL HNHYTQISLS HSPGK 215

SEQ ID NO: 133 moltype = DNA length = 867
FEATURE Location/Qualifiers
source 1..867
mol_type = unassigned DNA
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 133
atggggagcc ggccgggggccc ctggccgctc gtctggggcc tgctgcagct gggctggtgg 60
ccaggatggc tcctagagtc ccttgacagg ccttgagacc cgctcacctt ctecccgccg 120
cagctcacgg tgcaggaggg agagaacgcc acgttcacct gcagcctggc cgacatcccc 180
gacagcttcg tgctcaactg gtaccgctg agcccccgc accagacgga caagctggcc 240
gccttcacgg aggaccgcat cgagccgggc cgggacaggc gcttcgcgct catgcggtg 300
cccaacgggc gggacttcca catgagcatc gtccgctcgc gcctcaacga cagcgccatc 360
tacctgtgcg gggccatcta cctgcccccc aacacacaga tcaacgagag tccccgcgca 420
gagctctccg tgacggagag aacctgggag cccccacac agagccccag cccccaccc 480
agactcagcg gccagttgca ggggctggtc atcggcgcta cgagcgtgct ggtgggtgct 540
ctgctactgc tgcgtgtgac ctgggtcctg gccgctgtct tccccagggc caccgaggt 600
gcctgtgtgt ggggagcgca ggaacgagct ctgaaggagg gcccgatgc agcgcccgct 660
ttcaccctgg actacgggga gctggacttc cagtggcgag agaagacgcc ggagcccccg 720
gcgcctgtg ccccgaggca gaccgagtat gccaccatcg tcttccccgg caggccggcg 780
tccccgggcc gcaggggcctc ggccagcagc ctgcagggag cccagcctcc gagccccgag 840
gacggacccg gcctgtggcc cctctga 867

SEQ ID NO: 134 moltype = AA length = 288
FEATURE Location/Qualifiers
source 1..288
mol_type = protein
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 134
MGSRRGPWPL VWAVLQLGWW PGWLLDSPDR PWSPLTFSPA QLTVQEGENA TFTCSLADIP 60
DSFVLNMYRL SPRNQTDKLA AFQEDRIEPG RDRFRFVRL PNGRDFHMSI VAARLNDGI 120
YLCGAIYLP NQINESPRA ELSVTERTLE PPTQSPSPPP RLSGQLQGLV IGVTSLVGV 180
LLLLLLTWL AAVFPRATRG ACVCGSEDEP LKEGPDAAV FTLDYGELDF QWREKTPPEP 240
APCAPEQTEY ATIVFPGRPA SPGRRASASS LQGAQPPSPE DGPGLWPL 288

SEQ ID NO: 135 moltype = DNA length = 870
FEATURE Location/Qualifiers
source 1..870
mol_type = unassigned DNA
organism = Canis lupus
note = Canis familiaris

SEQUENCE: 135
atgagaatgt ttagtgtctt tacattcatg gcctactgcc atttgctaaa agcattttacg 60
atcacagttt ctaaggacct gtatgtggta gagtatggtg gcaatgtgac aatggaatgc 120
aaattcccg tggaaaaaca gttaaacttg tttgcactaa tcgtctactg ggaatggag 180
gataaaaaaa ttatacaatt tgtgaatgga aaggaagacc tgaagttca gcacagcagc 240
tacagccaga gggctcagct attgaaggac cagctcttct tggggaaggc tgcgcttcag 300
atcacagatg tgagattgca ggaatgcagg gtttactgct gcttgatcgg ctatggcggt 360
gctgactaca agcggattac tttgaaagt catgccccg accgcaacat cagccaaaga 420
attttctgtg atcctgtcac ctctgaacat gaactaatgt gtcaggctga gggttaccct 480
gaggctgaag tcactctggc aagcagtgac caccgagtc tgagtggcaa aaccaccatc 540
actaattcca atagggaaga gaagcttttc aatgtgacca gcacgctgaa catcaatgca 600
acagctaatg agattttcta ctgcactttt caaagatcag gtcctgagga aaacaatact 660
gccgagttgg tcattcccaga acgactgccc gttccagcaa gtgagaggac tcatttcatg 720
attctgggac ctttctgtgt gcttcttggg gtatgctctg cagtcacttt ctgtctaaaa 780
aaacatggga gaatgatgga tgtggaaaaa tgttgacccc gagataggaa ctcaaagaaa 840
cgaaatgata tacaatttga agagacataa 870

SEQ ID NO: 136 moltype = AA length = 289
FEATURE Location/Qualifiers
source 1..289
mol_type = protein
organism = Canis lupus
note = Canis familiaris

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SEQUENCE: 136
MRMFSVFTFM AYCHLLKAPT ITVSKDLYVV EYGGNVTMEC KFPVEKQLNL FALIVYWEME 60
DKKIIQFVNG KEDLKVQHSS YSQRALLKD QLFLGKAALQ ITDVRLQDAG VYCCCLIGYGG 120
ADYKRITLKV HAPYRNISQR ISVDPVTSEH ELMCQAEQYP EAEVIWTSSD HRVLSGKTTI 180
TNSNREEKLF NVTSTLNINA TANEIFYCTF QRSQPEENNT AELVIPERLP VPASERTHFM 240
ILGPFLLLG VVLAVTFCLK KHGRMDVEK CCTDRNSKK RNDIQFEET 289

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SEQ ID NO: 137      moltype = AA  length = 401
FEATURE            Location/Qualifiers
source             1..401
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

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SEQUENCE: 137
MNFLLSWVHW SLALLLYLHH AKWSQALDSP DRPWSPLTFS PAQLTVQEGE NATFTCSLAD 60
IPDSFVLNMY RLSPRNQTDK LAAFQEDRIE PGRDRRFRVM RLPNGRDFHM SIVAARLND 120
GIYLCGAIYL PPNTQINESP RAELSVTERT LEPPTQSPSP PPRLSGQLQG LVGTDKTHTC 180
PPCPAPELLG GPSVFLFPPK PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN 240
AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYCKVSNK ALPAPIEKTI SKAKGQPREP 300
QVYTLPPSRD ELTKNQVSLT CLVKGFPYPSD IAVEWESNGQ PENNYKTPPP VLDSDGSFFL 360
YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 401

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SEQ ID NO: 138      moltype = AA  length = 15
FEATURE            Location/Qualifiers
VARIANT            9
source             note = misc_feature - Xaa can be Met or Thr
                   1..15
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

```

```

SEQUENCE: 138
GRDRRFRVXR LPNGR 15

```

```

SEQ ID NO: 139      moltype = AA  length = 15
FEATURE            Location/Qualifiers
VARIANT            14
source             note = MISC_FEATURE - Xaa can be Met or Thr
                   1..15
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

```

```

SEQUENCE: 139
DRIEPGRDRR FRVXR 15

```

```

SEQ ID NO: 140      moltype = AA  length = 25
FEATURE            Location/Qualifiers
VARIANT            5
source             note = MISC_FEATURE - Xaa can be Met or Thr
                   1..25
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

```

```

SEQUENCE: 140
RFRVXRLPNG RPFHMSIVAA RLND 25

```

```

SEQ ID NO: 141      moltype = AA  length = 26
FEATURE            Location/Qualifiers
VARIANT            20
source             note = MISC_FEATURE - Xaa can be Met or Thr
                   1..26
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

```

```

SEQUENCE: 141
LAAFQEDRIE PGRDRRFRVX RLPNGR 26

```

```

SEQ ID NO: 142      moltype = AA  length = 31
FEATURE            Location/Qualifiers
VARIANT            15
source             note = MISC_FEATURE - Xaa can be Met or Thr
                   1..31
                   mol_type = protein
                   organism = Canis lupus
                   note = Canis familiaris

```

```

SEQUENCE: 142
EDRIEPGRDR RFRVXRLPNG RPFHMSIVAA R 31

```

-continued

```

SEQ ID NO: 143      moltype = AA  length = 31
FEATURE            Location/Qualifiers
VARIANT            25
                    note = MISC_FEATURE - Xaa can be Met or Thr
source             1..31
                    mol_type = protein
                    organism = Canis lupus
                    note = Canis familiaris

SEQUENCE: 143
NQTDKLAAFQ EDRIEGRDR RFRVXRLPNG R                               31

SEQ ID NO: 144      moltype = AA  length = 45
FEATURE            Location/Qualifiers
VARIANT            25
                    note = MISC_FEATURE - Xaa can be Met or Thr
source             1..45
                    mol_type = protein
                    organism = Canis lupus
                    note = Canis familiaris

SEQUENCE: 144
NQTDKLAAFQ EDRIEGRDR RFRVXRLPNG RDFHMSIVAA RLNDS                45

SEQ ID NO: 145      moltype = AA  length = 20
FEATURE            Location/Qualifiers
VARIANT            14
                    note = MISC_FEATURE - Xaa can be Met or Thr
source             1..20
                    mol_type = protein
                    organism = Canis lupus
                    note = Canis familiaris

SEQUENCE: 145
DRIEGRDRR FRVXRLPNGR                                           20

SEQ ID NO: 146      moltype = AA  length = 11
FEATURE            Location/Qualifiers
source             1..11
                    mol_type = protein
                    organism = Mus musculus

SEQUENCE: 146
IFYDYDGDDID V                                                  11

```

We claim:

1. An isolated nucleic acid that encodes a canine or caninized IgG heavy chain, wherein the heavy chain comprises a variable heavy chain (VH) region, the VH region comprising:

- a heavy chain complementary determining region 1 (VH CDR1) comprising the amino acid sequence of SEQ ID NO: 29;
- a heavy chain complementary determining region 2 (VH CDR2) comprising the amino acid sequence of SEQ ID NO: 33; and
- a heavy chain complementary determining region 3 (VH CDR3) comprising the amino acid sequence of SEQ ID NO: 38.

2. The isolated nucleic acid of claim 1, wherein the canine or caninized IgG heavy chain comprises the amino acid sequence of SEQ ID NO: 44, wherein SEQ ID NO: 44 comprises: a proline residue at amino acid position 244, a glycine or alanine residue at amino acid position 271, an asparagine or alanine residue at amino acid position 303, a glycine residue at amino acid position 304, a threonine residue at amino acid position 305 and an alanine residue at amino acid position 333.

3. The isolated nucleic acid of claim 2, wherein SEQ ID NO: 44 comprises an alanine residue at amino acid position 271 or an alanine residue at amino acid position 303 of SEQ

ID NO: 44, or an alanine residue at amino acid position 271 and an alanine residue at amino acid position 303 of SEQ ID NO: 44.

4. The isolated nucleic acid of claim 3, wherein the nucleic acid sequence comprises SEQ ID NO: 43 or a nucleic acid sequence having at least 85% sequence identity to SEQ ID NO: 43 and encoding the heavy chain having an amino acid sequence of SEQ ID NO: 44 having an alanine residue at amino acid position 271 or an alanine residue at amino acid position 303 of SEQ ID NO: 44, or an alanine residue at amino acid position 271 and an alanine residue at amino acid position 303 of SEQ ID NO: 44.

5. The isolated nucleic acid of claim 1, wherein the canine or caninized IgG heavy chain comprises the amino acid sequence of SEQ ID NO: 46, wherein SEQ ID NO: 46 comprises: a proline residue at amino acid position 242, a glycine or alanine residue at amino acid position 269, an asparagine or alanine residue at amino acid position 301, a glycine residue at amino acid position 302, a threonine residue at amino acid position 303 and an alanine residue at amino acid position 331.

6. The isolated nucleic acid of claim 5, wherein SEQ ID NO: 46 comprises an alanine residue at amino acid position 269 or an alanine residue at amino acid position 301 of SEQ ID NO: 46, or an alanine residue at amino acid position 269 and an alanine residue at amino acid position 301 of SEQ ID NO: 46.

7. The isolated nucleic acid of claim 6, wherein the nucleic acid sequence comprises SEQ ID NO: 45 or a nucleic acid sequence having at least 85% sequence identity to SEQ ID NO: 45 and encoding the heavy chain having an amino acid sequence of SEQ ID NO: 46 having an alanine residue at amino acid position 269 or an alanine residue at amino acid position 301 of SEQ ID NO: 46, or an alanine residue at amino acid position 269 and an alanine residue at amino acid position 301 of SEQ ID NO: 46.

8. An expression vector comprising a promoter operatively linked to the nucleic acid sequence of claim 4.

9. An isolated nucleic acid that encodes a canine or caninized light chain, wherein the light chain comprises three Variable light complementary determining regions (VL CDRs), which comprise:

- a light chain complementary determining region 1 (VL CDR1) comprising the amino acid sequence of SEQ ID NO: 15;
- a light chain complementary determining region 2 (VL CDR2) comprising the amino acid sequence of SEQ ID NO: 18; and
- a light chain complementary determining region 3 (VL CDR3) comprising the amino acid sequence of SEQ ID NO: 24.

10. The isolated nucleic acid of claim 9, wherein the canine or caninized light chain is a caninized kappa light chain.

11. The isolated nucleic acid of claim 9, wherein the canine or caninized light chain comprises the amino acid sequence of SEQ ID NO: 78 or at least 85% sequence identity to SEQ ID NO: 78.

12. The isolated nucleic acid of claim 9, wherein the nucleic acid comprises SEQ ID NO: 77 or a sequence at least 85% sequence identity to SEQ ID NO: 77.

13. An expression vector comprising a promoter operatively linked to the nucleic acid sequence of claim 9.

14. The expression vector of claim 8, further comprising the nucleic acid sequence that encodes a canine or caninized light chain, wherein the light chain comprises three Variable light complementary determining regions (VL CDRs), which comprise: a light chain complementary determining

region 1 (VL CDR1) comprising the amino acid sequence of SEQ ID NO: 15; a light chain complementary determining region 2 (VL CDR2) comprising the amino acid sequence of SEQ ID NO: 18; and a light chain complementary determining region 3 (VL CDR3) comprising the amino acid sequence of SEQ ID NO: 24.

15. A host cell comprising the expression vector of claim 8.

16. A host cell comprising the expression vector of claim 13.

17. The host cell of claim 15, further comprising a second expression vector, the second expression vector comprising the nucleic acid sequence that encodes a canine or caninized light chain, wherein the light chain comprises three Variable light complementary determining regions (VL CDRs), which comprise: a light chain complementary determining region 1 (VL CDR1) comprising the amino acid sequence of SEQ ID NO: 15; a light chain complementary determining region 2 (VL CDR2) comprising the amino acid sequence of SEQ ID NO: 18; and a light chain complementary determining region 3 (VL CDR3) comprising the amino acid sequence of SEQ ID NO: 24.

18. A host cell comprising the expression vector of claim 14.

19. The host cell of claim 15, wherein the host cell is selected from any of: an insect cell line, amphibian cell line, bacterial cell line, mammalian cell line, plant cell line and fungal cell line.

20. The host cell of claim 19, wherein the mammalian cell line is selected from any of the group consisting of: Chinese hamster ovary (CHO) cells, NSO, SP2 cells, HeLa cells, baby hamster kidney (BHK) cells, monkey kidney cells (COS), human hepatocellular carcinoma cells (e.g., Hep G2), A549 cells, 3T3 cells and HEK-293 cells.

21. A method comprising:

culturing the host cell of claim 17 in a culture medium for a sufficient amount of time to allow expression of the antibody in the host cell, and
collecting the antibody from the culture medium.

* * * * *